

ME3201 Machine Design Project

Semester 5

Gearbox for an Electrode Slurry Mixer

Final Report

By

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Date of submission	08/11/2023
Due date of submission	08/11/2023

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2 Introduction

As the world becomes more electrified and green energy becomes more prevalent, battery technologies must advance to meet the increased demand. In the automotive sector specifically, by 2030 more than half of new cars set to be electric. To meet this demand streamlined battery production procedure are required.

In battery manufacturing, uniformity of the battery content is crucial for its reliability and performance. To achieve the required homogeneity, electrode slurries are blended using large mixers over a long period of time.

This report consists of the design of a gearbox suitable for such mixers.



Figure 1 – An electrode slurry mixer used in large scale production.

3 Problem Description

With the growing demand for batteries, fuel cells and supercapacitors manufacturers are focused on expanding their production volume drastically. Also, there are investors trying to acquire a portion from this growing market. But the existing solutions available for battery manufacturing seems infeasible at the expected scale of production. Since the slurry mixing process is the most important and time-consuming production process, new designs for large scale electrode slurry mixers have emerged. The gearbox design is targeted to cater the requirements of such mixer.

3.1 Electrode slurry mixer requirements and constraints

To get a considerable improvement in production volume relative to existing mixers, production volume must be at least 500L (most commercially available mixers are from 30L to 200L). There for the gearbox is expected to handle the **maximum mixing capacity of 1000L**.

Electrode slurry mixing is done in multiple phases under different mixing parameters. Generalized mixing speeds for electrode slurries are as follows,

Electrode	RPM
Lithium-ion anode/cathode	300 – 800
Supercapacitor electrode	100 – 500
Fuel cell electrode	300 – 800

The viscosity of the slurry, the kind of mixing apparatus, and the required level of mixing uniformity all influence the mixing RPM selection. Also in between phases, mixing rpm is changed to get the desired rheological properties.

Battery production is done in batches. Each batch would take 4h – 8h hours depending on the type of the slurry being mixed. 2 – 3 batches are produced on average per day. There for estimated operation time would be around **16h a day**.^[1]

Overall, minimum maintenance and maintenance downtime is expected.

4 Background Study

First the slurry mixing process is examined.

4.1 The electrode slurry mixing process.

An electrode slurry consists of two types of materials. The **active material, conductive additives** that facilitates the electron conductivity and the **binder** which binds the active material and conductive additives.[2]

The slurry viscosity changes significantly regard to its composition. CMC (carboxymethylcellulose) is a typical polymer binder. Clearly, the addition of more CMC resulted in a considerable increase in viscosity in 2% CMC slurries over 1% CMC slurries.[3]

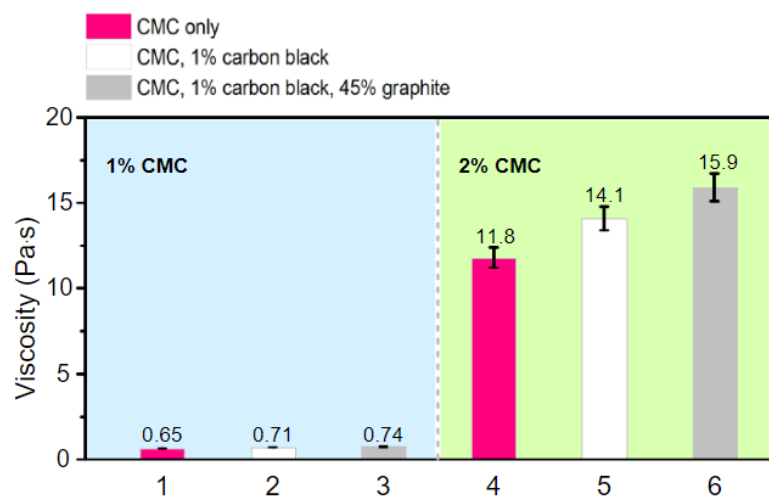


Figure 2 - Viscosities of various slurry compositions

The slurry mixing is carried out in multiple phases,

1. Wet mixing – initial low speed mixing to disperse solid components.
2. Intensive mixing – high speed mixing for maximum homogeneity.
3. Controlled shear rate mixing – controlled mixing to avoid excessive shear-thinning.

Each phase has its own rpm and mixing time depending on the slurry composition and the ultimate quality expected by the manufacturer. In between these phases, rheological tests and other quality control tests are carried out.

4.2 Mixing agitator geometry

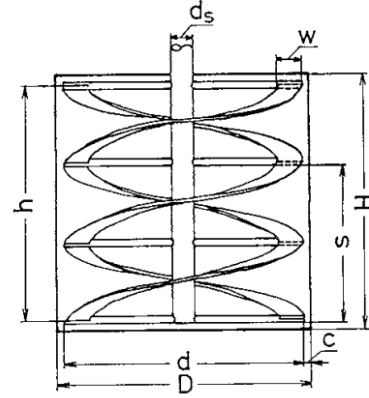
Given the electrode slurries can be highly viscous regards to its composition, double helical ribbon impeller is selected as the agitator. Once the mixing tank dimensions are calculated to match the expected mixing volume and impeller geometry is determined from it.[4]

$$d = 0.96 \times D \quad h = 0.96 \times H$$

$$w = 0.102 \times D \quad s = 0.6 \times h$$

$$\sin \theta_B = \frac{s}{\sqrt{(\pi d)^2 + s^2}} \quad (\theta_B - \text{blade angle})$$

$$L (\text{blade length}) = \frac{h}{\sin \theta_B}$$



Calculated agitator dimensions are then used in required power calculations further on in this report.

4.3 Agitator power calculation methodology

From research texts[4] that investigated the use of helical ribbon impellers to mix viscous non-Newtonian fluids, following equations were obtained to calculate the required power to rotate the impeller.

$$\text{Generalized Reynold's Number } (R_{eg}) = \frac{d^2 N \rho}{\eta_e}$$

$$\text{Power Number } (N_p) = \frac{P}{\rho N^3 d^5}$$

$$R_{eg} \times N_p = \frac{24n_b}{\left[\left(\frac{D}{d} \right)^{0.91} \times \left(\frac{d}{l} \right)^{1.23} \right]}$$

D – vessel diameter

n_b – number of blades

d – impellar diameter

l – blade length

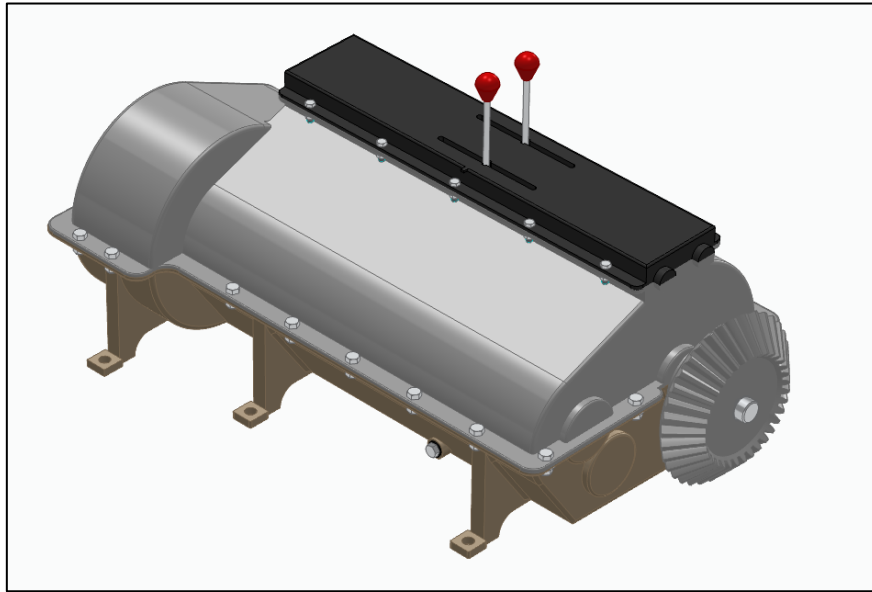
η_e – effective viscosity

ρ – density

N – rotational speed

Also, electrode slurries show non-Newtonian behaviour. Therefore the effective viscosity is used for all the calculations.

5 Gearbox overview



A sliding mesh gearbox with spur gears is implemented to get the required output speeds. Total of 4 gears are available, 3 forward gears and 1 reverse gear. All 3 forward gears are low gears (gear reductions). A pair of bevel gears is used to rotate the direction of power transmission by 90°.

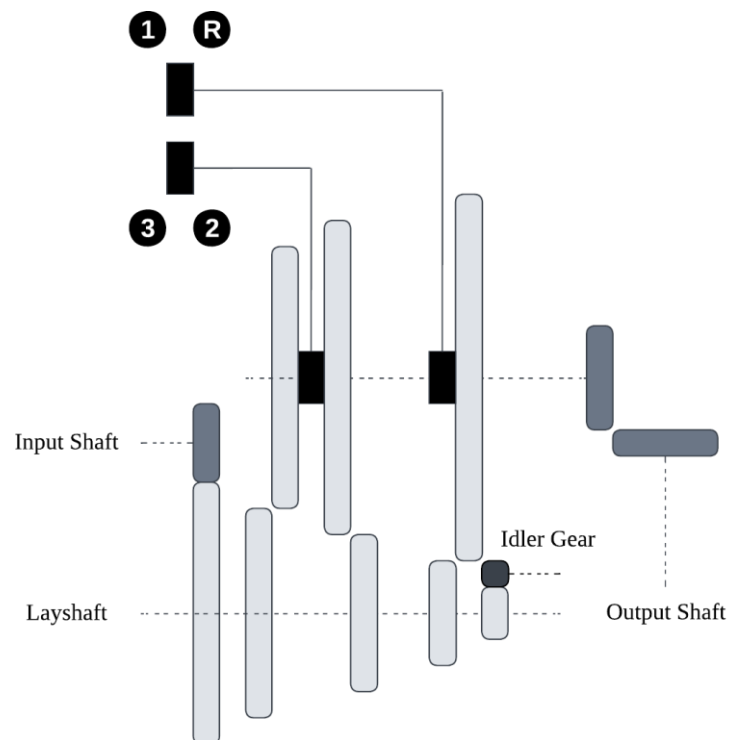


Figure 3 - High-level gearbox diagram

6 Design Calculations and Materials Selection

6.1 Initial Design

6.1.1 Required output calculations.

Power input is decided to be obtained from an A/C Induction motor, as the increased battery manufacturing is part of global electrification and green energy utilization trend. The motor shaft is connected to the input shaft of the gearbox via coupling. Then the power is transmitted to output shaft of the gearbox with relevant gear reduction. Output power from the gearbox is finally given to the mixing impeller with a bevel gear pair.

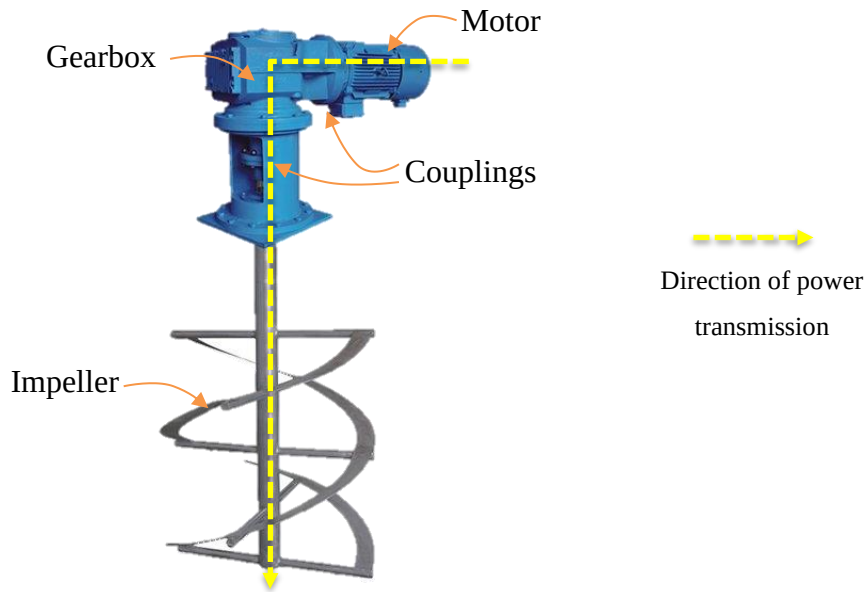


Figure 4 - Basic power flow

From the equations gathered in the background study, following equation is formed to calculate the power required by the mixing agitator. All the symbols have the same meaning as mentioned previously.

$$\text{Required Power } (P) = \frac{24n_b\rho N^3 d^5}{\left[\left(\frac{D}{d}\right)^{0.91} \times \left(\frac{d}{l}\right)^{1.23}\right]} \times \frac{\eta_e}{d^2 N \rho}$$

Varying parameters of rotation speed, fluid viscosity and density yield different required power results. Out of all the combinations of rpm, viscosity and density only the feasible values are selected. For an example rpms less than 300 ($100 < N < 300$) are neglected as its required power is only 1.5% of the maximum power requirement.

Table 1 - Required Power and Torque (for different electrodes, viscosities and speeds)

Tank dimensions						
Height	2m	Diameter	0.8m	Volume	1.0053m	1000l
Mixing agitator dimensions (double helical ribbon)						
Pitch	Diameter	Clearance	Height	Width	$\sin \theta_B$	Length
1.152	0.768m	0.016m	1.92m	0.0816m	0.4309	4.456m
Slurry	Electrode Slurry Properties		Mixing RPM	Power Number (N_p)	Required Power kW	Required Torque
	Effective Viscosity	Density kgm^{-3}				
500 L						
LI	100	1400	800	0.000248	54.50	650.52
LI	20	1400	800	0.000050	10.90	130.10
FC	100	1300	800	0.000267	54.50	650.52
FC	80	1300	800	0.000213	43.60	520.42
FC	50	1300	800	0.000133	27.25	325.26
SC	50	900	800	0.000193	27.25	325.26
SC	40	900	800	0.000154	21.80	260.21
SC	20	900	800	0.000077	10.90	130.10
LI	100	1400	500	0.000396	21.29	406.58
FC	100	1300	500	0.000427	21.29	406.58
FC	80	1300	500	0.000342	17.03	325.26
FC	50	1300	500	0.000213	10.64	203.29
SC	50	900	500	0.000308	10.64	203.29
1000 L						
LI	50	1400	800	0.000291	63.92	762.96
LI	20	1400	800	0.000116	25.57	305.18
FC	50	1300	800	0.000313	63.92	762.96
SC	50	900	800	0.000452	63.92	762.96
SC	40	900	800	0.000362	51.13	610.37
SC	20	900	800	0.000181	25.57	305.18
LI	100	1400	500	0.000930	49.94	953.70
LI	20	1400	500	0.000186	9.99	190.74
FC	100	1300	500	0.001002	49.94	953.70
FC	80	1300	500	0.000801	39.95	762.96
FC	50	1300	500	0.000501	24.97	476.85
SC	50	900	500	0.000723	24.97	476.85
SC	40	900	500	0.000579	19.97	381.48
SC	20	900	500	0.000289	9.99	190.74
LI	100	1400	300	0.000661	7.66	243.95
FC	100	1300	300	0.000712	7.66	243.95
FC	80	1300	300	0.000569	6.13	195.16

(LI – Lithium-ion, FC – fuel-cell, SC – Supercapacitor)

6.1.2 Prime Mover Selection

Considering the scenario of maximum power required to rotate the mixing impeller is around **65kW, 75kW** 3 phase 415v 50Hz cast iron industrial grade A/C motor is selected leaving room for losses.

Output kW	Full Load Speed [RPM]	Frame Size	Shaft Dia [mm]	Current @400v		Efficiency		Power Factor		Torque				Noise Level dB (A)	Weight Foot Mount [Kg]
				Current Full Load [A]	Locked Rotor [%]	Full Load [%]	3/4 Load [%]	Full Load [%]	3/4 Load [%]	Full Load [Nm]	Locked Rotor [%FLT]	Pull Up [%FLT]	Break Down [%FLT]		
55	2970	250M	60	93.3	892	94.5	94.3	0.91	0.90	176.8	302	142	299	88	420
	1480	250M	65	98.7	805	93.6	93.9	0.86	0.83	354.4	264	167	268	77	420
75	2970	280S	65	125.3	898	93.9	93.9	0.93	0.93	241.0	256	105	281	90	572
	1480	280S	75	127.7	756	94.9	95.2	0.90	0.89	482.5	254	211	237	80	570

There are two variants of the 75kw induction motor, one is 2 poles (2P) and the other one is 4 poles (4P). Out of the two, **2 pole motor was selected** after further calculations as there is a significant difference in permissible radial and axial loads. Calculations related to selection can be found in the detailed calculations further in this report.

Frame Size	Permissible Radial Load (N)				Permissible Axial Load (N)			
	2 Pole	4 Pole	6 Pole	8 Pole	2 Pole	4 Pole	6 Pole	8 Pole
250	3590	4650	5400	5980	2725	3780	4575	5225
280	3700	8100	9375	10300	3280	4560	5590	6375

6.1.3 Input Power, Torque and Speed

From motor specification,

Table 2 – Gearbox input parameters.

No. of poles	2
Input speed (RPM) [Rated speed]	2970
Maximum input torque (Nm) [Full load torque]	241.0
Maximum input power (kW) [Motor output]	75

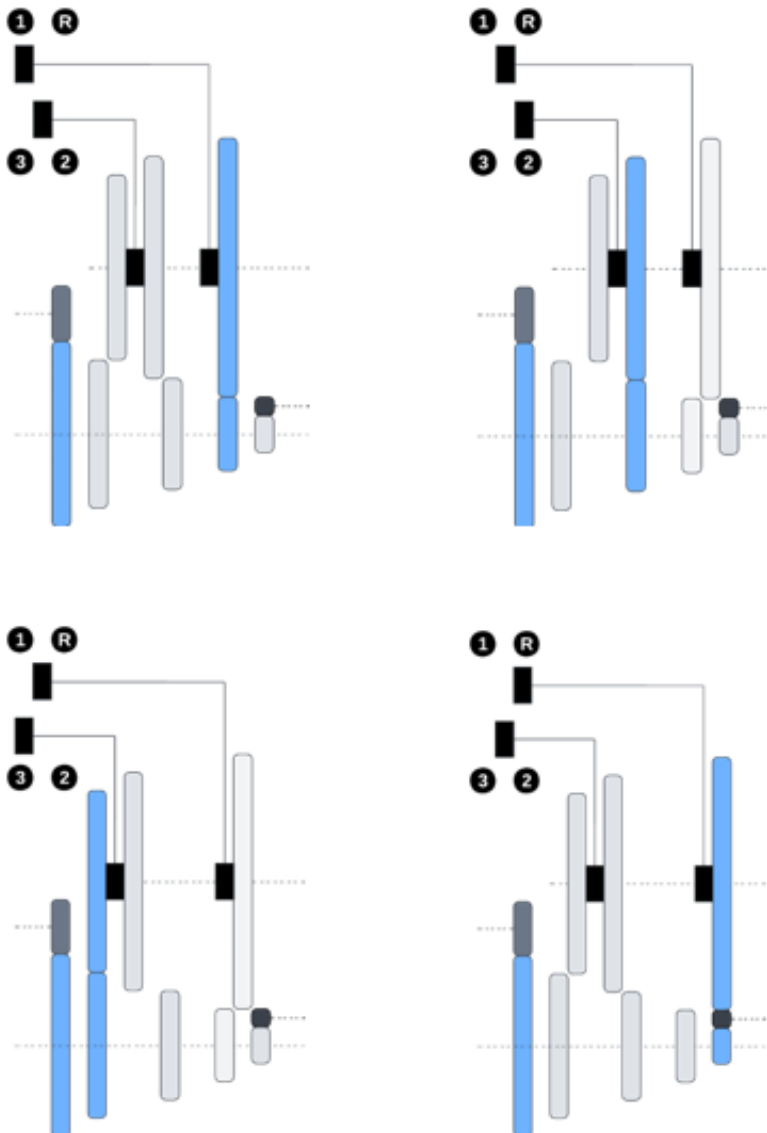
In the output power calculations, for different parameters (slurry viscosity, mixing speed) the power and the torque requirement are different. This requirement is fulfilled by the varying input torque provided by the motor.

6.1.4 Gear ratios

Table 3 - Gear ratios and output speeds

Motor RPM	Gear	Output/Impeller RPM	Gear Ratio
2975(2P)	3	800	2.975
	2	600	1.983
	1	400	1.487
	R	500	2.375

6.1.5 Power transmission in each gear



6.2 Detailed Calculations

6.2.1 Spur Gears

6.2.1.1 Assumptions:

The distance between lay shaft and the main shaft is 300mm.

The pressure angle is 20.

Velocity component was taken by,

$$C_v = \frac{3}{3 + v}$$

Table 28.10. Values of service factor.

Type of load	Type of service		
	Intermittent or 3 hours per day	8-10 hours per day	Continuous 24 hours per day
Steady	0.8	1.00	1.25
Light shock	1.00	1.25	1.54
Medium shock	1.25	1.54	1.80
Heavy shock	1.54	1.80	2.00

8 working hours per day. 300 days per year. The type of load is light shock. Therefore, the service factor is 1.25.

Due to 20-degree full depth involute system,

$$y = 0.154 - \frac{0.912}{T}$$

$$K = 0.111$$

6.2.1.2 Formulae:

$$\frac{T_1}{T_2} = \frac{\omega_1}{\omega_2}$$

$$m = \frac{D}{T}$$

$$P = \pi m$$

T – Teeth

D – pitch circle diameter

m – module

ω – angular velocity

The number of teeth on the pinion (TP) to avoid interference,

$$T_p = \frac{2 A_w}{G \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) (\sin \phi)^2} - 1 \right]}$$

Where,

A_w = Fraction by which the standard addendum for the wheel should be multiplied,

G = Gear ratio or velocity ratio = $TG / TP = DG / DP$,

ϕ = Pressure angle or angle of obliquity.

$$V = \omega \times \frac{D}{2}$$

$$F_t = \frac{P \times C_s}{v}$$

F_t - Actual tangential load

V - pitch line velocity

C_s – Service Factor

P - Power of motor

$$F = \sigma \cdot y \cdot C_v \cdot b \cdot m \cdot \pi$$

F - Allowable Tooth load

b - Face width of gears in mm

σ - Allowable static stress

$$W_I = \frac{21v(bC + W_T)}{21v + \sqrt{bC + W_T}}$$

C - Deformation dynamic factor

$$C = \frac{K_e}{\frac{1}{E_p} + \frac{1}{E_g}}$$

$$W_w = DbQK$$

W_w - Wear load

K - Wear load factor

Q - Ratio factor

$$Q = \frac{2 T_P}{T_P + T_G}$$

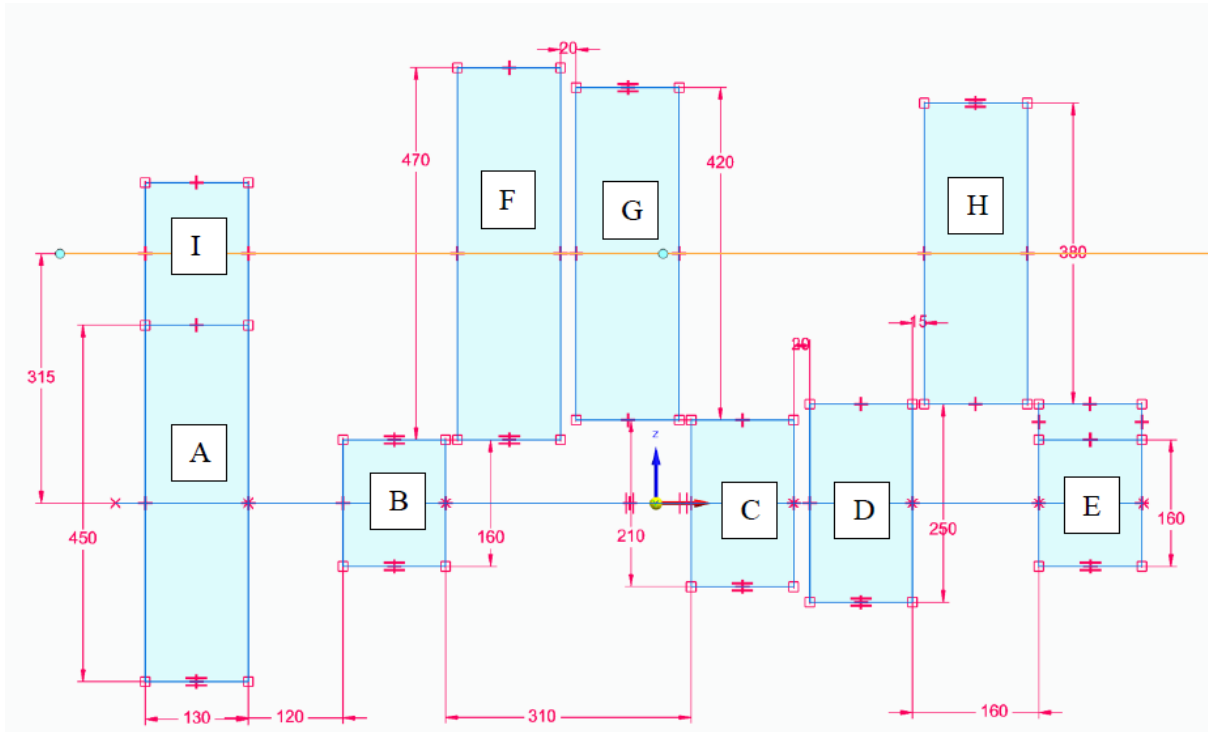
$$K = \frac{\sigma_{es}^2 \sin \phi}{1.4} \left(\frac{1}{E_P} + \frac{1}{E_G} \right)$$

σ_{es} – Surface endurance limit

$$W_D = F_T + W_I$$

W_D – Dynamic load

6.2.1.3 Calculations:



Consider gear ratio between input shaft and the lay shaft as 2.5.

Therefore, to avoid interference,

$$T_p = \frac{2 A_w}{G \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) (\sin \phi)^2} - 1 \right]}$$

$$T_I = \frac{2 \times 1}{2.5 \left[\sqrt{1 + \frac{1}{2.5} \left(\frac{1}{2.5} + 2 \right) (\sin 20)^2} - 1 \right]}$$

$$T_I = 14.68$$

According to the reference book,

Table 28.2. Minimum number of teeth on the pinion in order to avoid interference.

S. No.	Systems of gear teeth	Minimum number of teeth on the pinion
1.	$14\frac{1}{2}^\circ$ Composite	12
2.	$14\frac{1}{2}^\circ$ Full depth involute	32
3.	20° Full depth involute	18
4.	20° Stub involute	14

Therefore,

$$T_I = 18$$

$$T_A = 18 \times 2.5$$

$$T_A = 45$$

$$m = \frac{D}{T}$$

$$\frac{D_P}{2} + \frac{D_G}{2} = L$$

L – length between two shafts

$$D_P + D_G = 2 \times L$$

$$m (T_P + T_G) = 2 \times L$$

$$m = \frac{2 \times 300}{18 + 45}$$

$$m = 9.52$$

According to the standard module values, $m = 10$ mm.

Actual L = 315mm.

Therefore, $D_P = 10 \times 18 = 180$ mm

$$D_G = 10 \times 45 = 450$$
mm

According to the application,

Gear	Output RPM	Gear ratios
1	400	2.975
2	600	1.983333333
3	800	1.4875

Sample calculation for first gear,

$$m (T_P + T_G) = 2 \times L$$

$$\frac{T_G}{T_P} = 2.975$$

$$m (T_P + 2.975 \times T_P) = 2 \times L$$

$$T_P = 15.84$$

Gear teeth in lay shaft and main shaft for each gear,

Gear	Teeth lay shaft	Main shaft gear teeth
1	Gear B and E - 15.8490	Gear F - 47.1509
2	Gear C - 21.1173	Gear G - 41.8826
3	Gear D - 25.3266	Gear H - 37.6733

To avoid interference, pinion gear teeth ≥ 14.68 . Therefore, consider $T_B = T_E$.

First gear pinion wheel and reverse gear pinion wheel are the same. And reverse gear pinion wheel and reverse gear idle wheel are also same.

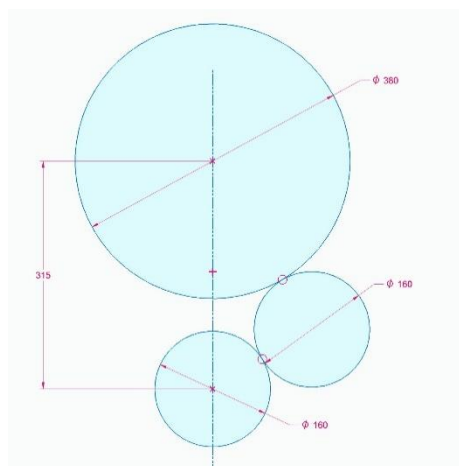


Figure 5 - Reverse gear.

Adjusted gear teeth in lay shaft and main shaft for each gear,

Gear	Main shaft gear teeth	Lay shaft gear teeth
1	Gear F - 47	Gear B and E - 16
2	Gear G - 42	Gear C - 21
3	Gear H - 38	Gear D - 25

Hence new gear ratio values,

Gear	Gear ratios
1	2.9375
2	2
3	1.52
Reverse	2.375

Therefore, new output RPM in main shafts in each gear,

Gear	Output RPM Actual
1	405.1063
2	595
3	782.894
Reverse	501.0526

$$\text{Lay shaft rotational speed} = \frac{2975}{2.5}$$

$$= 1190 \text{ rpm}$$

Pitch circle diameters and pitch line velocity in each gear,

$$v = \frac{2 \times \pi \times 1190}{60} \times \frac{0.16}{2}$$

$$v = 9.9693 \text{ ms}^{-1}$$

Gear	Pitch circle diameter (mm)	Pitch line velocity (ms^{-1})
I	180	28.0387
B & E	160	9.9693
C	210	13.0847
D	250	15.5770
F	470	9.9693
G	420	13.0847
H	380	15.5770

For first gear,

$$F_t = \frac{P \times C_s}{v}$$

$$F_t = \frac{15000 \times 1.25}{9.9693}$$

$$F_t = 9403.85 \text{ N}$$

C_s value is taken from the reference book.

Likewise, for all other gears,

Gear	F_t(N)
B and reverse	9403.85
C	7164.83
D	6018.46
I	3343.592963

For ordinary cut gears,

$$C_v = \frac{3}{3 + v}$$

For first gear,

$$C_v = \frac{3}{3 + 9.9693}$$

$$C_v = 0.2313$$

For all three gears,

Gear	C_v
B and reverse	0.2313
C	0.1865
D	0.1614
I	0.0966

According to the Lewis form factor for 20° full depth involute system,

$$y = 0.154 - \frac{0.912}{T}$$

For pinion gear in lay shaft of first gear,

$$y = 0.154 - \frac{0.912}{16}$$

$$y = 0.097$$

Hence for other pinion gears,

Gear	y
B & reverse	0.097
C	0.11057
D	0.11752
I	0.1033

Width of the gear,

$$b = 13 \text{ x m}$$

$$b = 13 \text{ x } 10$$

$$b = 130\text{mm}$$

Beam strength of the tooth is given by,

$$F_T = \sigma_0 \cdot y \cdot C_v \cdot b \cdot m \cdot \pi$$

$$\sigma_0 = \frac{F_T}{y \cdot C_v \cdot b \cdot m \cdot \pi}$$

For first gear pinion gear wheel,

$$\sigma_0 = \frac{9403.85}{0.097 \times 0.2313 \times 0.13 \times 0.01 \times \pi}$$

$$\sigma_0 = 102.6211141 \text{ Mpa}$$

Allowable static stress values for all three gears,

Gear	σ_0 values (MPa)
B & reverse	102.6211
C	85.0673
D	77.6491
I	81.9713

From the reference book,

Table 28.4. Values of allowable static stress.

Material	Allowable static stress (σ_0) MPa or N/mm ²
Cast iron, ordinary	56
Cast iron, medium grade	70
Cast iron, highest grade	105
Cast steel, untreated	140
Cast steel, heat treated	196
Forged carbon steel-case hardened	126
Forged carbon steel-untreated	140 to 210
Forged carbon steel-heat treated	210 to 245
Alloy steel-case hardened	350
Alloy steel-heat treated	455 to 472
Phosphor bronze	84
<i>Non-metallic materials</i>	
Rawhide, fabroil	42
Bakelite, Micarta, Celoron	56

We selected cast iron highest grade (105Mpa) as gear material.

6.2.1.4 Dynamic tooth load calculation

$$W_D = W_T + W_I$$

$$W_I = \frac{21v(bc + W_T)}{21v + \sqrt{bc + W_T}}$$

Table 28.6. Values of maximum allowable tooth error in action (e) verses pitch line velocity, for well cut commercial gears.

Pitch line velocity (v) m/s	Tooth error in action (e) mm	Pitch line velocity (v) m/s	Tooth error in action (e) mm	Pitch line velocity (v) m/s	Tooth error in action (e) mm
1.25	0.0925	8.75	0.0425	16.25	0.0200
2.5	0.0800	10	0.0375	17.5	0.0175
3.75	0.0700	11.25	0.0325	20	0.0150
5	0.0600	12.5	0.0300	22.5	0.0150
6.25	0.0525	13.75	0.0250	25 and over	0.0125
7.5	0.0475	15	0.0225		

Table 28.7. Values of tooth error in action (e) verses module.

Module (m) in mm	Tooth error in action (e) in mm		
	First class commercial gears	Carefully cut gears	Precision gears
Upto 4	0.051	0.025	0.0125
5	0.055	0.028	0.015
6	0.065	0.032	0.017
7	0.071	0.035	0.0186
8	0.078	0.0386	0.0198
9	0.085	0.042	0.021
10	0.089	0.0445	0.023
12	0.097	0.0487	0.0243
14	0.104	0.052	0.028
16	0.110	0.055	0.030
18	0.114	0.058	0.032
20	0.117	0.059	0.033

According to these two tables e values in pinion gear in each gear,

Gear	e
B & reverse	0.0375
C	0.025
D	0.0225
I	0.0125

$$C = \frac{K_e}{\frac{1}{E_p} + \frac{1}{E_g}}$$

$$K = \frac{\sigma_{es}^2 \sin \phi}{1.4} \left(\frac{1}{E_P} + \frac{1}{E_G} \right)$$

For first gear,

$K = 0.111$ (for 20° full depth involute system)

$$C = \frac{0.111 \times 0.0375}{\frac{1}{140000} + \frac{1}{140000}}$$

$$C = 291.375$$

Deformation factor (C) values for three gears,

Gear	C
B & reverse	291.375
C	194.25
D	174.825
I	97.125

W_D for first gear,

$$W_D = W_T + \frac{21v(bC + W_T)}{21v + \sqrt{bC + W_T}}$$

$$W_D = 9403.85 + \frac{21 \times 9.9693 (0.13 \times 291.375 + 9403.85)}{21 \times 9.9693 + \sqrt{0.13 \times 291.375 + 9403.85}}$$

$$W_D = 32597.03 \text{ N}$$

For all three gears,

Gear	W_D (N)
B & reverse	32597.03
C	26749.4407
D	24951.2864
I	16491.5929

6.2.1.5 Static tooth load calculation

$$W_S = \sigma_e \cdot b \cdot \pi \cdot m \cdot y$$

Table 28.8. Values of flexural endurance limit.

<i>Material of pinion and gear</i>	<i>Brinell hardness number (B.H.N.)</i>	<i>Flexural endurance limit (σ_e) in MPa</i>
Grey cast iron	160	84
Semi-steel	200	126
Phosphor bronze	100	168
Steel	150	252
	200	350
	240	420
	280	490
	300	525
	320	560
	350	595
	360	630
	400 and above	700

For gray cast iron $\sigma_e = 84$ Mpa.

Static tooth load for first gear,

$$W_S = 84 \times 130 \times \pi \times 10 \times 0.097$$

$$W_S = 41596.257 \text{ N}$$

For other gears,

Gear	W_S (N)
B & reverse	41596.257
C	47416.057
D	50395.795
I	44312.1643

For safety, statics tooth load should be greater than dynamic tooth load. For steady loads,

$$W_S \geq 1.25 W_D$$

Gear	$W_S - 1.25 W_D$	Safe or not
B & reverse	849.9615	Safe
C	13979.2570	Safe
D	19206.6876	Safe
I	23697.6732	Safe

6.2.1.6 Wear tooth load calculation

$$W_W = D. b. Q. K$$

Sample calculation for first gear,

$$Q = \frac{2 T_P}{T_P + T_G}$$

$$Q = \frac{2 \times 16}{16 + 47}$$

$$Q = 1.492$$

Table 28.9. Values of surface endurance limit.

Material of pinion and gear	Brinell hardness number (B.H.N.)	Surface endurance limit (σ_{es}) in N/mm^2
Grey cast iron	160	630
Semi-steel	200	630
Phosphor bronze	100	630
Steel	150	350
	200	490
	240	616
	280	721
	300	770
	320	826
	350	910
	400	1050

According to previous table surface endurance limit is 630 N/mm^2 .

$$K = \frac{\sigma_{es}^2 \sin \phi}{1.4} \left(\frac{1}{E_p} + \frac{1}{E_G} \right)$$

$$K = \frac{630^2 \sin 20}{1.4} \left(\frac{1}{140000} + \frac{1}{14000} \right)$$

$$K = 3.6974$$

$$W_W = 160 \times 130 \times 1.492 \times 3.6974$$

$$W_W = 114749.3928 \text{ N}$$

Wear tooth load for all three gears,

Gear	Q	K	W_W (N)
B & reverse	1.4920	3.6974	114749.3928
C	1.3333	3.6974	134586.3889
D	1.2063	3.6974	144962.6637
I	1.4285	3.6974	123599.7449

The wear load should not be less than the dynamic load.

Gear	W_D (N)	W_W (N)	
B & reverse	32597.03	114749.3928	Safe
C	26749.4407	134586.3889	Safe
D	24951.2864	144962.6637	Safe
I	16491.5929	123599.744	Safe

6.2.2 Bevel Gears

Mitre type bevel gears are used to rotate the power transmission direction by 90°.

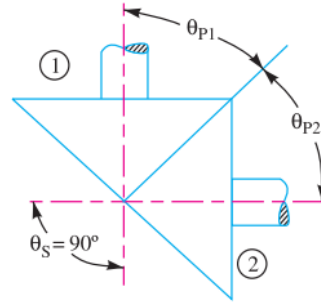


Figure 6 - Mitre gears.

Being the mitre type bevel gears,

$$\text{Velocity ratio (V.R)} = 1$$

$$\text{Pitch angle } (\theta_p) = 45^\circ$$

6.2.2.1 Gear Design Methodology

b = face width

L = slant height of pitch cone

m = module

T = number of teeth

For satisfactory operation of bevel gears, following constraints are satisfied.

$$6.3m \leq b \leq 9.5m$$

$$\frac{L}{b} \leq 3, \quad \therefore T \geq \frac{48}{\sqrt{1 + V.R}} = \frac{48}{\sqrt{2}} = 33.941 = 35$$

To select a suitable module for the bevel gears, the strength of bevel gears is considered. First the tangential load on bevel gear with regards to torque is calculated. Then it is equated with modified Lewis equation for bevel gears to obtain an expression only with module as the variable. Module is selected from the solutions of that equation and considering the standard gear modules.

Tangential load (W_T) obtained from torque,

$$W_T = \frac{\tau}{\frac{D}{2}}$$

$$\tau = \frac{60P}{2\pi N}$$

$\tau = \text{torque}$

$D = \text{diameter}$

From $m = \frac{D}{T}$ and first equation, W_T is adjusted as,

$$W_T = \frac{2\tau}{mT} \times 10^3 \text{ (as } m \text{ kept in mm).}$$

By substituting τ with second equation, $W_T = \frac{60P}{mT\pi N} \times 10^3$

Tangential load (W_T) obtained from Lewis equation,

$$W_T = \sigma_0 \times C_v \times b \times \pi \times m \times y_p \times \left[1 - \frac{b}{L}\right]$$

$\sigma_0 = \text{Allowable static stress}$

$$C_v = \text{Velocity factor} = \frac{3}{3 + v}$$

$$v = \text{peripheral speed} = \frac{\pi DN}{60} = \frac{\pi mTN}{60} \times 10^3$$

$y_p = \text{Lewis factor (for equivalent number of teeth, } T_E)$

$$T_E = T \sec \theta_p$$

For 20° involute teeth, $y_p = 0.154 - (0.912 \cos \theta_p) \frac{1}{T}$

$b = km$ ($k = \text{width factor}$)

$$L = \frac{D}{2 \sin \theta_p}, \quad \therefore \frac{b}{L} = \frac{2k \sin \theta_p}{T}$$

By equating the two equations for W_T , a cubic equation of m is obtained.

$$Am^3 + Bm^2 + Cm + D = 0$$

$$A = 3\sigma_0 N k \pi^2 \times (0.154T - 0.912 \cos \theta_p) \times \left(1 - \frac{2k \sin \theta_p}{T}\right)$$

$$B = 0$$

$$C = -\pi T N P$$

$$D = -180 \times 10^3 \times P$$

($k = 9.5$ is taken considering the constraints)

From the obtained solutions for m , closest highest module is selected from the standard modules list.

6.2.2.2 Force Calculations

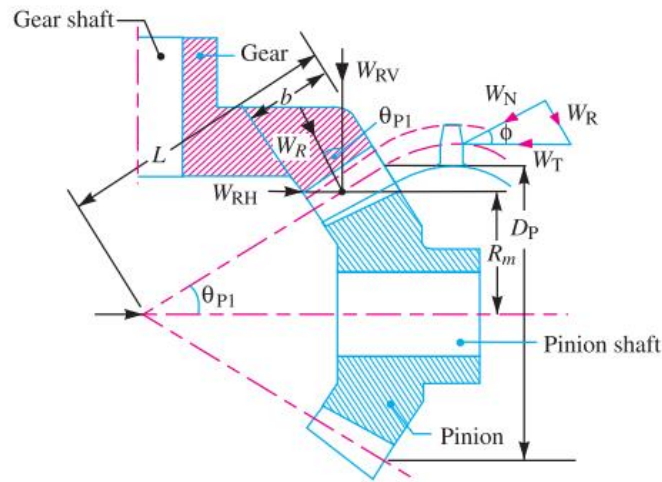


Figure 7 - Forces acting on a bevel gear.

$$W_T = W_N \cos \phi$$

$$W_R = W_T \tan \phi$$

$$W_{RH} = W_R \sin \theta_p$$

$$W_{RV} = W_R \cos \theta_p$$

$$R_m = \left(L - \frac{b}{2}\right) \sin \theta_p$$

6.2.2.3 Module and Forces Computation

Following MATLAB code snippet was used to get solutions for the module, select the best module with regards to standard modules and calculate the forces acting on the bevel gears.

```
T = 35;

% Inputs
N = 500; %rpm
P = 75 * 1000; %W
ast = 105; %MPa
k = 8.5; pressureAngle = deg2rad(14.5); pitchAngle = deg2rad(45);

d = -180*P*(10^3);
c = -1*pi*T*N*P;
a = 3*ast*N*(pi^2)*k*(0.154*T-0.912*cos(pitchAngle))*(1-((2*k*sin(pitchAngle))/T));

polyn = [a 0 c d];
disp(polyn);

R = roots(polyn);
disp("Roots:" + R);

RnP = R(real(R)>0&imag(R)==0);
sol = RnP(RnP >= 0);
m = min(sol);
disp("Module:" + m + "mm");

% Standard modules list
STD_MODULES = [1 1.5 2 2.5 3 4 6 8 10 12];

% Standard module selection
[dif, mIndex] = min(abs(STD_MODULES-m));
if m > STD_MODULES(mIndex)
    m = STD_MODULES(mIndex + 1);
else
    m = STD_MODULES(mIndex);
end
disp("Adjusted Module:" + m + "mm");

D = m * T;
disp("D:" + D + "mm");
L = D / (2*sin(pitchAngle));
disp("L:" + L + "mm");
b = k * m;
disp("b:" + b + "mm");
disp("L/b:" + (L/b));

Rm = (L - (b/2)) * sin(pitchAngle);
disp("Rm:" + Rm + "mm");

v = (pi * T * N * m * (10^-3))/60;
y = 0.154-((0.912*cos(pitchAngle))/T);
cv = 3 / (3+v);

% Maximum allowable tangential force
```



```

wT = ast * cv * k * pi * (m^2) * y * ((L-b)/L);
torque = (P * 60)/(2 * pi * N);
disp("Torque:" + torque + "Nm");

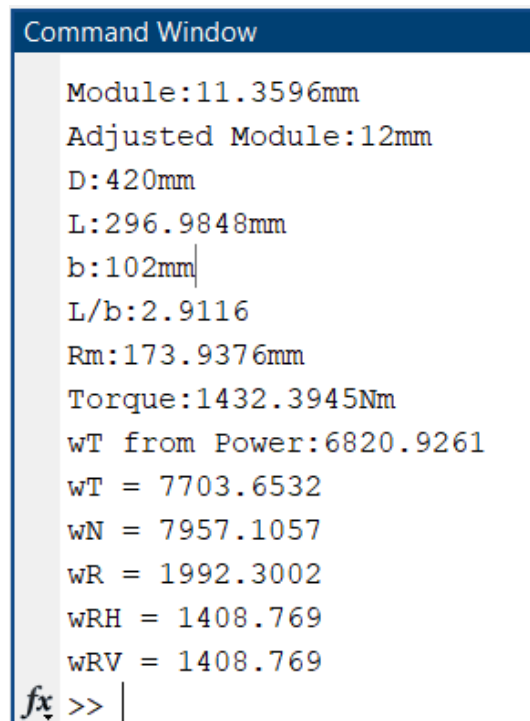
% Actual tangential force
wT2 = (torque * 1000) / (D/2);
disp("wT from Power:" + wT2);
disp("wT = " + wT);

% Calculated forces
wN = wT/cos(pressureAngle);
disp("wN = " + wN);
wR = wT * tan(pressureAngle);
disp("wR = " + wR);
wRH = wR * sin(pitchAngle);
disp("wRH = " + wRH);
wRV = wR * cos(pitchAngle);
disp("wRV = " + wRV);

```

Both maximum allowable tangential force and the actual tangential force is calculated and then compared to make sure that the gear will no fail after the standard module adjustment.

Sample output is shown below,



```

Command Window

Module:11.3596mm
Adjusted Module:12mm
D:420mm
L:296.9848mm
b:102mm
L/b:2.9116
Rm:173.9376mm
Torque:1432.3945Nm
wT from Power:6820.9261
wT = 7703.6532
wN = 7957.1057
wR = 1992.3002
wRH = 1408.769
wRV = 1408.769
fx >>

```

Figure 8 - MATLAB code output.

6.2.2.4 Computed Values

RPM	800	600	400	500 (R)
Module (mm)	10.907	11.1646	11.6362	11.3596
Adjusted Module (mm)	12	12	12	12
D (mm)	420	420	420	420
L (mm)	296.9848	296.9848	296.9848	296.9848
b (mm)	102	102	102	102
L/b	2.991	2.991	2.991	2.991
R _m (mm)	173.937	173.937	173.937	173.937
τ (Nm)	895.25	1193.66	1790.49	1432.39
W_T (N)	4263.08	5684.12	8526.16	6820.92
W_T maximum (N)	5235.63	6657.56	9139.78	7703.65
W_N (N)	5407.89	6876.59	9440.48	7957.12
W_R (N)	1354.02	1721.76	2363.71	1992.30
W_{RH} (N)	957.44	1217.47	1671.39	1408.77
W_{RV} (N)	957.44	1217.47	1671.39	1408.77

Since for all the output torque values, the suitable module(m) is 12 mm, it is selected. And the rest of the bevel gear properties were calculated.

Pitch circle diameter, $D = 420$ mm

Slant height, $L = 297$ mm

Face width, $b = 102$ mm

Addendum, $a = 1m = 12$ mm

Dedendum, $d = 1.2m = 14.2$ mm

Clearance = $0.2m = 2.4$ mm

Working depth = $2m = 24$ mm

Mean radius, $R_m = 174$ mm

Thickness of the tooth = $1.5708m = 18.849$ mm

6.2.3 Shafts

6.2.3.1 Lay shaft

Lay shaft configuration

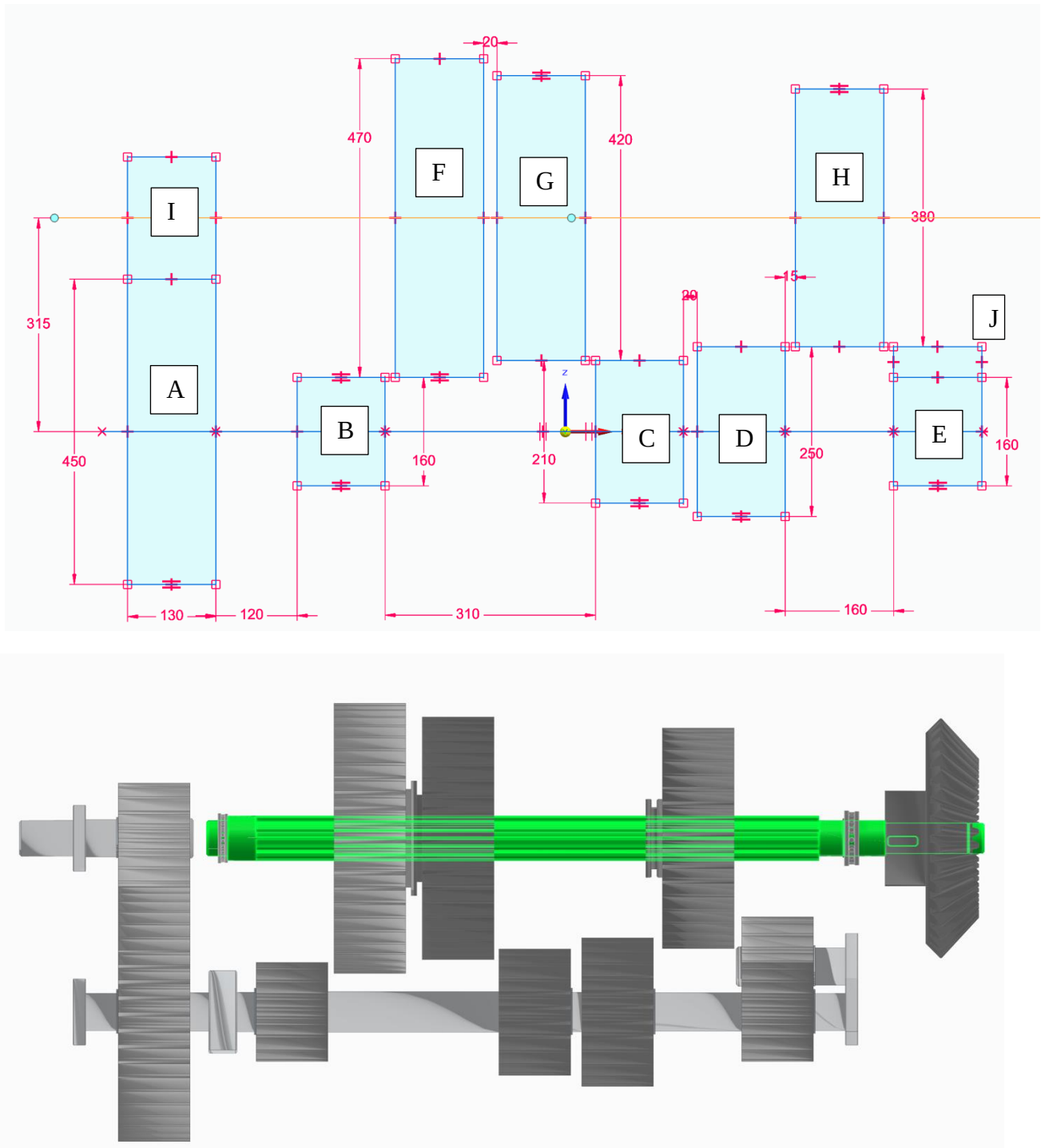


Figure 7: lay shaft configuration

Lay shaft calculation

Step 01: calculate the weight of each gear on layshaft.

The weight of gear is given by,

$$W_G = 0.00118 \times T_G \times b \times m^2$$

T_G = No. of teeth on the gear

b=Face width in mm

m=Module in mm

Gear	A	B	C	D	E
Weight (N)	690	245	322	384	245

Step 02: calculate the total bending moments applied on the layshaft at various gear positions (1st, 2nd, 3rd, reverse). For Calculating the bending moment at each gear position, we consider the moment generated by gears' weight and moment generated due tangential forces on meshing gear pair.

When meshing the first gear

Forces on gear A from the gear I:

$$W_T = 3343 \text{ N}$$

$$W_N = 8193 \text{ N}$$

$$W_G = 690 \text{ N}$$

$$W_R = 8498 \text{ N}$$

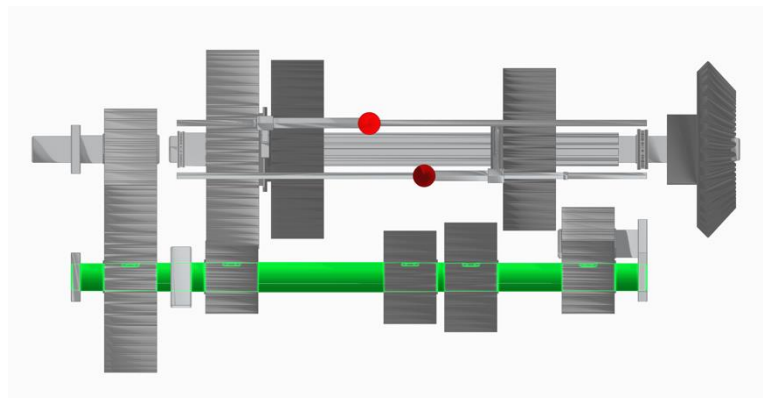


Figure 8: First gear arrangement

Forces on gear B from the gear F:

$$W_T = 9403 \text{ N}$$

$$W_N = 23044 \text{ N}$$

$$W_G = 245 \text{ N}$$

$$W_R = 23145 \text{ N}$$

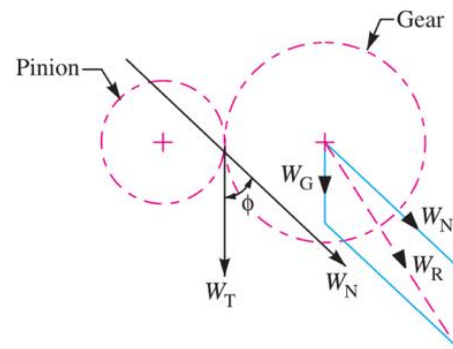


Figure 9: force diagram

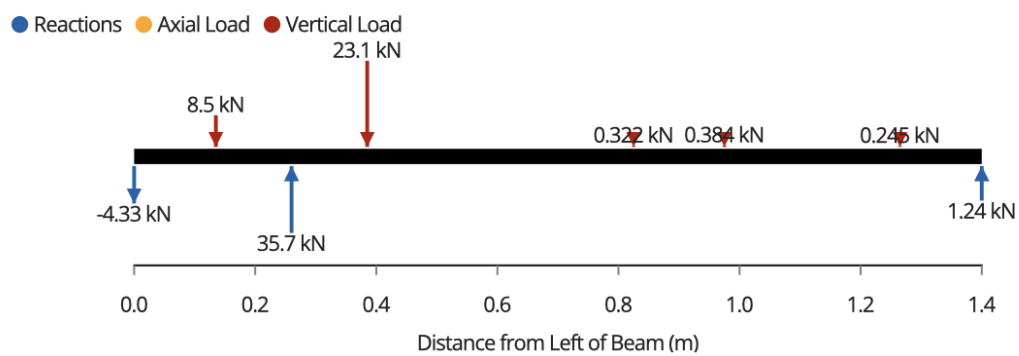


Figure 10: Force configuration on layshaft at first gear

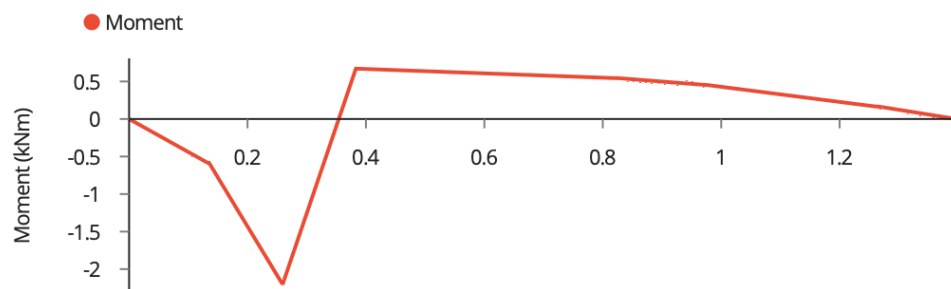


Figure 11: Bending moment diagram of layshaft

Bending moment = 2.19 kNm

When meshing the second gear

Forces on gear A from the gear I:

$$W_T = 3343 \text{ N}$$

$$W_N = 8193 \text{ N}$$

$$W_G = 690 \text{ N}$$

$$W_R = 8498 \text{ N}$$

Forces on gear C from the gear G:

$$W_T = 7164 \text{ N}$$

$$W_N = 17557 \text{ N}$$

$$W_G = 322 \text{ N}$$

$$W_R = 17690 \text{ N}$$

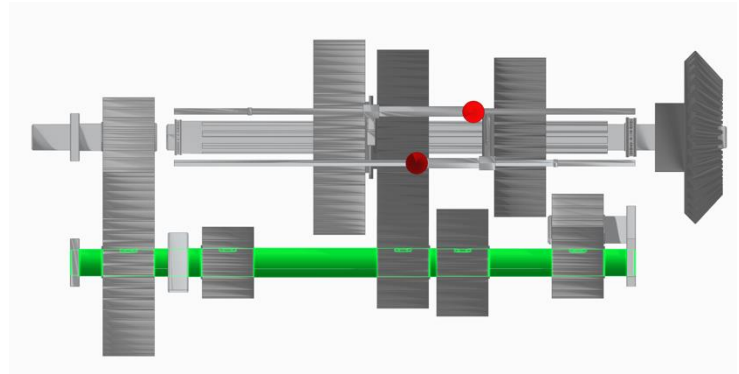


Figure 12: second gear arrangement

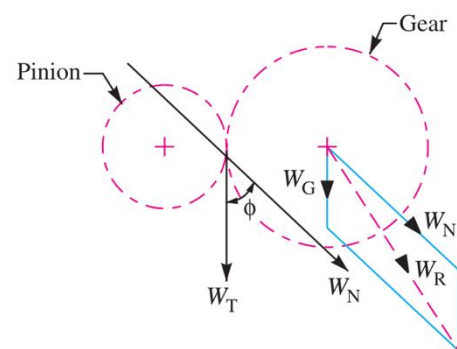


Figure 13: force diagram

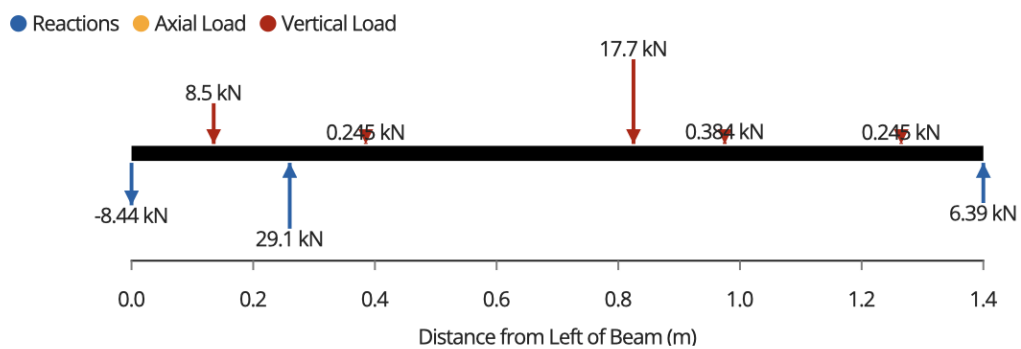


Figure 14: Force configuration on layshaft at second gear

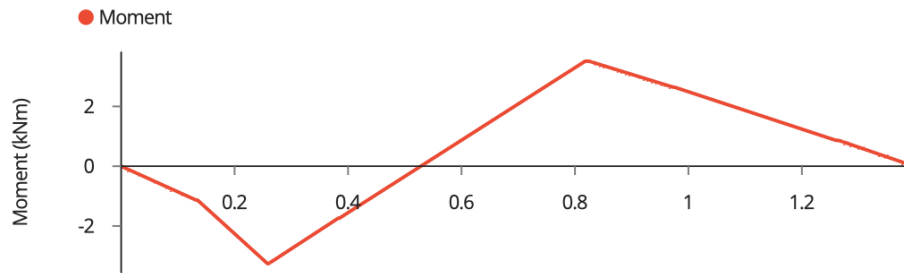


Figure 15: Bending moment diagram of layshaft

Bending moment = 3.51 kNm

When meshing the third gear

Forces on gear A from the gear I:

$$W_T = 3343 \text{ N}$$

$$W_N = 8193 \text{ N}$$

$$W_G = 690 \text{ N}$$

$$W_R = 8498 \text{ N}$$

Forces on gear D from the gear H:

$$W_T = 6018 \text{ N}$$

$$W_N = 14748 \text{ N}$$

$$W_G = 384 \text{ N}$$

$$W_R = 15132 \text{ N}$$

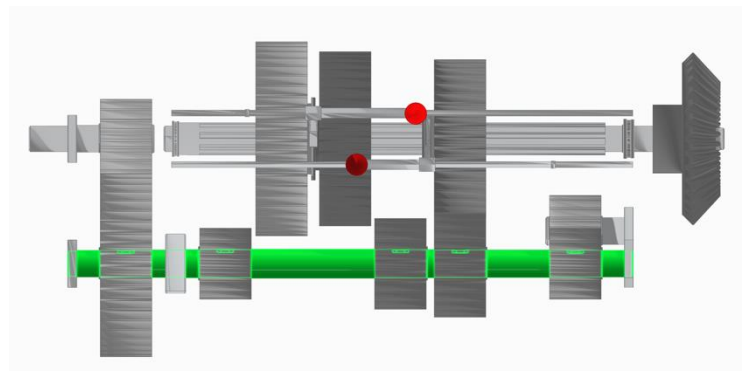


Figure 16: Third gear arrangement

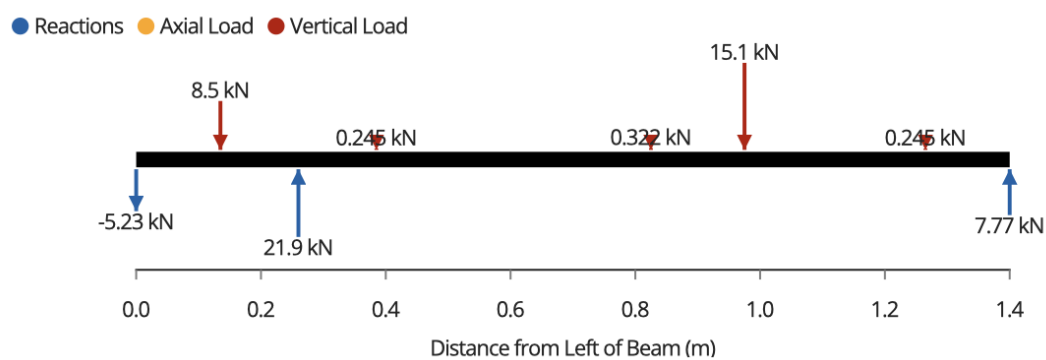


Figure 17: Force configuration on layshaft at third gear

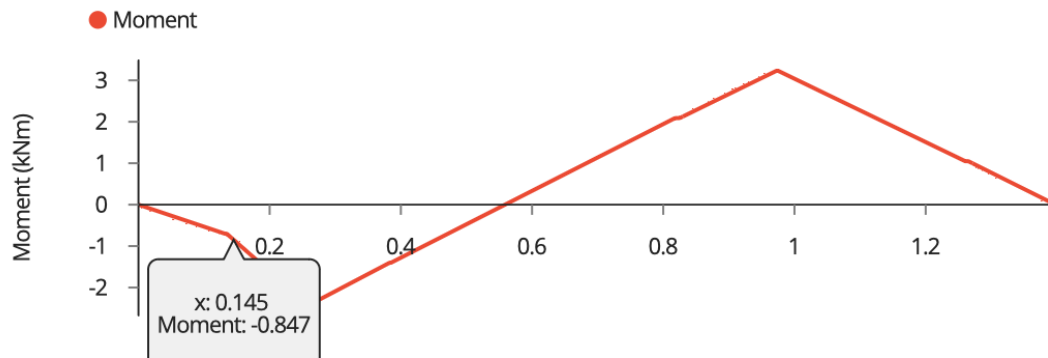


Figure 18: Bending moment diagram of layshaft

Bending moment = 3.23kNm

When meshing the reverse gear

Forces on gear A from the gear I:

$$W_T = 3343\text{N}$$

$$W_N = 8193\text{N}$$

$$W_G = 690\text{N}$$

$$W_R = 8498\text{N}$$

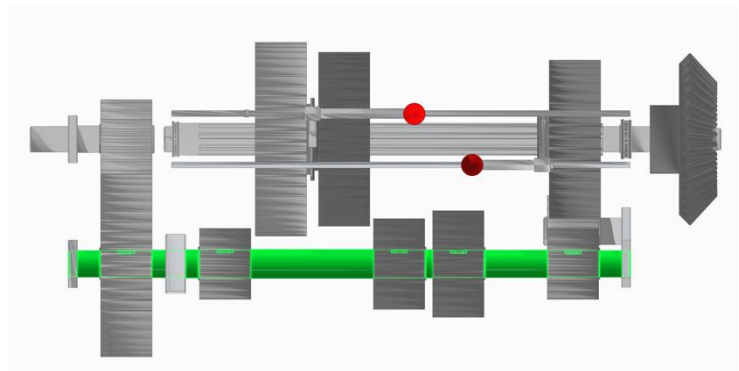


Figure19:reverse gear arrangement

Forces on gear E from the gear J:

$$W_T = 9403\text{N}$$

$$W_N = 23044\text{N}$$

$$W_G = 245\text{N}$$

$$W_R = 23145\text{N}$$

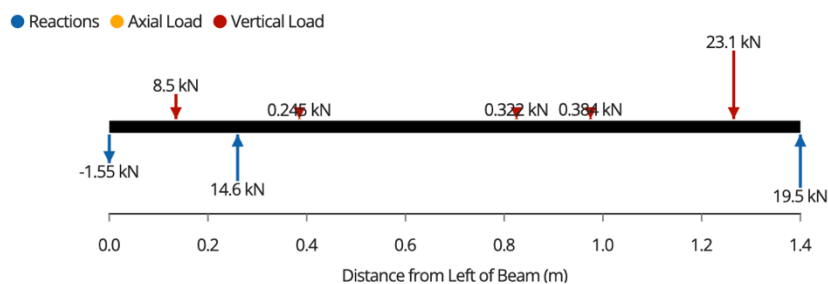


Figure 20:Force configuration on layshaft at reverse gear

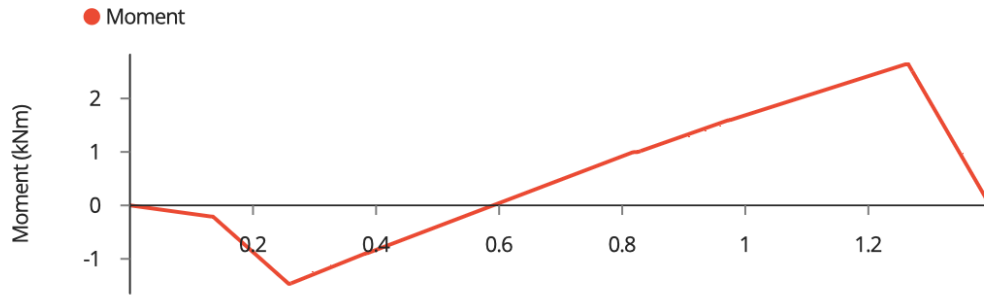


Figure 21: Bending moment diagram of layshaft

Bending moment = 2.64 kNm

Step 03: compare the bending moments obtained by each gear position and select the maximum bending moment value.

Gear Position	First Gear	Second gear	Third gear	Reverse gear
Bending Moment (kNm)	2.19	3.51	3.23	2.64

Maximum Bending moment is occurred at second gear.

$$M_{Max} = 3.51 \text{ kNm}$$

Therefore, the designing of the layshaft is done so that it can stand against the bending moment mentioned above.

Step 04: selecting a suitable shaft material.

Shaft material selection

Shaft material was selected as AISI 1040 carbon steel.

Properties of AISI 1040:

Shear stress	Tensile stress	Yield strength
111.6MPa	620MPa	415MPa

Chemical Composition

The following table shows the chemical composition of AISI 1040 carbon steel.

Element	Content (%)
Iron, Fe	98.6-99
Manganese, Mn	0.60-0.90
Carbon, C	0.370-0.440
Sulfur, S	≤ 0.050
Phosphorous, P	≤ 0.040

Step 05: calculate shaft diameter according to selected material.

Assuming a factor of safety as 1.5

Allowable shear stress,

$$\tau = \frac{111.6MPa}{1.5} = 74.4MPa$$

allowable tensile stress,

$$\sigma_b = \frac{620MPa}{1.5} = 413.33MPa$$

As we achieved maximum bending moment at second gear position, consider the torque that transmit through second gear for calculating total torque.

Torque transmitted through second gear = $T = 1193.6 \text{ Nm}$

According to maximum shear stress theory, equivalent twisting moment,

$$T_e = \sqrt{M_{Max}^2 + T^2}$$

$$T_e = \sqrt{3510^2 + 1193.6^2}$$

$$T_e = 3707 \text{ Nm}$$

We know,

$$T_e = \frac{\pi}{16} \times \tau \times d^3$$

$$3707 = \frac{\pi}{16} \times 74.4 \text{ MPa} \times d^3$$

$$d = 63.3 \text{ mm}$$

According to maximum normal stress theory, equivalent bending moment,

$$M_e = \frac{1}{2} \times (M + \sqrt{M^2 + T^2})$$

$$M_e = \frac{1}{2} \times (3510 + 3707) = 3608 \text{ Nm}$$

We know,

$$3608 = \frac{\pi}{32} \times \sigma_b \times d^3 = \frac{\pi}{32} \times 413.3 \text{ Mpa} \times d^3$$

$$d = 44.6 \text{ mm}$$

Taking the larger of the two values, we have

$$\text{layshaft diameter} = d = 63.3 \text{ mm say } 65 \text{ mm}$$

6.2.3.2 Power shaft

Power shaft configuration

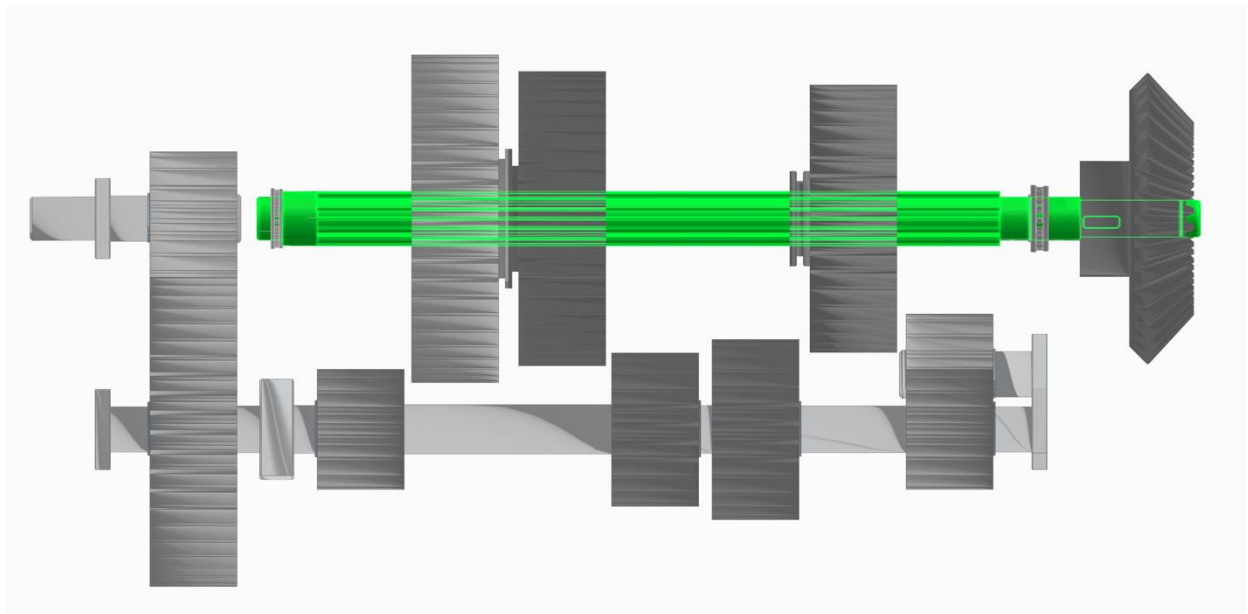
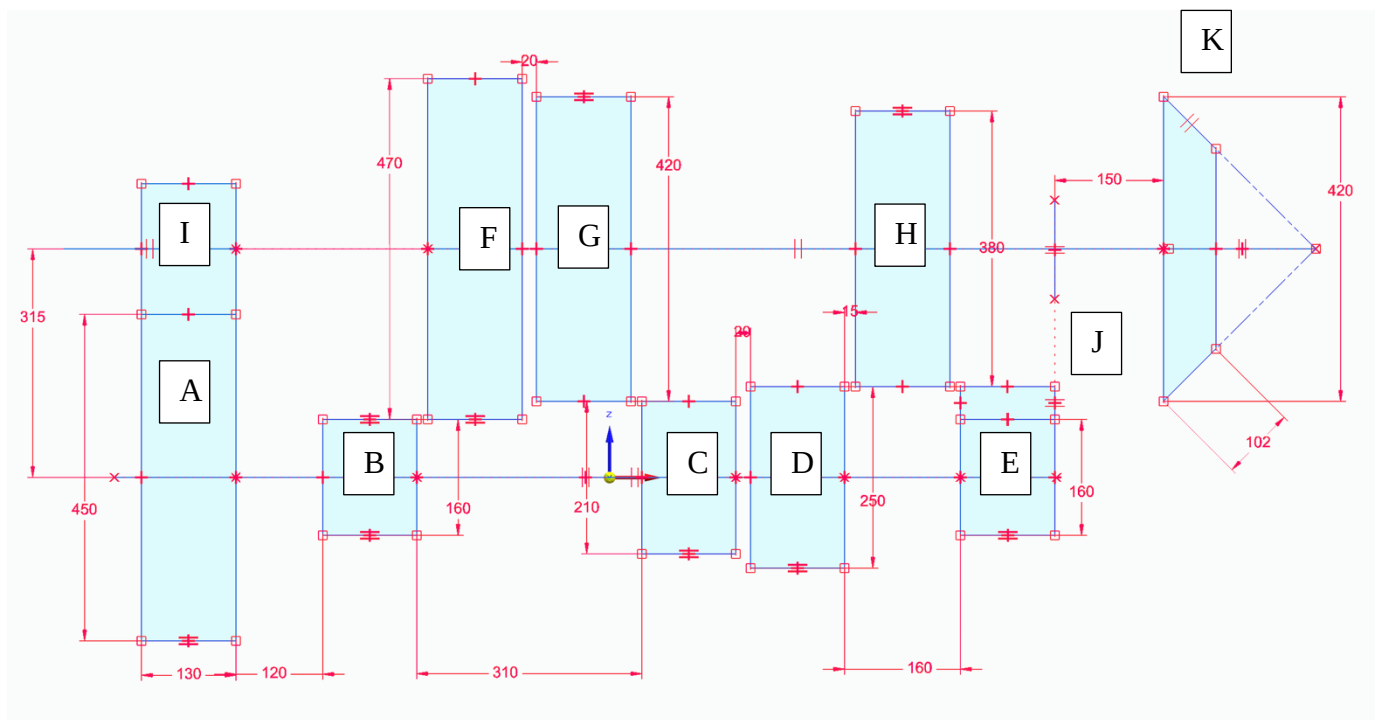


Figure 22: power shaft configuration

Power shaft calculation

All calculations are done for the power shaft are similar to which are done for layshaft.

The weight of gear is given by,

$$W_G = 0.00118 \times T_G \times b \times m^2$$

T_G = No. of teeth on the gear

b=Face width in mm

m=Module in mm

Table 4: weight of power shaft gears

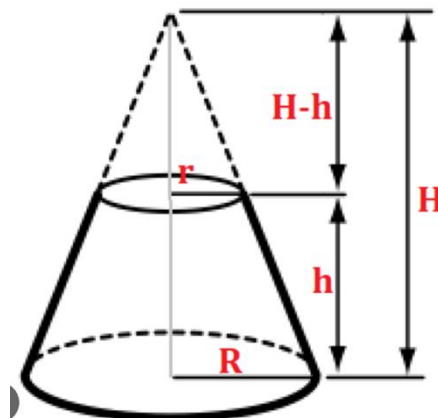
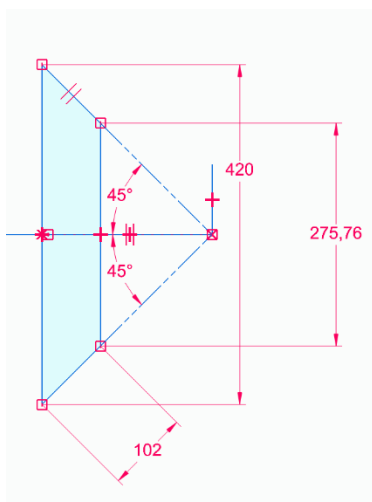
Gear	F	G	H
Weight (N)	721	644	583

Weight of the bevel gear [K]

Assume the volume of the bevel gear is same as the bottom volume of a cone as shown below.

$$\text{Volume} = \frac{1}{3} \times \pi \times (0.21^3 - 0.137^3) = 7 \times 10^{-3} m^3$$

$$\text{So, weight} = V \times \rho \times g = 7 \times 10^{-3} \times 7800 \times 9.81 = 536 N$$



When meshing with first gear

Forces on gear F from the gear B:

$$W_T = 9403 \text{ N}$$

$$W_N = 23044 \text{ N}$$

$$W_G = 721 \text{ N}$$

$$W_R = 23722 \text{ N}$$

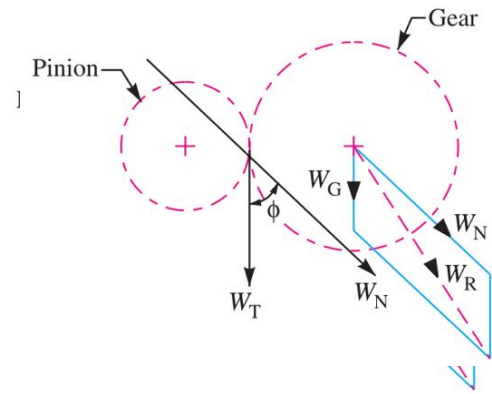


Figure 23: forces diagram

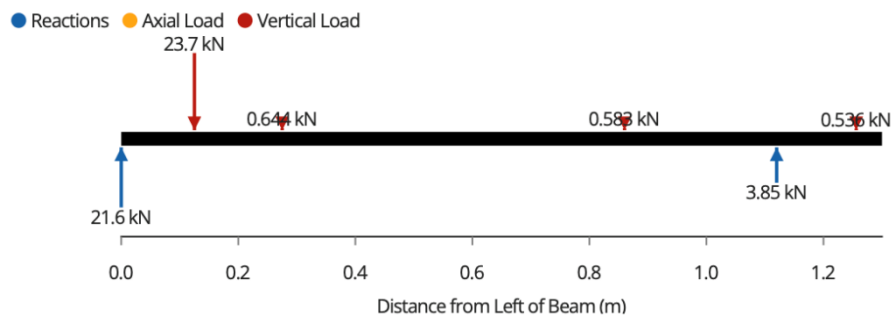


Figure 24: Forces configuration on power shaft at first gear

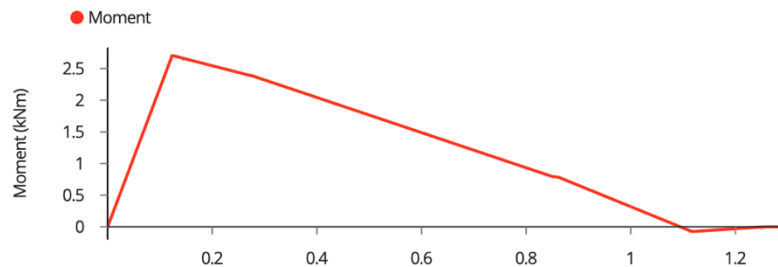


Figure 25: bending moment diagram

$$M_{\max} = 2.7 \text{ kNm}$$

When meshing with second gear

Forces on gear G from the gear C:

$$W_T = 7164 \text{ N}$$

$$W_N = 17557 \text{ N}$$

$$W_G = 644 \text{ N}$$

$$W_R = 18163 \text{ N}$$

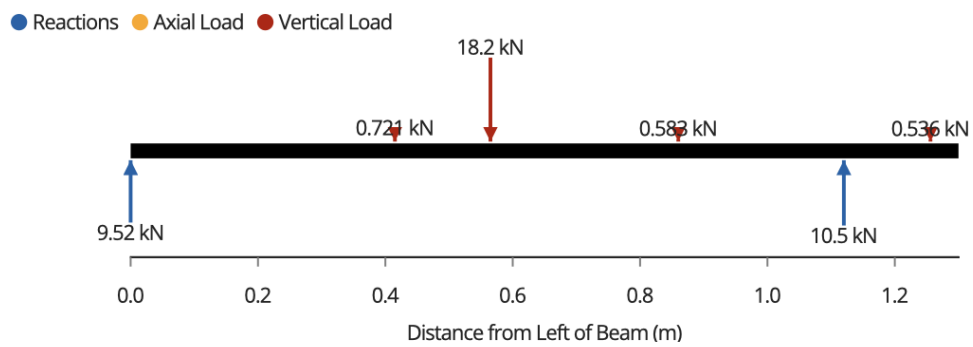
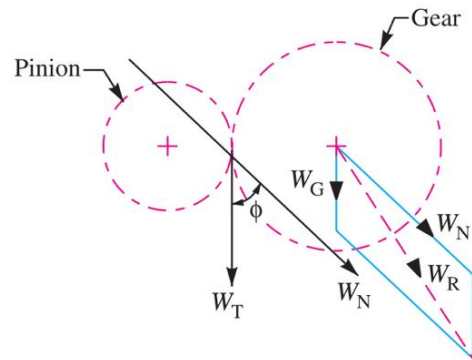


Figure 26: Force configuration on power shaft at second gear

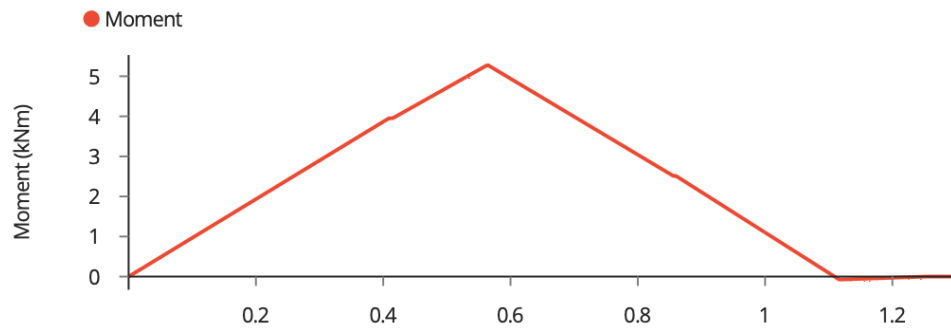


Figure 27: bending moment diagram

$$M_{\max} = 5.27 \text{ kNm}$$

When meshing with third gear

Forces on gear H from the gear D:

$$W_T = 6018 \text{ N}$$

$$W_N = 14748 \text{ N}$$

$$W_G = 583 \text{ N}$$

$$W_R = 15297 \text{ N}$$

● Reactions ● Axial Load ● Vertical Load

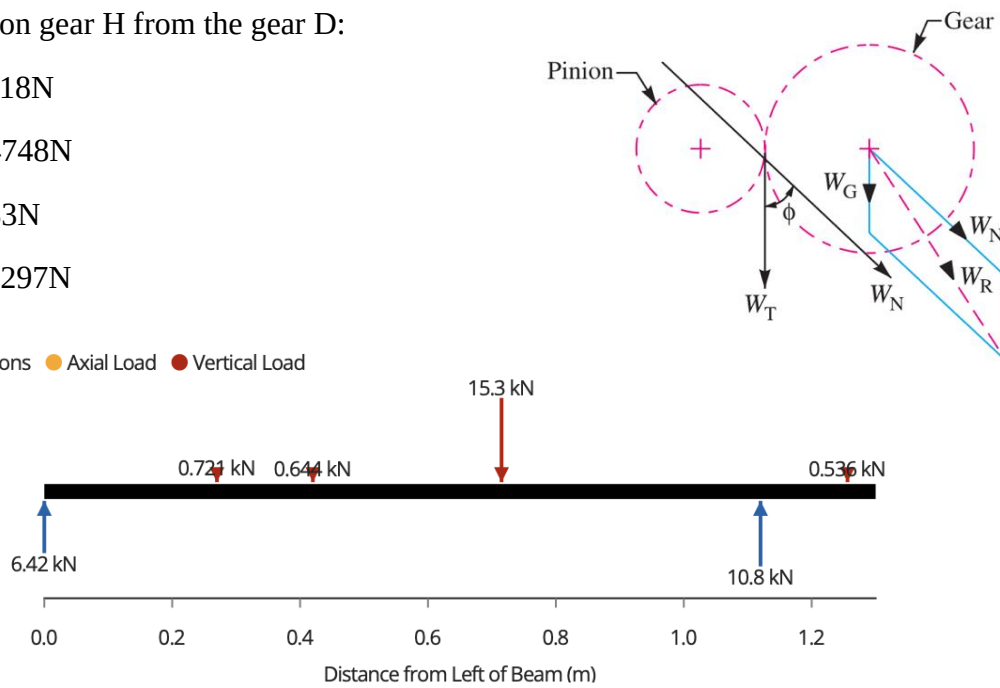


Figure 28: Force configuration on power shaft at third gear

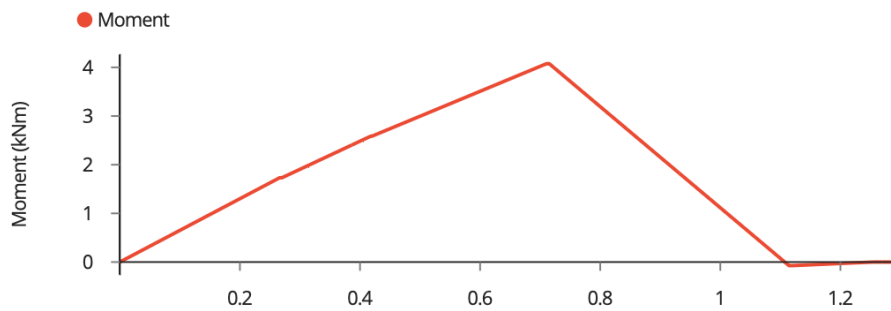


Figure 29: bending moment diagram

$$M_{\max} = 4.08 \text{ kNm}$$

When meshing with reverse gear

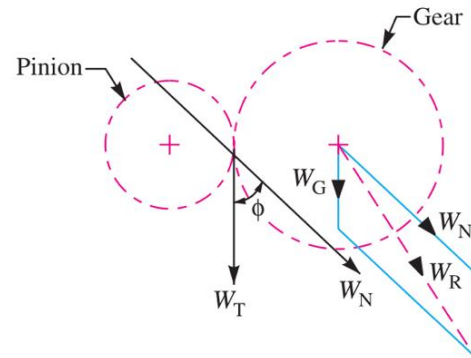
Forces on gear H from the gear J:

$$W_T = 9403 \text{ N}$$

$$W_N = 23044 \text{ N}$$

$$W_G = 583 \text{ N}$$

$$W_R = 23592 \text{ N}$$



● Reactions ● Axial Load ● Vertical Load

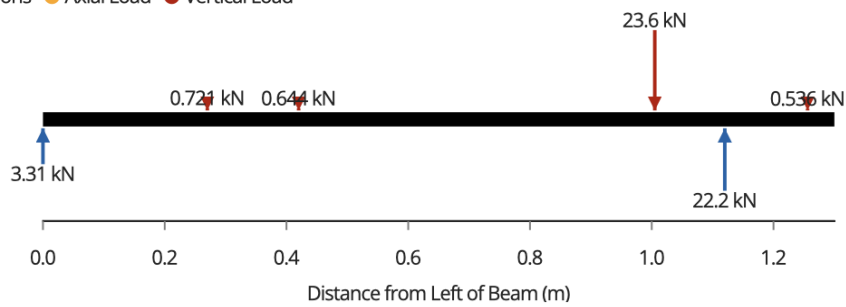


Figure 30: Force configuration on power shaft at reverse gear

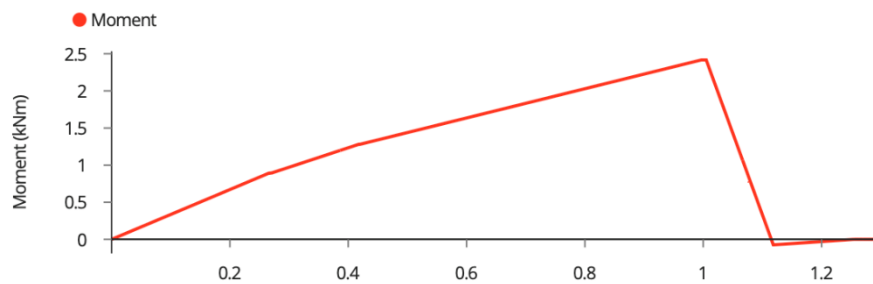


Figure 31: bending moment diagram

$$M_{\max} = 2.42 \text{ kNm}$$

Gear Position	First Gear	Second gear	Third gear	Reverse gear
Bending Moment (kNm)	2.7	5.27	4.08	2.42

Maximum Bending moment is occurred at the second gear.

$$M_{max} = 5.27 \text{ kNm}$$

Therefore, the designing of the layshaft is done so that it can stand against the bending moment mentioned above.

Shaft material selection

Shaft material was selected as AISI 4340 Alloy Steel

Properties of AISI 4340:

Shear stress	Tensile stress	Yield strength
134.1MPa	745MPa	470MPa

Chemical Composition

The following table shows the chemical composition of AISI 4340 alloy steel.

Element	Content (%)
Iron, Fe	95.195 - 96.33
Nickel, Ni	1.65 - 2.00
Chromium, Cr	0.700 - 0.900
Manganese, Mn	0.600 - 0.800
Carbon, C	0.370 - 0.430
Molybdenum, Mo	0.200 - 0.300
Silicon, Si	0.150 - 0.300
Sulfur, S	0.0400
Phosphorous, P	0.0350

Assuming a factor of safety as 1.2

Allowable shear stress,

$$\tau = \frac{134.1 \text{ MPa}}{1.2} = 111.75 \text{ MPa}$$

Consider the bending moments generated by bevel gear,

$$R_m = 173.93 \text{ mm}$$

$$\text{Torque} = 1193.6 \text{ Nm}$$

$$W_T = 6657.5 \text{ N}$$

$$W_N = 6876.5 \text{ N}$$

$$W_R = 1721.76 \text{ N}$$

$$W_{RH} = 1217.46 \text{ N}$$

$$W_{RV} = 1217.46 \text{ N}$$

$$\text{Overhang length} = 86 \text{ mm}$$

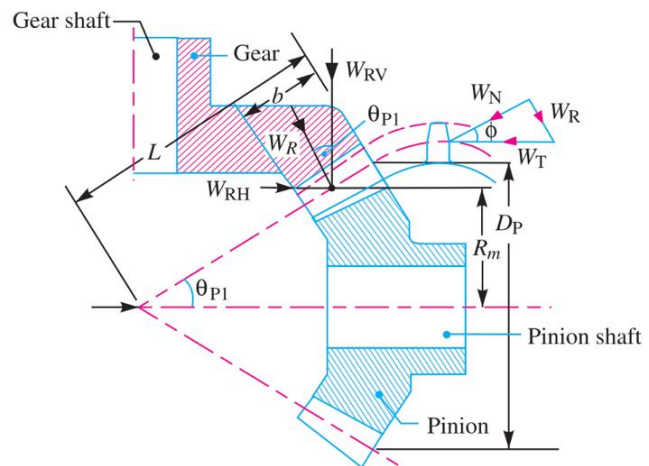


Figure 32: bevel gear force diagram

The bending moment due to W_{RH} and W_{RV} is given by,

$$M_1 = W_{RV} \times \text{Overhang} - W_{RH} \times R_m$$

$$M_1 = -105.91 \text{ Nm}$$

and bending moment due to W_T ,

$$M_2 = W_T \times \text{Overhang}$$

$$M_2 = 572.5 \text{ Nm}$$

$\therefore$ Resultant bending moment,

$$M = \sqrt{(M_1)^2 + (M_2)^2}$$

$$M = 582 \text{ Nm}$$

Total resultant bending moment on the whole power shaft,

$$M_T = \sqrt{(M)^2 + (M_{max})^2}$$

$$M_T = \sqrt{(582)^2 + (5270)^2} = 5300 \text{ Nm}$$

Since the shaft is subjected to twisting moment (T) and resultant bending moment (M), therefore, equivalent twisting moment,

$$T_e = \sqrt{(M_T)^2 + T^2}$$

$$T_e = \sqrt{5300^2 + 1193.6^2} = 5432.7 \text{ Nm}$$

We know,

$$T_e = \frac{\pi}{16} \times \tau \times d^3$$

$$5432.7 = \frac{\pi}{16} \times 111.75 \text{ MPa} \times d^3$$

$$d = 62.8 \text{ mm say } 65 \text{ mm}$$

Power shaft (Output shaft) diameter is = 65mm

6.2.4 Input shaft

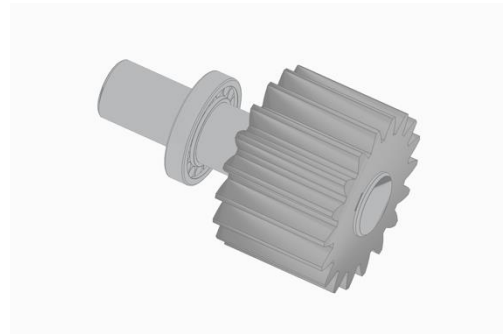
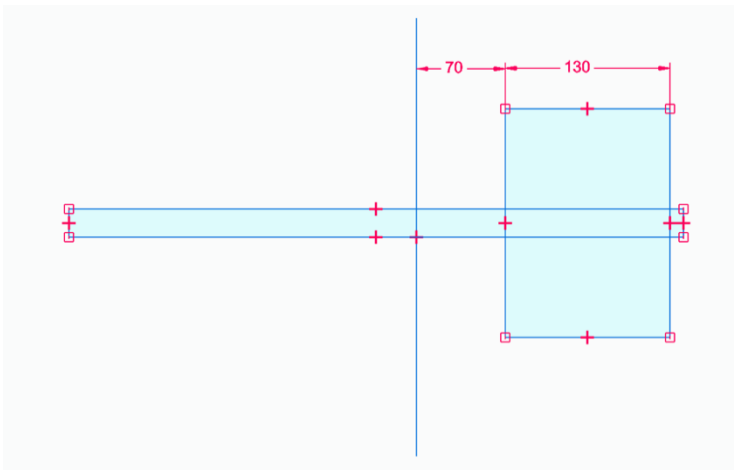


Figure 33:input shaft

All calculations are done for the power shaft are similar to which are done for layshaft.

Loads apply on the input shaft gear,

$$W_T = 3343\text{N}$$

$$W_N = 8193\text{N}$$

$$W_G = 276\text{N}$$

$$W_R = 8452\text{N}$$

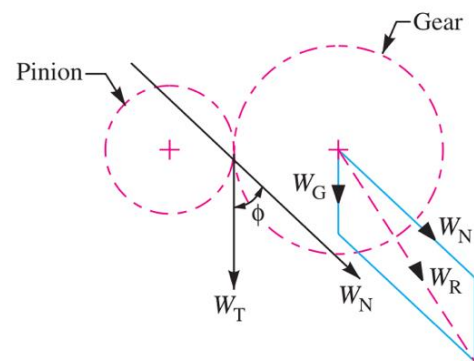


Figure 34:force diagram

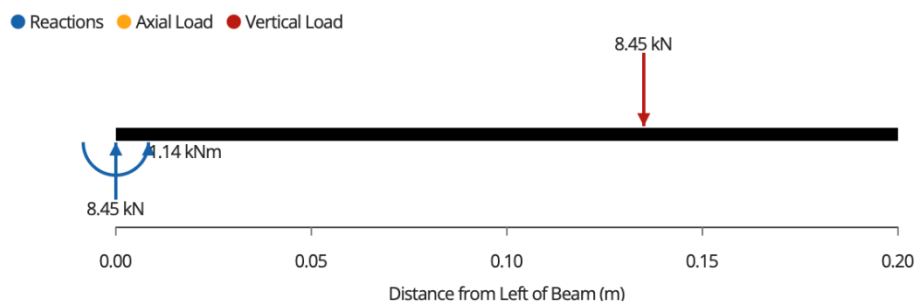


Figure 35:force diagram on input shaft

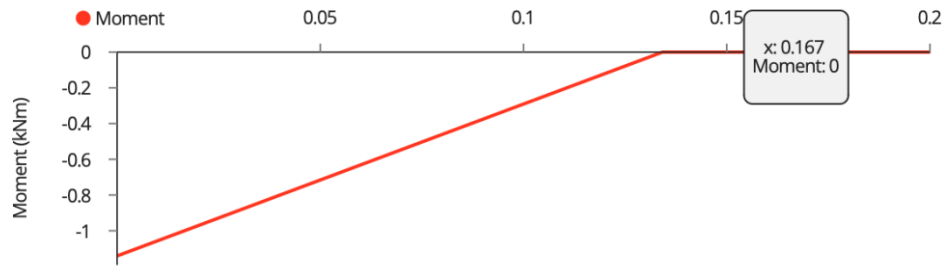


Figure 36:bending moment diagram

Bending moment =M = 1.14 kNm

Shaft material selection

Shaft material was selected as AISI 1040 carbon steel.

Properties of AISI 1040:

Shear stress	Tensile stress	Yield strength
111.6MPa	620MPa	415MPa

Assuming a factor of safety as 1.5

Allowable shear stress,

$$\tau = \frac{111.6MPa}{1.5} = 74.4MPa$$

allowable tensile stress,

$$\sigma_b = \frac{620MPa}{1.5} = 413.33MPa$$

Torque transmitted through input shaft =T

Motor's shaft speed = input shaft speed = 2970 RPM

$$T = \frac{75000 \times 60}{2 \times \pi \times 2970} = 241 Nm$$

According to maximum shear stress theory, equivalent twisting moment,

$$T_e = \sqrt[2]{M^2 + T^2}$$

$$T_e = \sqrt[2]{1140^2 + 241^2}$$

$$T_e = 1165.2Nm$$

We know,

$$T_e = \frac{\pi}{16} \times \tau \times d^3$$

$$1165.2 = \frac{\pi}{16} \times 74.4 \text{ MPa} \times d^3$$

$$d = 43 \text{ mm}$$

According to maximum normal stress theory, equivalent bending moment,

$$M_e = \frac{1}{2} \times (M + \sqrt{M^2 + T^2})$$

$$M_e = \frac{1}{2} \times (1140 + 1165.2) = 1152.6Nm$$

We know,

$$1152.6 = \frac{\pi}{32} \times \sigma_b \times d^3 = \frac{\pi}{32} \times 413.3 \text{ Mpa} \times d^3$$

$$d = 30.5mm$$

Taking the larger of the two values, we have

$$\text{Input shaft diameter} > 43mm$$

The motor shaft diameter is 65mm. Therefore, the Input shaft diameter is also same as motor shaft diameter = 65mm.

6.2.5 Bearings

6.2.5.1 Sample calculation for input shaft bearing,

Forces in horizontal direction = 7.48 kN

Forces in vertical direction = 3.62 kN

Resultant force = 8309.92 N

Working hours per day = 8

Working days per year = 300

Working years = 2

Rotational speed in input shaft = 2975 RPM

$$L_{10} = 2975 \times 8 \times 300 \times 2$$

$$L_{10} = 856.8 \text{ million revolutions}$$

$$C_r = P \times \sqrt[p]{L_{10}}$$

C_r = Basic dynamic load rating

P = Equivalent dynamic bearing load

$p = 10/3$ for roller bearings

$$C_r = 8309.92 \times \sqrt[10/3]{856.8}$$

$$C_r = 63017.442 \text{ N}$$

Bore diameter = 65mm

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	Alternative
d	D	B	dynamic	static	P _u	Reference speed	Limiting speed		Bearing with standard cage	standard cage ¹⁾
mm			kN		kN	r/min		kg	–	
55 cont.	120	29	156	143	18,6	6 000	7 000	1,5	► NUP 311 ECP	J, M, ML
	120	43	232	232	30,5	6 000	7 000	2,25	► NJ 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,25	► NU 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,3	► NUP 2311 ECP	ML, PH
	140	33	142	140	18,6	5 600	6 300	2,5	► NU 411	–
	140	33	142	140	18,6	5 600	6 300	2,55	► NJ 411	–
60	95	18	37,4	44	5,3	8 000	13 000	0,5	► NU 1012 ML	–
	110	22	108	102	13,4	6 700	7 500	0,79	► N 212 ECP	M
	110	22	108	102	13,4	6 700	7 500	0,8	► NU 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,82	► NJ 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,86	► NUP 212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,05	► NU 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NJ 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NUP 2212 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,75	► N 312 ECP	J, M
	130	31	173	160	21,2	5 600	6 700	1,75	► NU 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,85	► NJ 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,9	► NUP 312 ECP	J, M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,75	► NU 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,8	► NJ 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,85	► NUP 2312 ECP	M, ML, PH
	150	35	168	173	22	5 000	6 000	3	► NU 412	–
	150	35	168	173	22	5 000	6 000	3,05	► NJ 412	–
65	100	18	38	46,5	5,5	7 500	12 000	0,51	► NU 1013 ML	–
	100	18	62,7	81,5	9,8	7 500	7 500	0,45	► NU 1013 ECP	PH
	120	23	122	118	15,6	6 300	6 700	1	► NU 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► N 213 ECP	–
	120	23	122	118	15,6	6 300	6 700	1,05	► NJ 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► NUP 213 ECP	J, M, ML, PH

According to the bearing catalogue, N 213 ECP cylindrical roller bearing is suitable. [7]

6.2.5.2 Sample calculation for lay shaft bearings

There are three bearings in the lay shaft.

Length of the lay shaft is 1400mm. Two bearings are placed in both ends and one is placed at 260mm from the left edge.

In first gear,

Vertical forces acting on bearings,

Tangential load(W_T) and normal load(W_N)

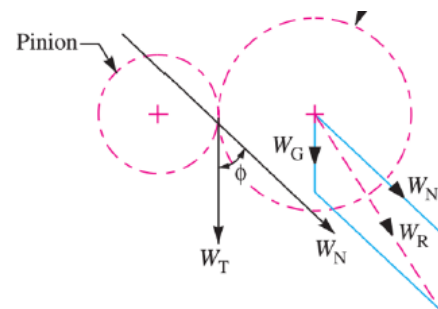
On engaging gear pair (I and A) :

$$W_T = 3343 \text{ N}$$

$$W_N = 8193 \text{ N}$$

$$W_G = 690 \text{ N}$$

$$W_R = 8498 \text{ N}$$



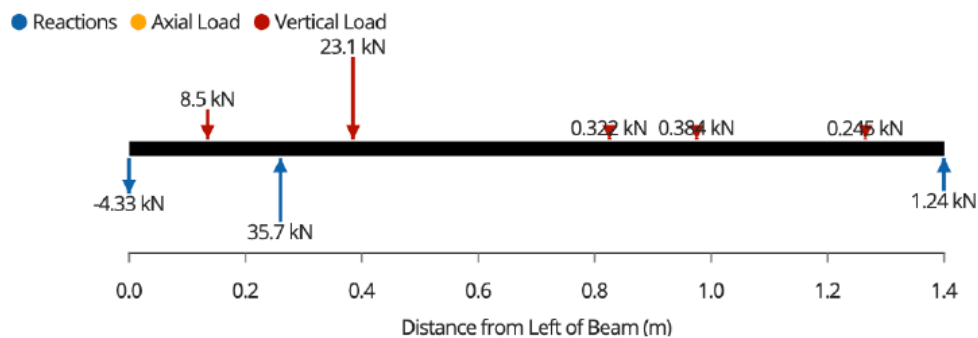
Tangential load(W_T) and normal load(W_N) on engaging gear pair (B and F) :

$$W_T = 9403 \text{ N}$$

$$W_N = 23044 \text{ N}$$

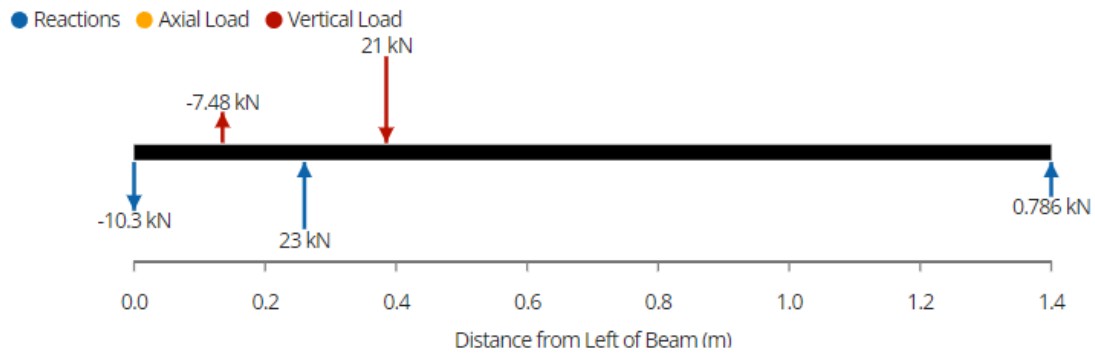
$$W_G = 245 \text{ N}$$

$$W_R = 23145 \text{ N}$$



Bearing	Vertical forces (KN)
lay shaft 1st bearing	4.33
lay shaft 2nd bearing	35.7
lay shaft 3rd bearing	1.24

Due to W_R forces in gears, there are horizontal forces acting on bearings.



Bearing	Horizontal force (kN)
lay shaft 1st bearing	10.3
lay shaft 2nd bearing	23
lay shaft 3rd bearing	0.786

Rotational speed in lay shaft = 1190 RPM

$$L_{10} = 1190 \times 8 \times 300 \times 2$$

$$L_{10} = 342.72 \text{ million revolutions}$$

Basic dynamic load rating in each bearing,

Bearing	Basic dynamic load rating
lay shaft 1st bearing	64366.04157
lay shaft 2nd bearing	244646.3294
lay shaft 3rd bearing	8457.565522

For second gear,

Tangential load(W_T) and normal load(W_N)

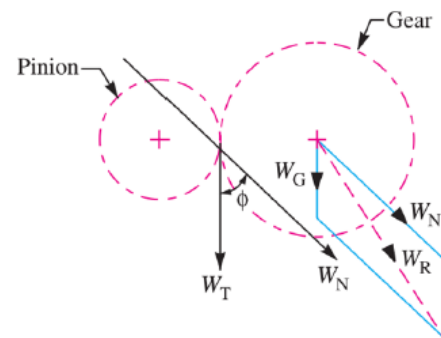
On engaging gear pair (I and A) :

$$W_T=3343\text{N}$$

$$W_N=8193\text{N}$$

$$W_G=690\text{N}$$

$$W_R=8498\text{N}$$



Tangential load(W_T) and normal load(W_N) on engaging gear pair (C and G) :

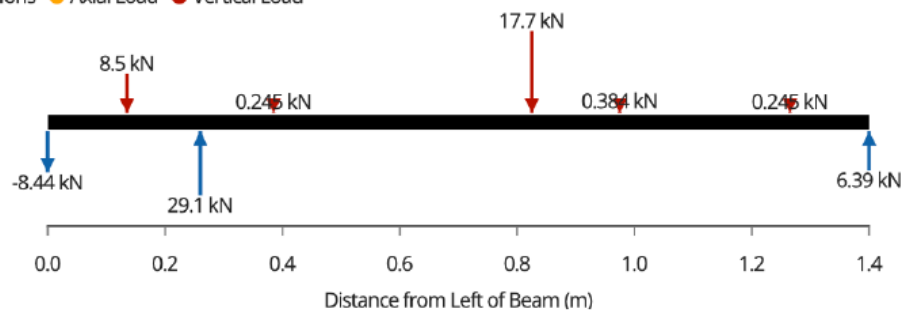
$$W_T=7164\text{N}$$

$$W_N=17557\text{N}$$

$$W_G=322\text{N}$$

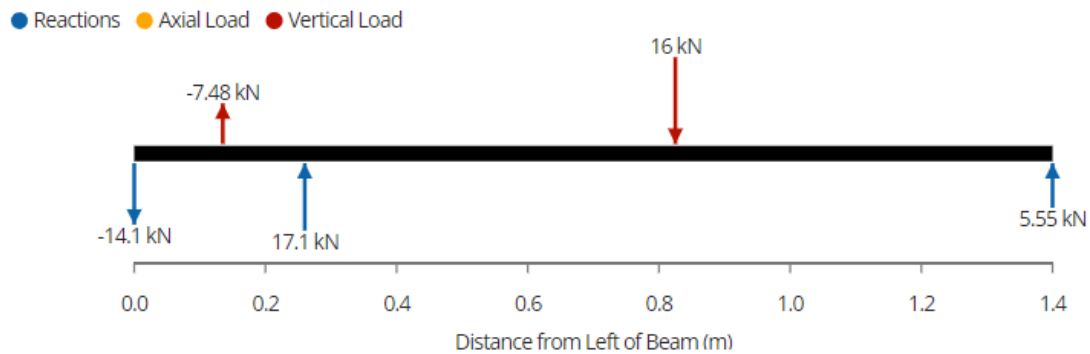
$$W_R=17690\text{N}$$

● Reactions ● Axial Load ● Vertical Load



Bearing	Vertical force (kN)
lay shaft 1st bearing	8.44
lay shaft 2nd bearing	29.1
lay shaft 3rd bearing	6.39

For horizontal forces,



Bearing	Horizontal force (kN)
lay shaft 1st bearing	14.1
lay shaft 2nd bearing	17.1
lay shaft 3rd bearing	5.55

Bearing	Resultant force (N)	Basic dynamic load rating
lay shaft 1st bearing	16433.00	94667.03591
lay shaft 2nd bearing	33752.33	194440.0102
lay shaft 3rd bearing	8463.72	48757.70494

For third gear,

Tangential load(W_T) and normal load(W_N)

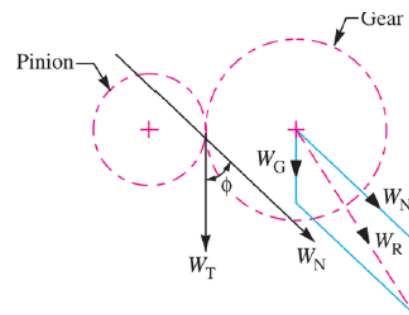
On engaging gear pair (I and A) :

$$W_T = 3343 \text{ N}$$

$$W_N = 8193 \text{ N}$$

$$W_G = 690 \text{ N}$$

$$W_R = 8498 \text{ N}$$



Tangential load(W_T) and normal load(W_N) on engaging gear pair (D and H) :

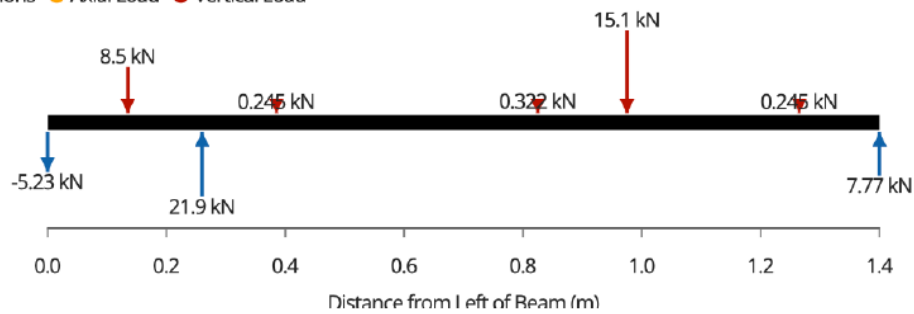
$$W_T = 6018 \text{ N}$$

$$W_N = 14748 \text{ N}$$

$$W_G = 384 \text{ N}$$

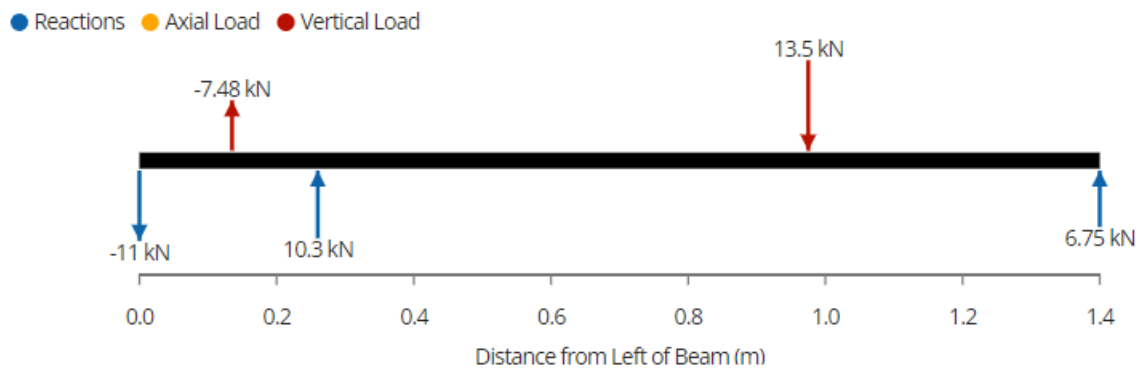
$$W_R = 15132 \text{ N}$$

● Reactions ● Axial Load ● Vertical Load



Bearing	Vertical force (kN)
lay shaft 1st bearing	5.23
lay shaft 2nd bearing	21.9
lay shaft 3rd bearing	7.77

Horizontal forces,



Bearing	Horizontal force (kN)
lay shaft 1st bearing	11
lay shaft 2nd bearing	10.3
lay shaft 3rd bearing	6.75

Bearing	Resultant force (N)	Basic dynamic load rating
lay shaft 1st bearing	12180.02053	70166.5066
lay shaft 2nd bearing	24201.23964	139418.1921
lay shaft 3rd bearing	10292.49241	59292.8587

For reverse gear,

Tangential load(W_T) and normal load(W_N)

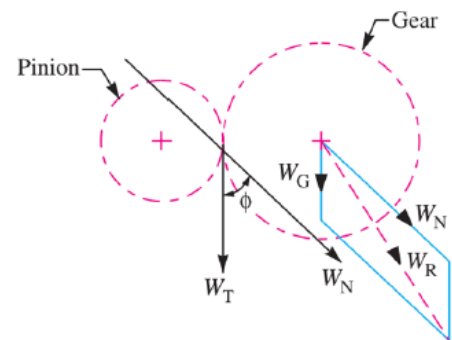
On engaging gear pair (I and A) :

$$W_T=3343\text{N}$$

$$W_N=8193\text{N}$$

$$W_G=690\text{N}$$

$$W_R=8498\text{N}$$



Tangential load(W_T) and normal load(W_N) on engaging gear pair (E and H) :

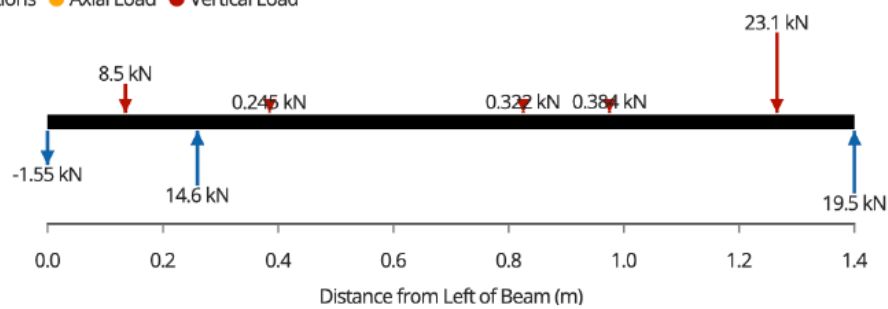
$$W_T=9403\text{N}$$

$$W_N=23044\text{N}$$

$$W_G=245\text{N}$$

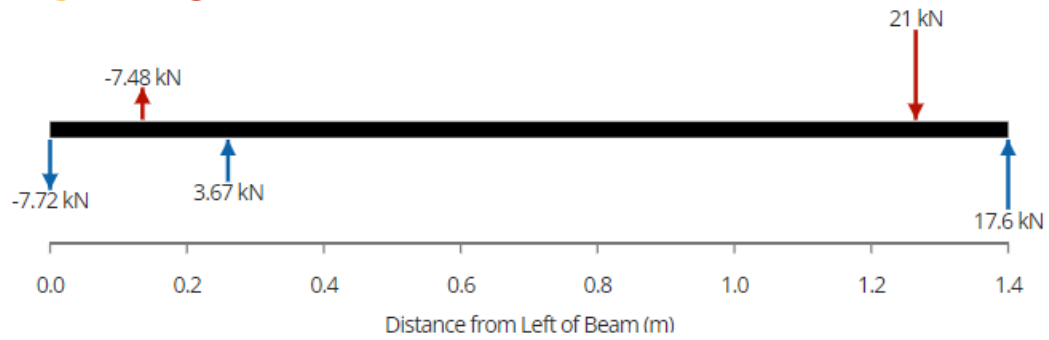
$$W_R=23145\text{N}$$

● Reactions ● Axial Load ● Vertical Load



Bearing	Vertical force (kN)
lay shaft 1st bearing	1.55
lay shaft 2nd bearing	14.6
lay shaft 3rd bearing	19.5

● Reactions ● Axial Load ● Vertical Load



Bearing	Horizontal force (kN)
lay shaft 1st bearing	7.72
lay shaft 2nd bearing	3.67
lay shaft 3rd bearing	17.6

Bearing	Resultant Force (N)	Basic dynamic load rating
lay shaft 1st bearing	7874.065	45360.81319
lay shaft 2nd bearing	15054.1987	86724.03582
lay shaft 3rd bearing	26268.0414	151324.5974

- Maximum basic dynamic load rating for first bearing in lay shaft = 94667.0359
- Maximum basic dynamic load rating for second bearing in lay shaft = 244646.3294.
- Maximum basic dynamic load rating for third bearing in lay shaft = 151324.5974.

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	Alternative
d	D	B	dynamic	static	P_u	Reference speed	Limiting speed		Bearing with standard cage	standard cage ¹⁾
mm			kN	C_0	kN	r/min		kg	–	
55 cont.	120	29	156	143	18,6	6 000	7 000	1,5	► NUP 311 ECP	J, M, ML
	120	43	232	232	30,5	6 000	7 000	2,25	► NJ 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,25	► NU 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,3	► NUP 2311 ECP	ML, PH
60	140	33	142	140	18,6	5 600	6 300	2,5	► NU 411	–
	140	33	142	140	18,6	5 600	6 300	2,55	► NJ 411	–
	95	18	37,4	44	5,3	8 000	13 000	0,5	► NU 1012 ML	–
	110	22	108	102	13,4	6 700	7 500	0,79	► N 212 ECP	M
	110	22	108	102	13,4	6 700	7 500	0,8	► NU 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,82	► NJ 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,86	► NUP 212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,05	► NU 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NJ 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NUP 2212 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,75	► N 312 ECP	J, M
	130	31	173	160	21,2	5 600	6 700	1,75	► NU 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,85	► NJ 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,9	► NUP 312 ECP	J, M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,75	► NU 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,8	► NJ 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,85	► NUP 2312 ECP	M, ML, PH
65	150	35	168	173	22	5 000	6 000	3	► NU 412	–
	150	35	168	173	22	5 000	6 000	3,05	► NJ 412	–
	100	18	38	46,5	5,5	7 500	12 000	0,51	► NU 1013 ML	–
	100	18	62,7	81,5	9,8	7 500	7 500	0,45	► NU 1013 ECP	PH
	120	23	122	118	15,6	6 300	6 700	1	► NU 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► N 213 ECP	–
	120	23	122	118	15,6	6 300	6 700	1,05	► NJ 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► NUP 213 ECP	J, M, ML, PH

According to the bearing catalogue, NU 213 ECP cylindrical roller bearing is suitable for first bearing in lay shaft. [7]

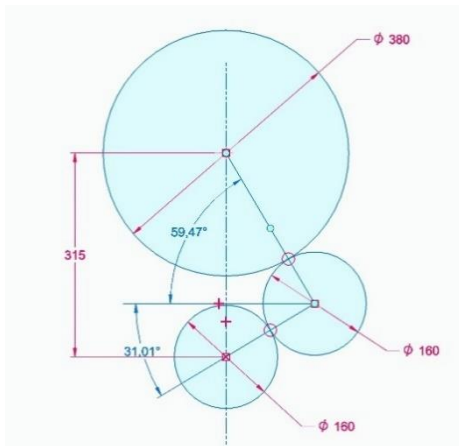
Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	Alternative
d	D	B	dynamic	static	P_u	Reference speed	Limiting speed		Bearing with standard cage	standard cage ¹⁾
mm			kN	C_0	kN	r/min		kg	–	
65 cont.	140	33	212	196	25,5	5 300	6 000	2,35	► NUP 313 ECP	J, M, ML, PH
	140	48	285	290	38	5 300	6 000	3,2	► NU 2313 ECP	ML, PH
	140	48	285	290	38	5 300	6 000	3,35	► NJ 2313 ECP	ML, PH
	140	48	285	290	38	5 300	6 000	3,45	► NUP 2313 ECP	ML, PH
	160	37	183	190	24	4 800	5 600	3,55	► NU 413	–
	160	37	183	190	24	4 800	5 600	3,65	► NJ 413	–

According to the bearing catalogue, NU 2313 ECP cylindrical roller bearing is suitable for second bearing in lay shaft. [7]

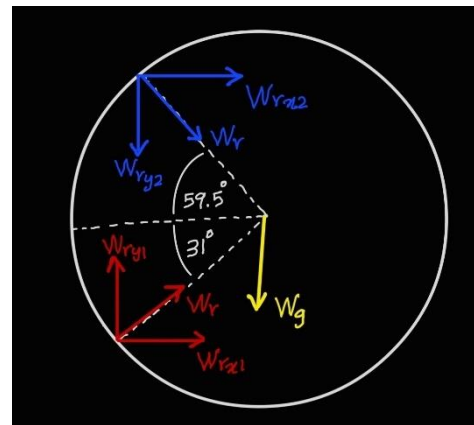
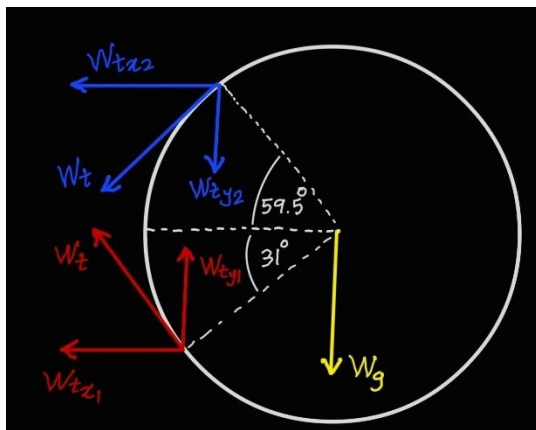
Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	Alternative
d	D	B	dynamic	static		Reference	Limiting		Bearing with	standard cage ¹⁾
mm			C	C ₀	P _u	speed	speed	kg	standard cage	
			kN		kN	r/min			–	
65 cont.	140	33	212	196	25,5	5 300	6 000	2,35	► NUP 313 ECP	J, M, ML, PH
	140	48	285	290	38	5 300	6 000	3,2	► NU 2313 ECP	ML, PH
	140	48	285	290	38	5 300	6 000	3,35	► NJ 2313 ECP	ML, PH
	140	48	285	290	38	5 300	6 000	3,45	► NUP 2313 ECP	ML, PH
	160	37	183	190	24	4 800	5 600	3,55	► NU 413	–
	160	37	183	190	24	4 800	5 600	3,65	► NJ 413	–

According to the bearing catalogue, NUP 13 ECP cylindrical roller bearing is suitable for third bearing in lay shaft. [7]

6.2.5.3 Sample calculation for idler shaft bearing



Forces acting in idler gear,



According to them,

Resultant vertical load = 4.25 kN

Resultant horizontal load = 15.7732 kN

Teeth in E gear = 16

Teeth in idler gear = 16

Therefore, rotational speed of idler gear = 1190 RPM

Basic rating life in millions of revolutions,

$$L_{10} = 1190 \times 8 \times 300 \times 2$$

$$L_{10} = 342.72 \text{ million revolutions}$$

Basic dynamic load rating = 94106.7111

Principal dimensions			Basic load ratings		Fatigue load limit	Speed ratings		Mass	Designations	Alternative
d	D	B	dynamic	static	P _u	Reference speed	Limiting speed		Bearing with standard cage	standard cage ¹⁾
mm			kN		kN	r/min		kg	–	
55 cont.	120	29	156	143	18,6	6 000	7 000	1,5	► NUP 311 ECP	J, M, ML
	120	43	232	232	30,5	6 000	7 000	2,25	► NJ 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,25	► NU 2311 ECP	ML, PH
	120	43	232	232	30,5	6 000	7 000	2,3	► NUP 2311 ECP	ML, PH
	140	33	142	140	18,6	5 600	6 300	2,5	► NU 411	–
	140	33	142	140	18,6	5 600	6 300	2,55	► NJ 411	–
60	95	18	37,4	44	5,3	8 000	13 000	0,5	► NU 1012 ML	–
	110	22	108	102	13,4	6 700	7 500	0,79	► N 212 ECP	M
	110	22	108	102	13,4	6 700	7 500	0,8	► NU 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,82	► NJ 212 ECP	J, M, ML
	110	22	108	102	13,4	6 700	7 500	0,86	► NUP 212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,05	► NU 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NJ 2212 ECP	J, M, ML, PH
	110	28	146	153	20	6 700	7 500	1,1	► NUP 2212 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,75	► N 312 ECP	J, M
	130	31	173	160	21,2	5 600	6 700	1,75	► NU 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,85	► NJ 312 ECP	J, M, ML, PH
	130	31	173	160	21,2	5 600	6 700	1,9	► NUP 312 ECP	J, M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,75	► NU 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,8	► NJ 2312 ECP	M, ML, PH
	130	46	260	265	34,5	5 600	6 700	2,85	► NUP 2312 ECP	M, ML, PH
	150	35	168	173	22	5 000	6 000	3	► NU 412	–
	150	35	168	173	22	5 000	6 000	3,05	► NJ 412	–
65	100	18	38	46,5	5,5	7 500	12 000	0,51	► NU 1013 ML	–
	100	18	62,7	81,5	9,8	7 500	7 500	0,45	► NU 1013 ECP	PH
	120	23	122	118	15,6	6 300	6 700	1	► NU 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► N 213 ECP	–
	120	23	122	118	15,6	6 300	6 700	1,05	► NJ 213 ECP	J, M, ML, PH
	120	23	122	118	15,6	6 300	6 700	1,05	► NUP 213 ECP	J, M, ML, PH

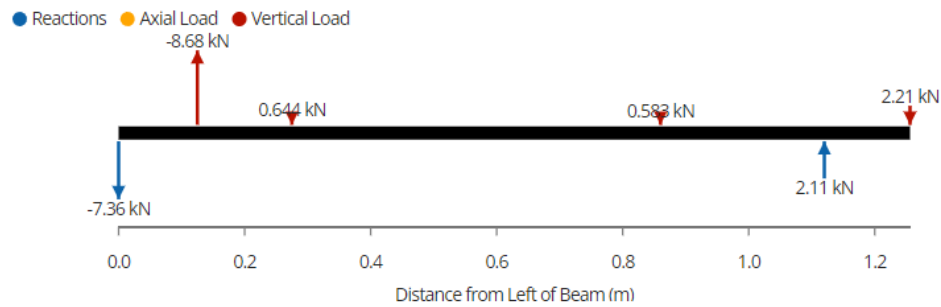
According to the bearing catalogue, NU 213 ECP cylindrical roller bearing is suitable. [7]

6.2.5.4 Sample calculation for main shaft bearing

There are two bearings in output shaft. Length of the shaft is 1256mm.

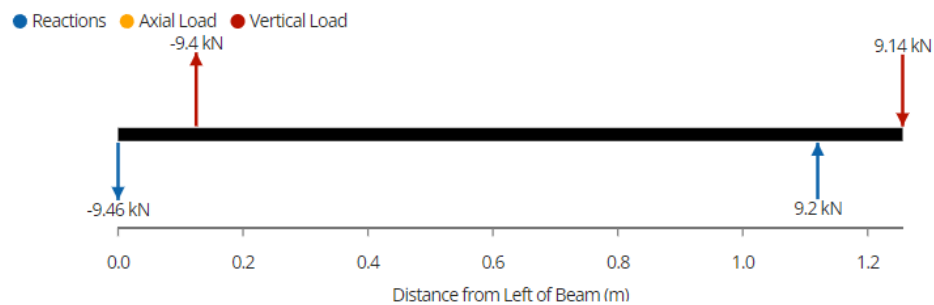
In first gear,

Vertical loads acting on two bearings,



Bearing	Vertical load (kN)
main shaft 1st bearing	7.36
main shaft 2nd bearing	2.11

Horizontal loads,



Bearing	Horizontal load (kN)
main shaft 1st bearing	9.46
main shaft 2nd bearing	9.2

Resultant load,

Bearing	Resultant load (N)
main shaft 1st bearing	11985.87
main shaft 2nd bearing	9438.86

Axial load in both bearings = 1671.3939 N

1Therefore, in one bearing = 835.696 N

The dynamic equivalent radial load,

$$W = X.V. W_R + Y. W_A$$

V = 1 (for all types of bearings when the inner race is rotating)

Table 27.4. Values of X and Y for dynamically loaded bearings.

Type of bearing	Specifications	$\frac{W_A}{W_R} \leq e$		$\frac{W_A}{W_R} > e$		e
		X	Y	X	Y	
Deep groove ball bearing	$\frac{W_A}{C_0} = 0.025$	1	0	0.56	2.0	0.22
	$= 0.04$				1.8	0.24
	$= 0.07$				1.6	0.27
	$= 0.13$				1.4	0.31
	$= 0.25$				1.2	0.37
	$= 0.50$				1.0	0.44
Angular contact ball bearings	Single row	1	0	0.35	0.57	1.14
	Two rows in tandem		0	0.35	0.57	1.14
	Two rows back to back		0.55	0.57	0.93	1.14
	Double row		0.73	0.62	1.17	0.86
Self-aligning bearings	Light series : for bores	1	1.3	6.5	2.0	0.50
	10 – 20 mm				2.6	0.37
	25 – 35				3.1	0.31
	40 – 45				3.5	0.28
	50 – 65				3.8	0.26
	70 – 100				3.5	0.28
	105 – 110	0.65	1.0	0.65	1.6	0.63
	Medium series : for bores				1.9	0.52
	12 mm				2.3	0.43
	15 – 20				2.5	0.39
Spherical roller bearings	For bores :	1	2.1	0.67	3.1	0.32
	25 – 35 mm				3.7	0.27
	40 – 45				4.4	0.23
	50 – 100				3.9	0.26
	100 – 200					

Bearing	$\frac{W_A}{W_R}$
main shaft 1st bearing	0.069
main shaft 2nd bearing	0.088

Therefore, X = 0.67 and Y = 4.4.

$$W = 1 \times 1 \times 11985.87 + 2.3 \times 835.69695$$

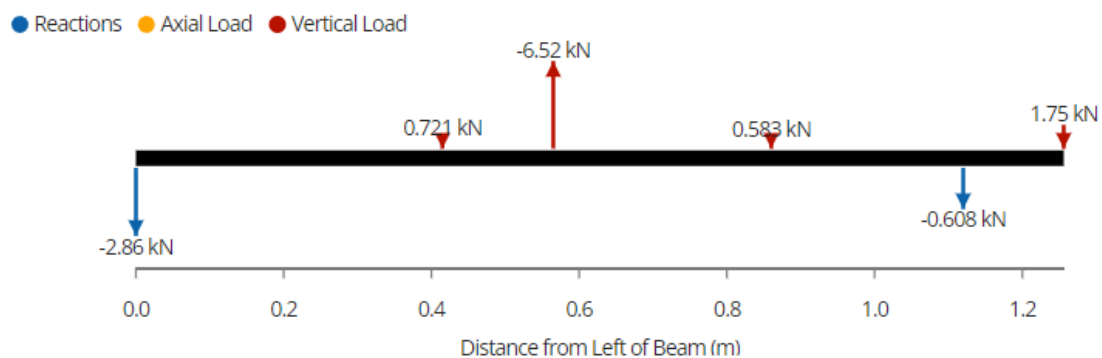
$$W = 14409.39618 \text{ N}$$

Therefore,

Bearing	Basic dynamic load rating
main shaft 1st bearing	59852.3939
main shaft 2nd bearing	49272.8474

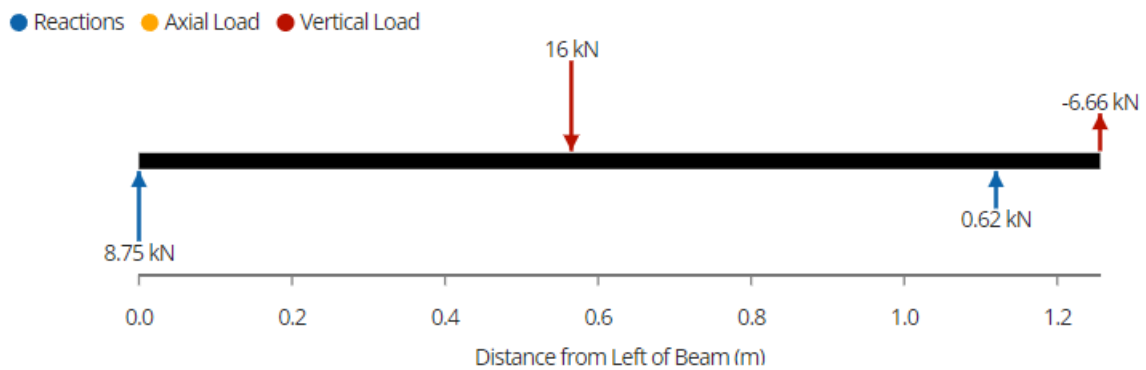
For second gear,

Vertical loads



Bearing	Vertical load (kN)
main shaft 1st bearing	2.86
main shaft 2nd bearing	0.608

Horizontal loads



Bearing	Horizontal load (kN)
main shaft 1st bearing	8.75
main shaft 2nd bearing	0.62

Resultant load,

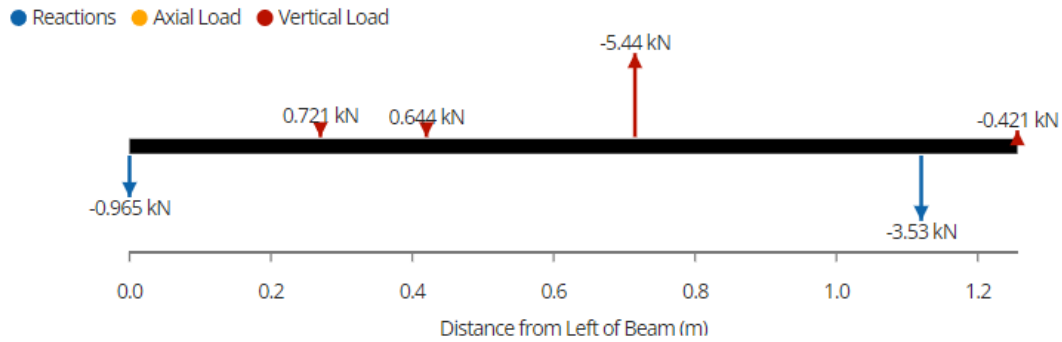
Bearing	Resultant load (N)
main shaft 1st bearing	9205.547
main shaft 2nd bearing	868.368

Axial load in one bearing = 608.7344 N

Bearing	Basic dynamic load rating
main shaft 1st bearing	51464.1107
main shaft 2nd bearing	15293.6975

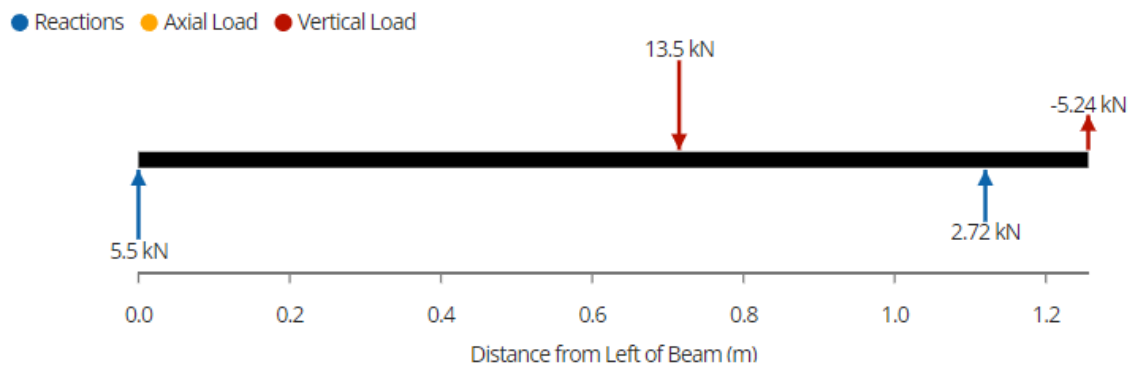
For third gear

Vertical loads,



Bearing	Vertical load (kN)
main shaft 1st bearing	0.965
main shaft 2nd bearing	3.53

Horizontal loads,



Bearing	Horizontal load (kN)
main shaft 1st bearing	5.5
main shaft 2nd bearing	2.72

Axial load in one bearing = 478.72115 N

Dynamic equivalent radial load,

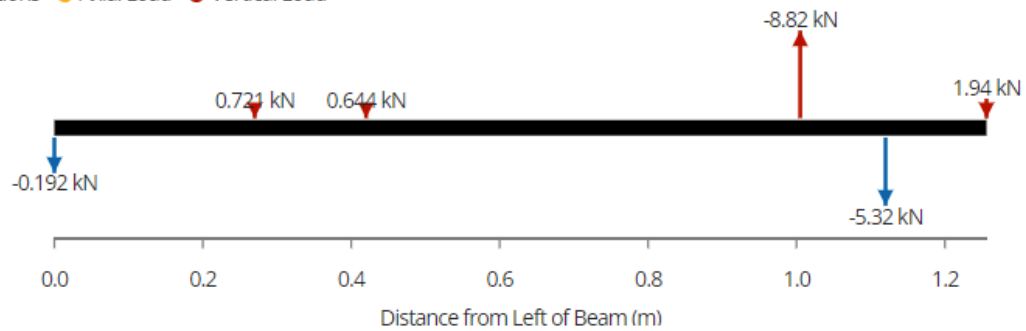
Bearing	dynamic equivalent radial load (N)
main shaft 1st bearing	6972.306468
main shaft 2nd bearing	5844.668788

Bearing	Basic dynamic load rating
main shaft 1st bearing	35655.0632
main shaft 2nd bearing	29888.5363

For reverse gear,

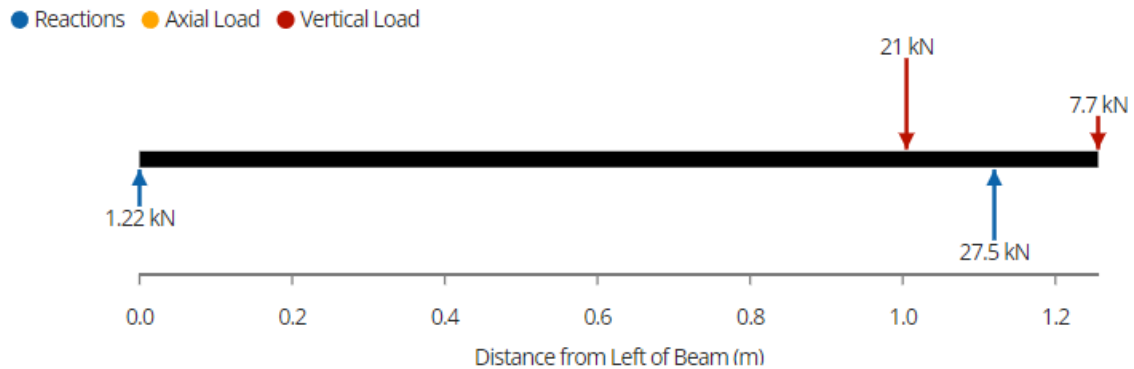
Vertical loads

● Reactions ● Axial Load ● Vertical Load



Bearing	Vertical load (kN)
main shaft 1st bearing	0.192
main shaft 2nd bearing	5.32

Horizontal load



Bearing	Horizontal load (kN)
main shaft 1st bearing	1.22
main shaft 2nd bearing	27.5

Axial load in one bearing = 704.3845 N

Dynamic equivalent radial load,

Bearing	dynamic equivalent radial load (N)
main shaft 1st bearing	3926.752379
main shaft 2nd bearing	30052.5776

Bearing	Basic dynamic load rating
main shaft 1st bearing	17439.83064
main shaft 2nd bearing	133472.0943

- Maximum basic dynamic load rating for first bearing in output shaft = 59852.39395.
- Maximum basic dynamic load rating for second bearing in output shaft=133472.0943.

Principal dimensions					Basic load ratings		Fatigue load limit	Minimum load factor	Speed ratings		Mass	Designation
d	D	H	E _a	E _b	C	C ₀	P _u	A	Reference speed	Limiting speed		
mm					kN		kN	–	r/min		kg	–
15	28	9	16	27	11,2	27	2,45	0,000 058	4 300	8 500	0,024	► 81102 TN
17	30	9	18	29	12,2	31,5	2,85	0,000 079	4 300	8 500	0,027	► 81103 TN
20	35	10	21	34	18,6	48	4,65	0,00018	3 800	7 500	0,037	► 81104 TN
25	42	11	26	41	25	69,5	6,8	0,00039	3 200	6 300	0,053	► 81105 TN
30	47	11	31	46	27	78	7,65	0,00049	3 000	6 000	0,057	► 81106 TN
	52	16	31	50	50	134	13,4	0,0014	2 400	4 800	0,12	► 81206 TN
35	52	12	36	51	29	93	9,15	0,00069	2 800	5 600	0,073	► 81107 TN
	62	18	39	58	62	190	19,3	0,0029	2 000	4 000	0,21	► 81207 TN
40	60	13	42	58	43	137	13,7	0,0015	2 400	5 000	0,11	► 81108 TN
	68	19	43	66	83	255	26,5	0,0052	1 900	3 800	0,25	► 81208 TN
	78	22	44	77	95	365	36,5	0,011	2 000	4 000	0,48	89308 TN
45	65	14	47	63	45	153	15,3	0,0019	2 200	4 500	0,13	► 81109 TN
	73	20	48	70	83	255	26,5	0,0052	1 800	3 600	0,29	► 81209 TN
50	70	14	52	68	47,5	166	16,6	0,0022	2 200	4 300	0,14	► 81110 TN
	78	22	53	75	91,5	300	31	0,0072	1 700	3 400	0,36	► 81210 TN
55	78	16	57	77	69,5	285	29	0,0065	1 900	3 800	0,23	► 81111 TN
	90	25	59	85	122	390	40	0,012	1 400	2 800	0,57	► 81211 TN
60	85	17	62	82	80	300	30,5	0,0072	1 800	3 600	0,27	► 81112 TN
	95	26	64	91	137	465	47,5	0,017	1 400	2 800	0,65	► 81212 TN
	110	30	66	108	153	640	65,5	0,033	1 400	2 800	1,25	89312 TN
65	90	18	67	87	83	320	32,5	0,0082	1 700	3 400	0,31	► 81113 TN
	100	27	69	96	140	490	50	0,019	1 300	2 600	0,72	► 81213 TN
	115	30	71	113	153	640	65,5	0,033	1 400	2 800	1,35	89313 TN

According to the bearing catalogue, 81113 TN cylindrical roller thrust bearing is suitable for main shaft first bearing. [7]

Principal dimensions					Basic load ratings		Fatigue load limit	Minimum load factor	Speed ratings		Mass	Designation
d	D	H	E _a	E _b	C	C ₀	P _u	A	Reference speed	Limiting speed		
mm					kN		kN	–	r/min		kg	–
15	28	9	16	27	11,2	27	2,45	0,000 058	4 300	8 500	0,024	► 81102 TN
17	30	9	18	29	12,2	31,5	2,85	0,000 079	4 300	8 500	0,027	► 81103 TN
20	35	10	21	34	18,6	48	4,65	0,00018	3 800	7 500	0,037	► 81104 TN
25	42	11	26	41	25	69,5	6,8	0,00039	3 200	6 300	0,053	► 81105 TN
30	47	11	31	46	27	78	7,65	0,00049	3 000	6 000	0,057	► 81106 TN
	52	16	31	50	50	134	13,4	0,0014	2 400	4 800	0,12	► 81206 TN
35	52	12	36	51	29	93	9,15	0,00069	2 800	5 600	0,073	► 81107 TN
	62	18	39	58	62	190	19,3	0,0029	2 000	4 000	0,21	► 81207 TN
40	60	13	42	58	43	137	13,7	0,0015	2 400	5 000	0,11	► 81108 TN
	68	19	43	66	83	255	26,5	0,0052	1 900	3 800	0,25	► 81208 TN
	78	22	44	77	95	365	36,5	0,011	2 000	4 000	0,48	89308 TN
45	65	14	47	63	45	153	15,3	0,0019	2 200	4 500	0,13	► 81109 TN
	73	20	48	70	83	255	26,5	0,0052	1 800	3 600	0,29	► 81209 TN
50	70	14	52	68	47,5	166	16,6	0,0022	2 200	4 300	0,14	► 81110 TN
	78	22	53	75	91,5	300	31	0,0072	1 700	3 400	0,36	► 81210 TN
55	78	16	57	77	69,5	285	29	0,0065	1 900	3 800	0,23	► 81111 TN
	90	25	59	85	122	390	40	0,012	1 400	2 800	0,57	► 81211 TN
60	85	17	62	82	80	300	30,5	0,0072	1 800	3 600	0,27	► 81112 TN
	95	26	64	91	137	465	47,5	0,017	1 400	2 800	0,65	► 81212 TN
	110	30	66	108	153	640	65,5	0,033	1 400	2 800	1,25	89312 TN
65	90	18	67	87	83	320	32,5	0,0082	1 700	3 400	0,31	► 81113 TN
	100	27	69	96	140	490	50	0,019	1 300	2 600	0,72	► 81213 TN
	115	30	71	113	153	640	65,5	0,033	1 400	2 800	1,35	89313 TN

According to the bearing catalogue, 81213 TN cylindrical roller thrust bearing is suitable for main shaft second bearing. [7]

6.2.6 Keys and Keyways

Given the scale and the power requirement parallel sunk keys were selected as the best fit for this application due to following reasons,

- Parallel sunk keys are relatively efficient in torque transmission as it closely fit between the shaft and the gear. As high torques are required to rotate the mixer blades it is necessary to have efficient torque transmission.
- Installation and removal process is simpler, reduce effort in maintenance and repairs.
- Given the size of the shaft and gears, cost-effectiveness is also considered. Due to simplicity in design parallel sunk keys cost relatively less to manufacture.

A rectangular sunk key is designed instead of a square type as the increased thickness of the key will impact the strength of the shaft. Increasing the diameter to accommodate a square type key seems impractical as it would significantly increase costs given the shaft dimensions.

6.2.6.1 Design Calculations Methodology

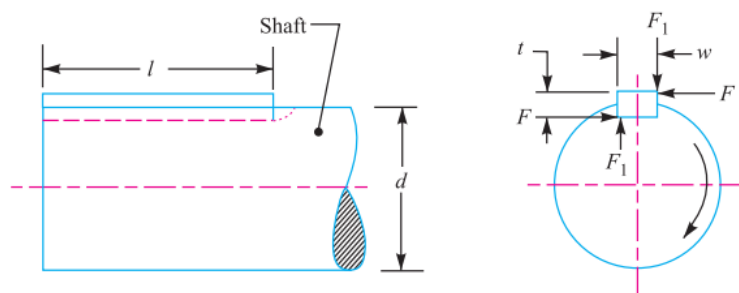


Figure 9 - Forces acting on a sunk key.

$$F = \text{Tangetial force}$$

$$F_1 = \text{Force due to fit of key and keyway}$$

$$T = \text{Torque transmitted}$$

$$\tau = \text{Shear stress}$$

$$\sigma_c = \text{Crushing stress}$$

Since key is subjected F and F_1 forces, both shearing and crushing conditions are considered during design calculation process.

The key considered to be made from same material as the shaft.

For rectangular sunk keys,

$$w = \frac{d}{4}$$

$$t = \frac{d}{6}$$

Shearing

$$F = \text{Area resisting shearing} \times \text{shear stress}$$

$$F = lw \times \tau$$

$$T = F \times \left(\frac{d}{2}\right) = lw \times \tau \times \left(\frac{d}{2}\right)$$

$$l = \frac{2T}{d\tau w}$$

Crushing

$$F = \text{Area resisting crushing} \times \text{crushing stress}$$

$$F = l \left(\frac{t}{2}\right) \times \sigma_c$$

$$T = F \times \left(\frac{d}{2}\right) = l \left(\frac{t}{2}\right) \times \sigma_c \times \left(\frac{d}{2}\right)$$

$$l = \frac{4T}{dt\sigma_c}$$

After considering both shearing and crushing conditions, maximum length out of two is selected. Then it is rounded up to a suitable value for manufacturing.

6.2.6.2 Examining the effect of keyway on shaft strength

Weakening effect of the shaft due to material removal to seat keys is examined using the shaft strength factor.

$$\text{Shaft strength factor } (e) = 1 - 0.2 \left(\frac{w}{d} \right) - 1.1 \left(\frac{h}{d} \right)$$

$$h = \text{Depth of keyway} = \frac{t}{2}$$

Once the strength factor is obtained from above equation, the maximum allowable torque transmitted by the shaft is calculated again including e and it is compared with the actual torque transmitted by the shaft to verify safe operation.

6.2.6.3 Computed values

	Shaft Diameter (mm)	Gears	RPM	Torque		width (mm)	t (mm)	h (mm)	L (mm)			e	Maximum allowable torque - Keyed
				Nm	Nmm				Shearing	Crushing	Result		
Input Shaft	65	I	2975	240.7	2.41E+05	16	10	5	4.46	7.14	8	0.866	4.85E+06
Lay Shaft	65	A	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
	65	B	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
	65	C	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
	65	D	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
	65	E	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
Idler Shaft	65	RI	1190	601.8	6.02E+05	16	10	5	11.16	17.85	18	0.866	4.85E+06
Output Shaft	65	F	405	1768.4	1.77E+06	16	10	5	28.94	46.31	48	0.866	5.49E+06
	65	G	595	1203.7	1.20E+06	16	10	5	19.70	31.52	32	0.866	5.49E+06
	65	H	782	915.9	9.16E+05	16	10	5	14.99	23.98	24	0.866	5.49E+06
	65	HR	501	1429.5	1.43E+06	16	10	5	23.40	37.43	38	0.866	5.49E+06
Bevel Gears	65	BV	405	1768.4	1.77E+06	16	10	5	32.78	52.45	54	0.866	4.85E+06
	65	BH	405	1768.4	1.77E+06	16	10	5	32.78	52.45	54	0.866	4.85E+06

Gear references are like the previous sections.

HR – H gear in reverse BV, BR – bevel gears, horizontal and vertical.

*Since the required values are smaller than the dimension standards of keys, dimensions are taken as L = 40 mm, b = 20 mm and t = 12 mm to match the standards.

6.2.6.4 Limits, Fit and Tolerances for Keys and Keyways

Since the operation speeds are high (400 – 800 rpm), *free running fits* are preferred. Also, the dimensions of components are relatively large. Hence accuracy is not essential.

Section x-x

Enlarged Detail of Key and Keyways

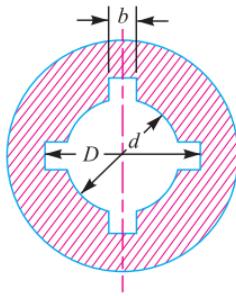
Shaft		Key	Keyway												
Nominal Diameter d		Size, $b \times h$	Nominal	Width b					Depth				Radius r		
Over	Up to and Incl			Free Fit		Normal Fit		Close Fit	Shaft t_1		Hub t_2		Max.	Min.	
				Shaft (H9)	Hub (D10)	Shaft (N9)	Hub (J ₅ /9) ^a	Shaft and Hub (P9)	Nominal	Tolerance	Nominal	Tolerance			
				Tolerances					Nominal	Tolerance	Nominal	Tolerance			
Keyways for Square Parallel Keys															
6	8	2 × 2	2	+0.025	+0.060	-0.004	+0.012	-0.006	1.2	+0.1 0	1	+0.1 0	0.16	0.08	
8	10	3 × 3	3	0	+0.020	-0.029	-0.012	-0.031	1.8		1.4		0.16	0.08	
10	12	4 × 4	4	+0.030	+0.078	0	+0.015	-0.012	2.5		1.8		0.16	0.08	
12	17	5 × 5	5	0	+0.030	-0.030	-0.015	-0.042	3		2.3		0.25	0.16	
17	22	6 × 6	6						3.5		2.8		0.25	0.16	

Keyways for Rectangular Parallel Keys															
22	30	8 × 7	8	+0.036	+0.098	0	+0.018	-0.015	4	+0.2 0	3.3	+0.2 0	0.25	0.16	
30	38	10 × 8	10	0	+0.040	-0.036	-0.018	-0.051	5		3.3		0.40	0.25	
38	44	12 × 8	12						5		3.3		0.40	0.25	
44	50	14 × 9	14	+0.043	+0.120	0	+0.021	-0.018	5.5		3.8		0.40	0.25	
50	58	16 × 10	16		+0.050	-0.043	-0.021	-0.061	6		4.3		0.40	0.25	
58	65	18 × 11	18						7	4.4	0.40	0.25			
65	75	20 × 12	20						7.5	4.9	0.60	0.40			
75	85	22 × 14	22	+0.052	+0.149	0	+0.026	-0.022	9	5.4	0.60	0.40			
85	95	25 × 14	25		+0.065	-0.052	-0.026	-0.074	9	5.4	0.60	0.40			
95	110	28 × 16	28						10	6.4	0.60	0.40			

Figure 10 - Tolerance limits - BS4500 ISO Limits and Fits

From ISO standard tables highlighted tolerances are selected considering the shaft diameter of 65mm.

6.2.7 Splines in output shaft



$$D = 1.25 d, b = 0.25 d \text{ (Standard)}$$

$$d = 65 \text{ mm}$$

$$D = 81.25 \text{ mm} = 82 \text{ mm}, b = 16.25 \text{ mm}$$

From ISO standard texts number of splines is selected as 8.

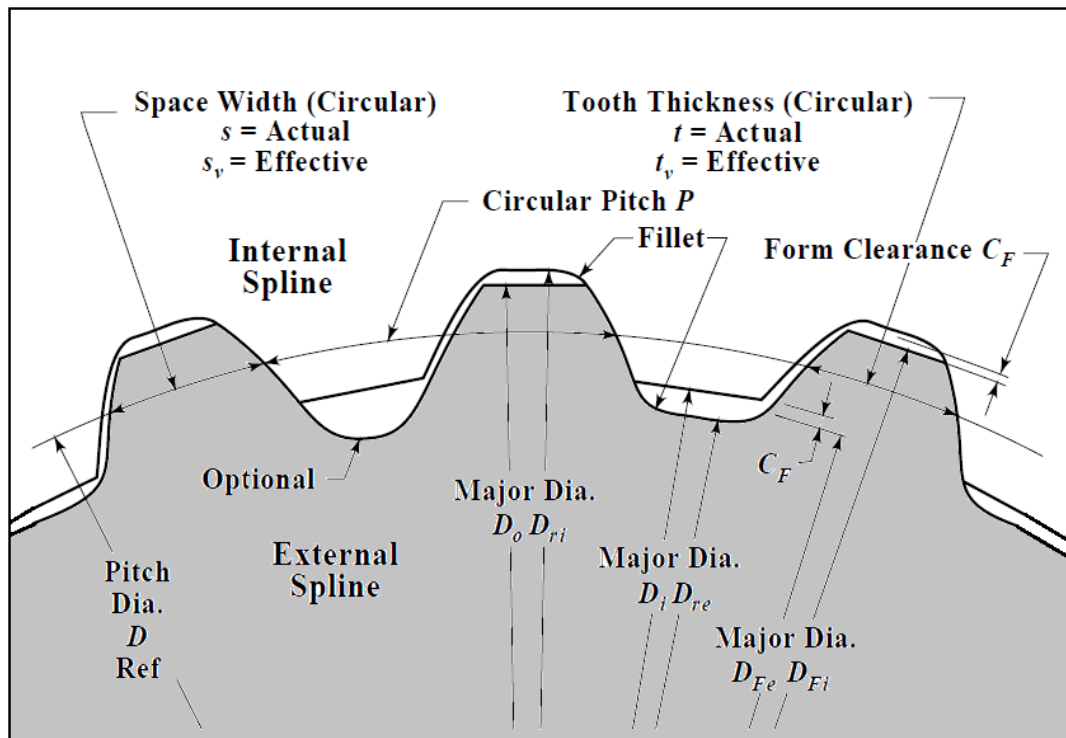


Figure 11 - Splines terminology

6.2.7.1 Spline Dimensions

In order to calculate spline strength pitch diameter of the splines need to be calculated.

$$D_{Ref} = D_{re} + 2 \times h_f \text{ (} h_f \text{ - Reference dedendum)}$$

$$h_f = 0.6m \text{ (} m \text{ - Module)}$$

$$m = \frac{d}{1.05} \text{ (} d \text{ - Depth)}$$

$$d = h_k + h_f = (D_o - D_{re})/2$$

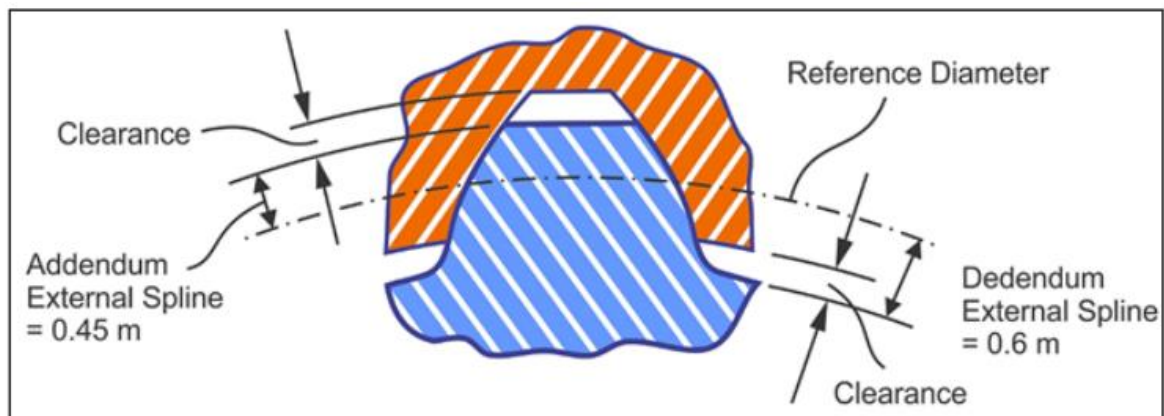


Figure 12 - Flank centred spline connection

$$d = \frac{81.25 - 65}{2} = 8.125 \text{ mm}$$

$$m = \frac{8.125}{1.05} = 7.738 = 8 \text{ mm}$$

$$h_f = 0.6 \times 8 = 4.8 \text{ mm}$$

$$D_{Ref} = 65 + (2 \times 4.8) = 74.6 \text{ mm}$$

Minimum clearance selection according to reference diameter (d_B) and module (m),

d_B mm	Minimum form clearance c_{Fmin} μm		
	Module 0,5 to 1,5	Module 1,75 to 4	Module 5 to 10
Up to and including 12	25	—	—
13 to 25	28	30	—
26 to 50	30	35	40
51 to 100	35	40	45
101 to 200	40	45	50
201 to 400	—	50	55
Greater than 400	—	—	65

Figure 13 - Design clearance for splines

$$clearance = 45 \mu m$$

6.2.7.2 Spline Strength Calculations

Once the spline dimensions are selected to standards, shear stresses and compressive stresses acting on spline teeth are evaluated to further confirm its suitability for the given operation.

According to the AISC code, the allowable stresses on spline teeth should be under the following conditions with respect to its minimum yield strength.

$$\text{Allowable Shear Stress } (S_s^a) = 0.4 \times S_y$$

$$\text{Allowable Compressive Stress } (S_c^a); 0.45 S_y \leq S_c^a \leq 0.6 S_y$$

S_y – Material yield strength

6.2.7.3 Shear Stress

To avoid failure, following condition must be satisfied. [5]

$$S_s^a \geq S_s \times (K_a/L_f)$$

K_a – Application factor

L_f – Life factor

S_s – Shear stress in spline

$$S_s = \frac{4TK_m}{DNF_e t_e}$$

T – Torque

D – Pitch diameter

N – Number of splines

K_m – Load distribution factor

$$t_e - \text{chordal thickness at pitch line} = \frac{D}{2N} = 4.66 \text{ mm}$$

F_e – Effective face width ~ 16 mm

Load distribution factor, $K_m = 1.5$

	Load distribution factor for splines K_m			
	Effective face width (Fe)			
Misalignment	½-in. (12.7 mm)	1-in. (25.4 mm)	2-in. (50.8 mm)	4-in. (101.6)
0.001 in. / in. (mm/mm)	1	1	1	1 ½
0.002 in. / in. (mm/mm)	1	1	1 ½	2
0.004 in. / in. (mm/mm)	1	1 ½	2	2 ½
0.008 in. / in. (mm/mm)	1 ½	2	2 ½	3

Figure 14 - Load distribution factor with face width[6]

Spline application factor, $K_a = 1$

Input source category (Driving input or machine)	Type of Load			
	Uniform	Light shock	Intermittent Shock	heavy shock
	Generators, Fans	Oscillating pumps	Actuators	Presses, Shears
Uniform (Turbine, Motor)	1	1.2	1.5	1.8
light shock, (Hydraulic Motor)	1.2	1.3	1.8	2.1
Medium shock, (Internal combustion engine)	2	2.2	2.4	2.8

Figure 15 - Spline application factor.[6]

Fatigue life factor, $L_f = 0.5$ (for 50,000 cycles)

Number of torque cycles	Fatigue Life Factor, L_f	
	Unidirectional	Fully – reversed
1,000	1.8	1.8
10,000	1.0	1.0
100,000	0.5	0.4
1,000,000	0.4	0.3
10,000,000	0.3	0.2

Figure 16 - Fatigue life factor for splines[6]

6.2.7.4 Compressive Stress

To avoid failure, following condition must be satisfied. (All the symbols have the same meaning as the shear stress section above)[5]

$$S_c^a \geq S_c \times (K_a/L_f)$$

S_c – Compressive stress in spline

$$S_s = \frac{2TK_m}{DNF_e h}$$

h – Radial height = 8.125 mm

6.2.7.5 Computed results

With computed results, required number of splines were selected as 8.

No of Splines	Shear Stress (MPa)	Shear Failure Test	Compressive Stress (MPa)	Compressive Failure Test
4	246.76	-120.44	70.76	123.05
6	164.50	-17.63	47.17	152.53
8	123.38	33.78	35.38	167.27
10	98.70	64.62	28.30	176.12
12	82.25	85.19	23.59	182.02

6.2.8 Couplings

6.2.8.1 Coupling for motor shaft - gearbox input shaft

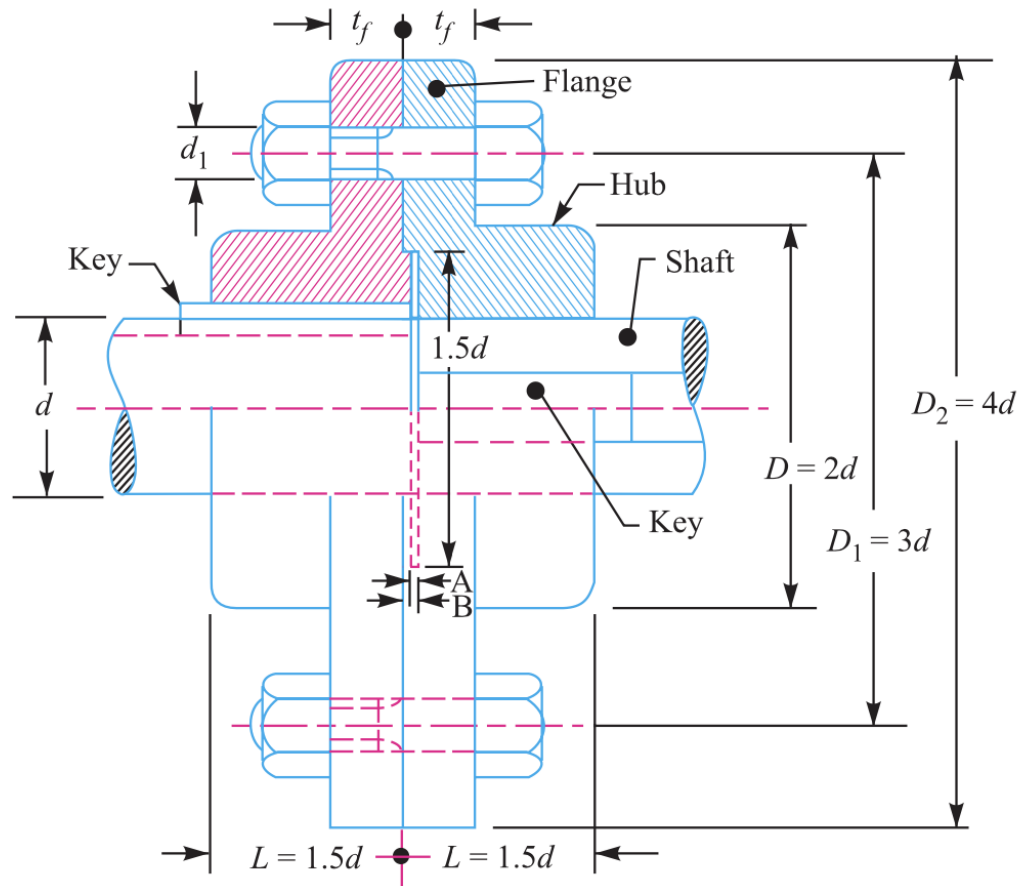


Figure 17: cross section of flange coupling

Motor's output shaft diameter = $d=65\text{mm}$

Outside diameter of hub = $D = 2 \times d$

$$D = 2 \times 65\text{mm}$$

$$D = 130\text{mm}$$

Length of the hub = $L = 1.5 \times d$

$$L = 97.5\text{mm}$$

Pitch circle diameter of bolts = $D_1 = 3 \times d$

$$D_1 = 195\text{mm}$$

Outside diameter of flange = $D_2 = 4 \times d$

$$D_2 = 260 \text{ mm}$$

Thickness of flange = $t_f = 0.5 \times d$

$$= 32.5 \text{ mm}$$

Design for hub

Maximum input torque = $T_{\max} = 241 \text{ Nm}$

$$T_{\max} = \frac{\pi}{16} \times \tau_c \times \left(\frac{D^4 - d^4}{D} \right)$$

τ_c = shear stress of the hub

$$241 = \frac{\pi}{16} \times \tau_c \times \left(\frac{0.13^4 - 0.065^4}{0.13} \right)$$

$$\tau_c = 0.6 \text{ Mpa}$$

Cast iron is chosen as the material for the hub.

Shear strength of Cast iron = $[\tau] = 15 \text{ MPa}$

$$0.6 < 15$$

$$\tau_c < [\tau]$$

$\therefore$ The hub is safe.

Design for key

Diameter of the shaft = $d = 65 \text{ mm}$

Width of the key = $w = 20 \text{ mm}$

Thickness of the key = $t = 12 \text{ mm}$

Length of the key = $l = L = 97.5 \text{ mm}$

Shaft diameter (mm) upto and including	Key cross-section		Shaft diameter (mm) upto and including	Key cross-section	
	Width (mm)	Thickness (mm)		Width (mm)	Thickness (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

$$T = l \times w \times \tau_s \times \frac{d}{2}$$

$$T = l \times \frac{t}{2} \times \tau_{cs} \times \frac{d}{2}$$

τ_s = Shear stress in the key

σ = Direct stress in the key when crushing

$$\tau_s = \frac{2T}{lw}$$

$$\tau_s = \frac{2 \times 241}{0.0975 \times 0.020}$$

$$\tau_s = 0.25 \text{ MPa}$$

$$\sigma = \frac{4T}{ltd}$$

$$\sigma = \frac{4 \times 241}{0.0975 \times 0.012 \times 0.065}$$

$$\sigma = 12.67 \text{ MPa}$$

AISI 1040 carbon steel is chosen as the material for the hub key.

$$\text{Allowable shear stress} = [\tau] = 111.6 \text{ MPa}$$

$$\text{Allowable direct stress} = [\sigma] = 620 \text{ MPa}$$

To avoid failure by shearing,

$$[\tau] > \tau_s$$

$$111.6 > 0.25$$

∴ The key is safe from shearing.

To avoid failure by crushing,

$$[\sigma] > \sigma$$

$$620 > 12.67$$

∴ The key is safe from crushing.

Design for flange

$$T_{max} = \frac{\pi \times D^2}{2} \times \tau_c \times t_f$$

$$\text{Thickness of flange} = t_f = 0.5 \times d = 32.5 \text{ mm}$$

τ_c = shear stress of the flange

$$241 = \frac{\pi \times 0.13^2}{2} \times \tau_c \times 0.0325$$

$$\tau_c = 0.27 \text{ MPa}$$

$$\text{Shear strength of Cast iron} = [\tau] = 15 \text{ MPa}$$

$$0.27 < 15$$

$$\tau_c < [\tau]$$

∴ The flange is safe.

Design for bolts

$$T_{max} = \frac{\pi}{4} \times d_1^2 \times \tau_b \times n \times \frac{D_1}{2}$$

Nominal Diameter of the bolts = d1

Since the diameter of the shaft is 65mm, n = 4

Pitch circle diameter = D1 = 3d = 3x65 = 195 mm

Bolt material was selected as AISI 1018 mild/low carbon steel.

τ_b = shear stress of AISI 1018 = 79.2 MPa

Assume safety factor of bolts = 2

Allowable shear stress = $79.2/2 = 39.6$ MPa

$$T_{max} = \frac{\pi}{4} \times d_1^2 \times \tau_b \times n \times \frac{D_1}{2}$$

$$241 = \frac{\pi}{4} \times d_1^2 \times 39.6 \text{ MPa} \times 4 \times \frac{0.195}{2}$$

$$d_1 = 4.457 \text{ mm}$$

Assuming coarse threads, the nearest standard size of bolt is M 6

Considering bolt in crushing,

$$T_{max} = (n \times d_1 \times t_f \times \sigma_{cb}) \times \frac{D_1}{2}$$

σ_{cb} = crushing stress of the bolt

$$T_{max} = (n \times d_1 \times t_f \times \sigma_{cb}) \times \frac{D_1}{2}$$

$$241 = (4 \times 4.773 \text{ mm} \times 32.5 \text{ mm} \times \sigma_{cb}) \times \frac{195 \text{ mm}}{2}$$

$$\sigma_{cb} = 3.98 \text{ MPa}$$

Crushing stress of AISI 1018 mild/low carbon steel = $[\sigma] = 230 \text{ MPa}$

$$3.98 < 230$$

$$\sigma_{cb} < [\sigma]$$

$\therefore$ bolt is safe from crushing.

6.2.8.2 Coupling for gearbox output shaft - Impeller shaft

Gearbox output shaft(power shaft) diameter = $d = 65 \text{ mm}$

Outside diameter of hub = $D = 2 \times d$

$$D = 2 \times 65 \text{ mm}$$

$$D = 130 \text{ mm}$$

Length of the hub = $L = 1.5 \times d$

$$L = 97.5 \text{ mm}$$

Pitch circle diameter of bolts = $D_1 = 3 \times d$

$$D_1 = 195 \text{ mm}$$

Outside diameter of flange = $D_2 = 4 \times d$

$$D_2 = 260 \text{ mm}$$

Thickness of flange = $t_f = 0.5 \times d$

$$= 32.5 \text{ mm}$$

Design for hub

Maximum output torque = $T_{\max} = 1790.5 \text{ Nm}$

$$1790.5 = \frac{\pi}{16} \times \tau_c \times \left(\frac{D^4 - d^4}{D} \right)$$

τ_c = shear stress of the hub

$$1790.5 = \frac{\pi}{16} \times \tau_c \times \left(\frac{0.13^4 - 0.065^4}{0.13} \right)$$

$$\tau_c = 4.42 \text{ Mpa}$$

Cast iron is chosen as the material for the hub.

Shear strength of Cast iron = $[\tau] = 15 \text{ MPa}$

$$4.42 < 15$$

$$\tau_c < [\tau]$$

$\therefore$ The hub is safe.

Design for key

Diameter of the shaft = $d = 65 \text{ mm}$

Width of the key = $w = 20 \text{ mm}$

Thickness of the key = $t = 12 \text{ mm}$

Shaft diameter (mm) upto and including	Key cross-section		Shaft diameter (mm) upto and including	Key cross-section	
	Width (mm)	Thickness (mm)		Width (mm)	Thickness (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

Length of the key = $l = L = 97.5 \text{ mm}$

$$T = l \times w \times \tau_s \times \frac{d}{2}$$

$$T = l \times \frac{t}{2} \times \tau_{cs} \times \frac{d}{2}$$

τ_s = Shear stress in the key

σ = Direct stress in the key when crushing

$$\tau_s = \frac{2T}{lw}$$

$$\tau_s = \frac{2 \times 1790.5}{0.0975 \times 0.020}$$

$$\tau_s = 1.83 \text{ MPa}$$

$$\sigma = \frac{4T}{ltd}$$

$$\sigma = \frac{4 \times 1790.5}{0.0975 \times 0.012 \times 0.065}$$

$$\sigma = 94.17 \text{ MPa}$$

AISI 1040 carbon steel is chosen as the material for the hub key.

Allowable shear stress = $[\tau] = 111.6 \text{ MPa}$

Allowable direct stress = $[\sigma] = 620 \text{ MPa}$

To avoid failure by shearing,

$$[\tau] > \tau_s$$

$$111.6 > 1.83$$

$\therefore$ The key is safe from shearing.

To avoid failure by crushing,

$$[\sigma] > \sigma$$

$$620 > 94.17$$

$\therefore$ The key is safe from crushing.

Design for flange

$$T_{max} = \frac{\pi \times D^2}{2} \times \tau_c \times t_f$$

Thickness of flange = $t_f = 0.5 \times d = 32.5 \text{ mm}$

τ_c = shear stress of the flange

$$1790.5 = \frac{\pi \times 0.13^2}{2} \times \tau_c \times 0.0325$$

$$\tau_c = 2.07 \text{ MPa}$$

Shear strength of Cast iron = $[\tau] = 15 \text{ MPa}$

$$2.07 < 15$$

$$\tau_c < [\tau]$$

∴ The flange is safe.

Design for bolts

$$T_{max} = \frac{\pi}{4} \times d_1^2 \times \tau_b \times n \times \frac{D_1}{2}$$

Nominal Diameter of the bolts = d_1

Since the diameter of the shaft is 65mm, $n = 4$

Pitch circle diameter = $D_1 = 3d = 3 \times 65 = 195 \text{ mm}$

Bolt material was selected as AISI 1018 mild/low carbon steel.

τ_b = shear stress of AISI 1018 = 79.2 MPa

Assume safety factor of bolts = 2

Allowable shear stress = $79.2/2 = 39.6 \text{ MPa}$

$$T_{max} = \frac{\pi}{4} \times d_1^2 \times \tau_b \times n \times \frac{D_1}{2}$$

$$1790.5 = \frac{\pi}{4} \times d_1^2 \times 39.6 \text{ MPa} \times 4 \times \frac{0.195}{2}$$

$$d_1 = 12.14 \text{ mm}$$

Assuming coarse threads, the nearest standard size of bolt is M 16

Considering bolt in crushing,

$$T_{max} = (n \times d_1 \times t_f \times \sigma_{cb}) \times \frac{D_1}{2}$$

σ_{cb} = crushing stress of the bolt

$$T_{max} = (n \times d_1 \times t_f \times \sigma_{cb}) \times \frac{D_1}{2}$$

$$1790.5 = (4 \times 13.546 \text{ mm} \times 32.5 \text{ mm} \times \sigma_{cb}) \times \frac{19.5 \text{ mm}}{2}$$

$$\sigma_{cb} = 10.4 \text{ MPa}$$

Crushing stress of AISI 1018 mild/low carbon steel = $[\sigma] = 230 \text{ MPa}$

$$3.8 < 370$$

$$\sigma_{cb} < [\sigma]$$

$\therefore$ bolt is safe from crushing.

7 Lubrication

In the world of technology and mechanical systems, the importance of implementing lubricating techniques for a gearbox cannot be overstated. The gearbox must be properly lubricated in order for it to function consistently and efficiently. By releasing heat, it not only reduces wear and friction but also preserves the integrity of the internal components. Here, we'll look at the main reasons that choosing an effective lubrication technique is so important to the reliability and longevity of the gearbox.

Purpose of Lubrication in a Gearbox

- Reduce Friction
- Heat dissipation
- Wear prevention
- Noise reduction
- Sealing and contaminate exclusion.
- Enhanced efficiency
- Corrosion protection
- Temperature control

There is main 3 types of lubrication methods.

1. Grease lubrication
2. Splash (Oil bath) lubrication
3. Forced lubrication.

Grease lubrication is advised for situations where the pitch line velocities are low, while the other two methods are typically suggested for applications involving higher pitch line velocities.

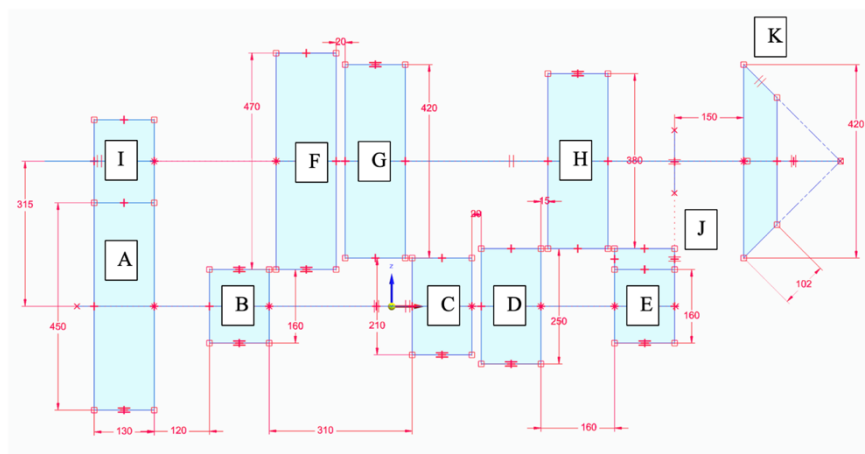


Figure 18: gearbox configuration

Gear	Maximum Pitch line velocity (m/s)
A	28
B	9.95
C	13
D	15.6
E	9.95
F	9.95
G	13
H	15.6
I	28
J	9.95

No.	Lubrication	Range of tangential speed v (m/s)
1	Grease lubrication	
2	Splash lubrication	
3	Forced oil circulation lubrication	

In this design tangential speeds are exceeding the 5 m/s for all gears. Also, most the speeds except 28m/s are varying in the region of 5-15 m/s. Therefore, Splash lubrication is selected as the lubrication method of the design.

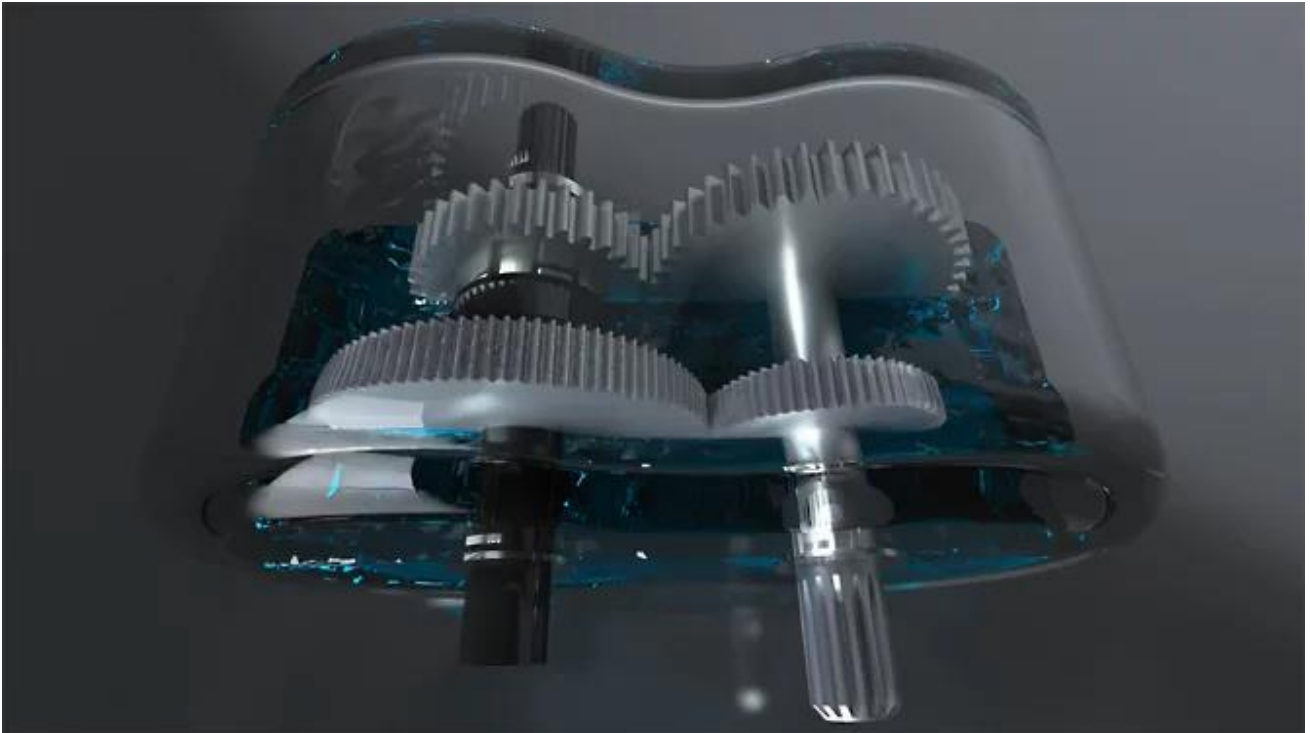


Figure 19 – Lubrication inside the gear

8 Final Design

8.1 Spur gear construction

Once the basic gear dimensions are finalized, gear construction is carried out. The main goal behind the gear construction process is to reduce gear volume, hence reduce the cost. To do that, some portion from the middle of large gears are removed.

The requirement of gear construction was decided, if the pitch circle diameter of the gear is more than 250mm as mentioned in the reference texts.[5]

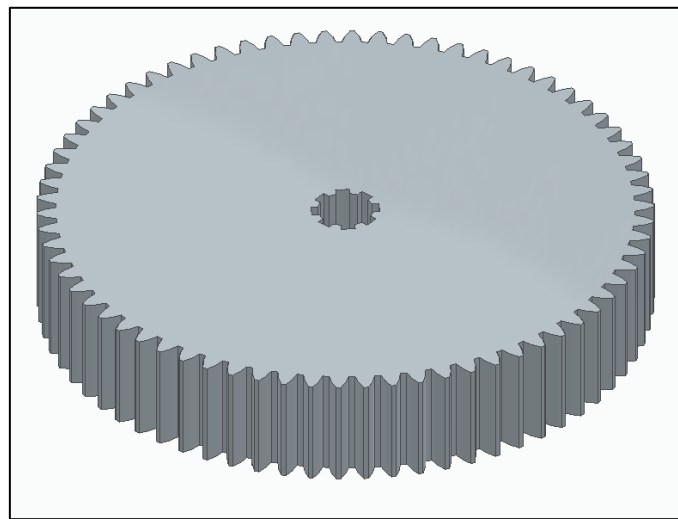


Figure 20 - Before gear construction

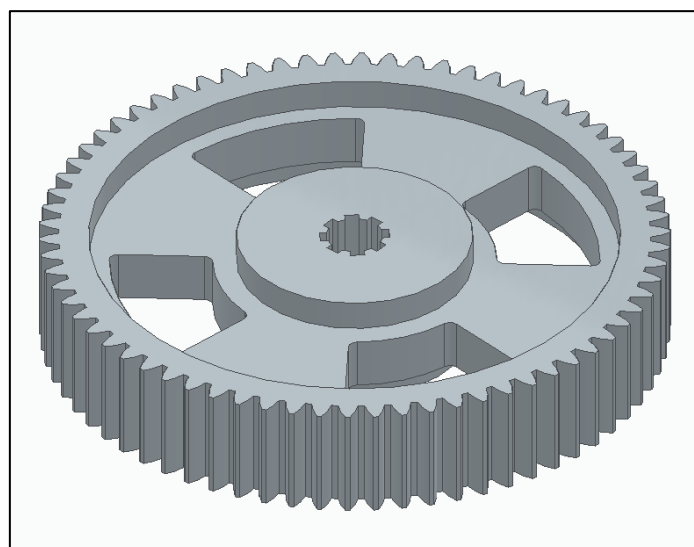


Figure 21 - After gear construction

8.2 Stepped shafts

All the shafts are stepped on the places where it is necessary to have guide for gears and bearings. Starting from the minimum diameter obtained from the design calculations the shaft is stepped at each place where there should be a gear or a bearing.

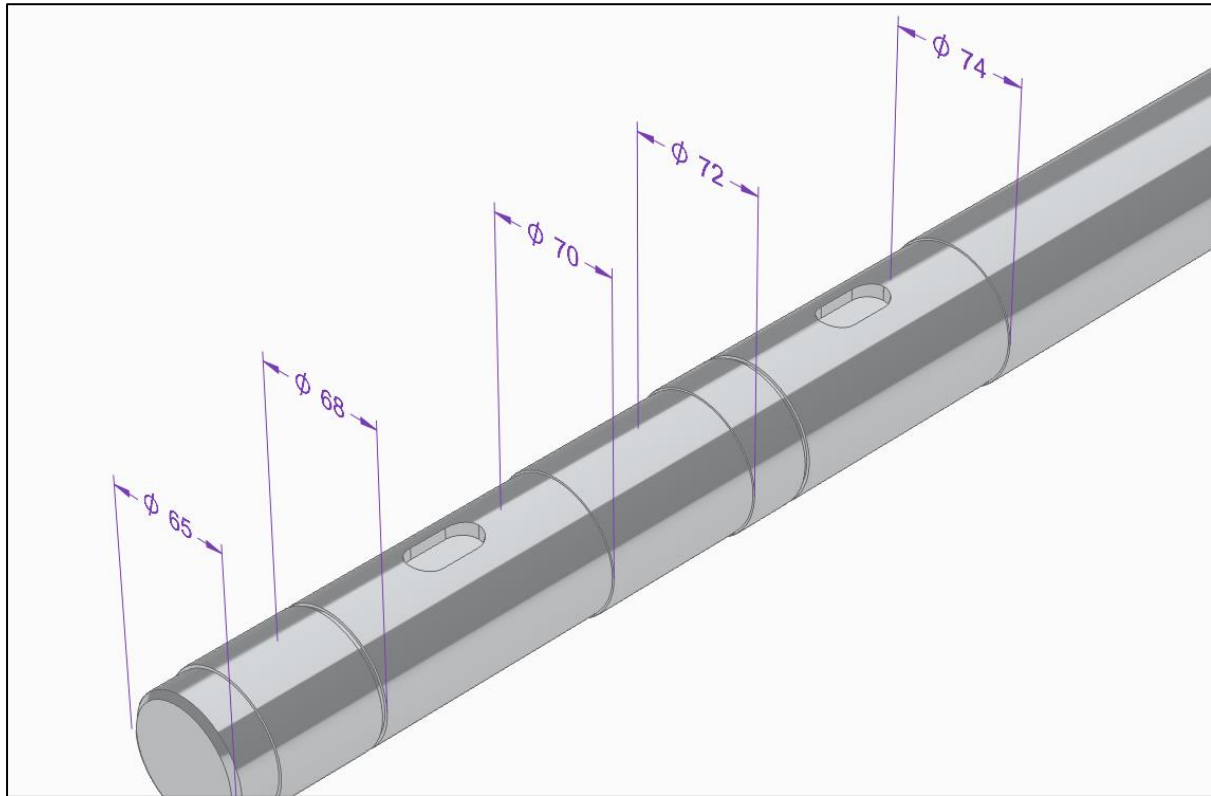


Figure 22 - Steps in shaft

Since the shaft diameter is changed at places, previous bearing selection must be altered to match the new dimensions. Previous bearing load calculations are used, bearings that satisfy previous load conditions and new bore diameter are selected.

Bearings					Load Rating		Calculated Dynamic Load
		Shaft Diameter	Outer Diameter	Width	Static	Dynamic	
A	Initial	65	120	23	118	122	95
	Final	68	120	23	107	118	
B	Initial	65	140	48	290	285	245
	Final	70	150	51	335	287	
C	Initial	65	140	33	196	212	152
	Final	68	140	33	180	190	

8.3 Keys and circlips

Keys are only used for torque transmission in all the shafts. To avoid axial movement of the gears, circlips are placed on one side as the other side is seated with the step in the shaft.



Figure 23- Circlip

8.4 Gearbox Assembly

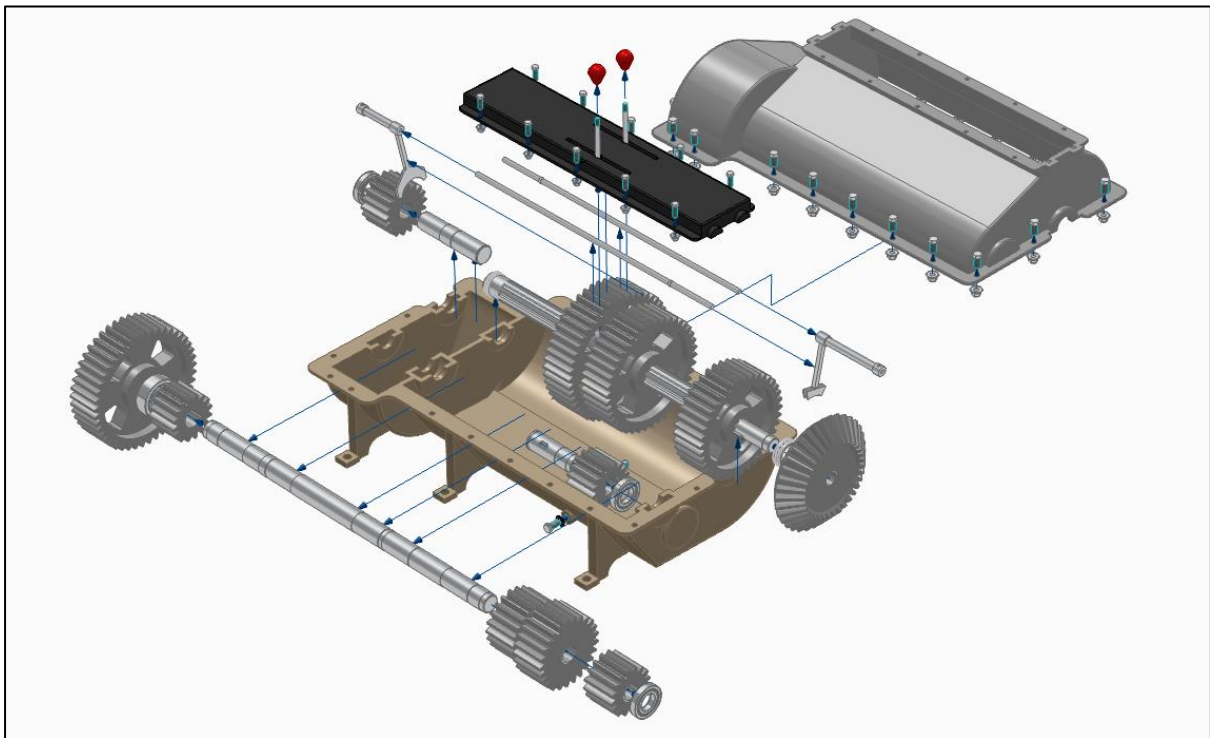


Figure 24 - Exploded view

First the idler shaft is assembled and placed in the gearbox bottom housing. Gear, bearing and the circlip is inserted from right side.

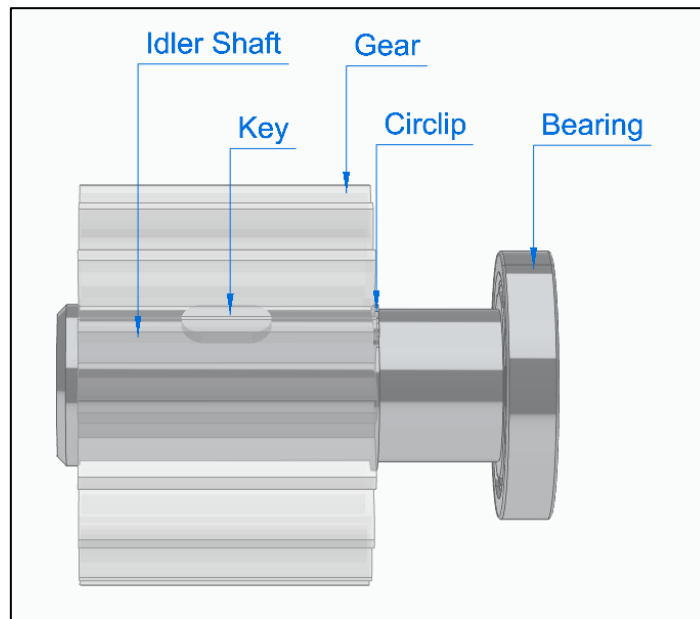


Figure 25 - Idler shaft assembly

Next the layshaft is assembled and placed. First two gears, bearings and circlips are inserted from the left-hand side while the rest of the components are inserted from the right-hand side.

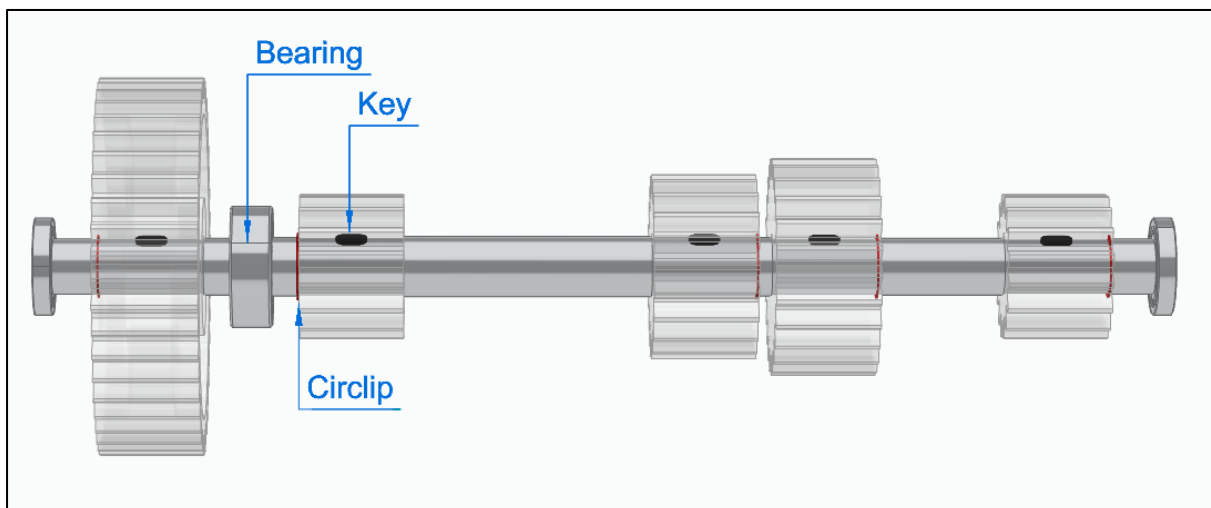


Figure 26 - Layshaft assembly

Then the input shaft is assembled and placed. Gear, bearing and the circlip is inserted from right side.

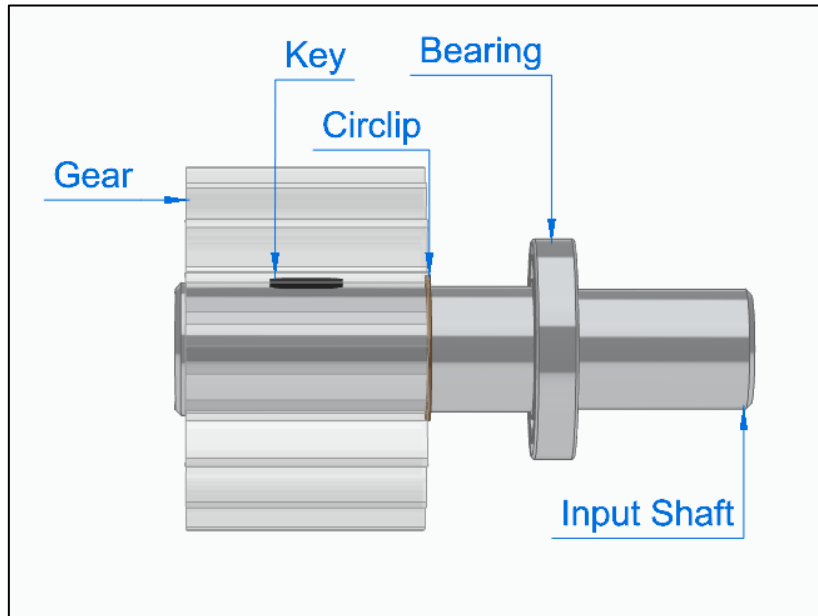


Figure 27 - Input shaft assembly

Finally, the output shaft is placed completing the gear assembly. Only left bearing is inserted from that side. Rest of the components are inserted from right side. A circlip is placed at the right end of the shaft to keep the bevel gear in place.

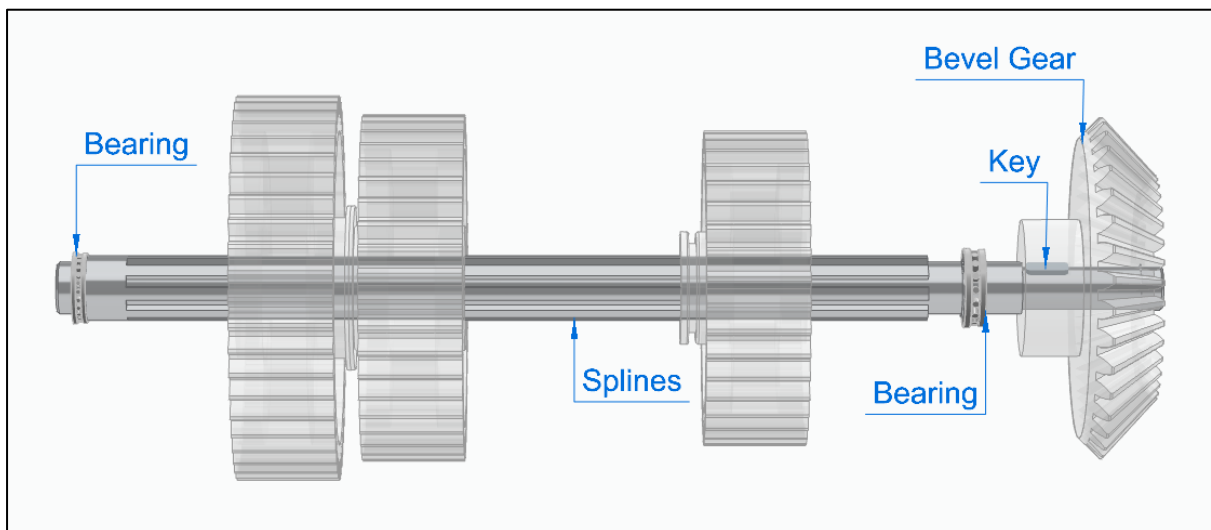


Figure 28 - Output shaft assembly

8.5 Gear shifting mechanism.

We have decided to use two gear levers for shifting the gears.

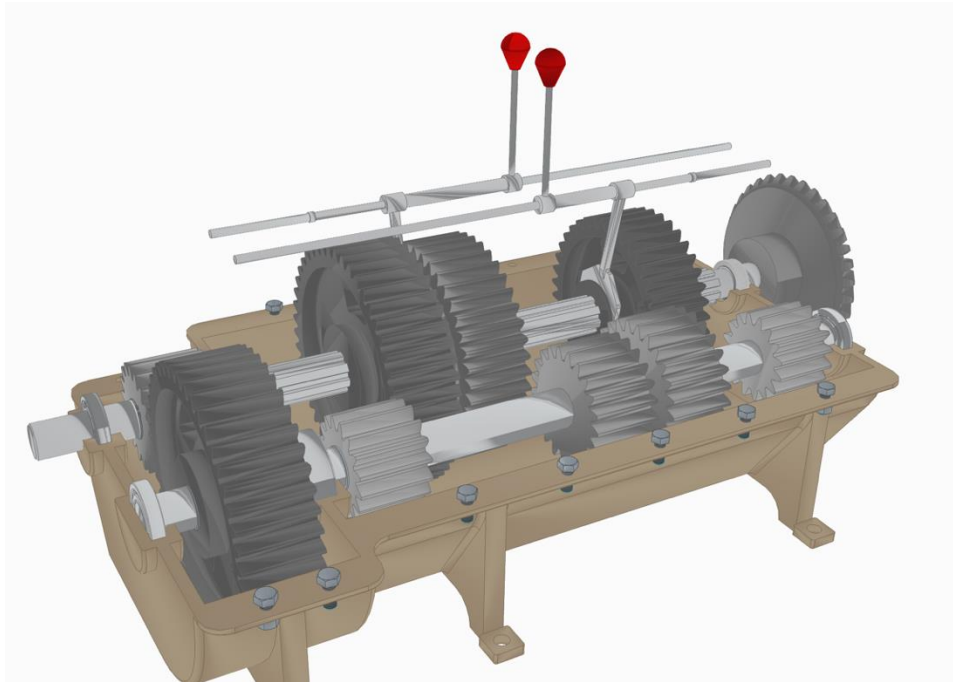


Figure 29:gearbox configuration

Gear forks are held by two metal rods that attached to top housing.



Figure 31:left gear fork



Figure 30: right gear fork

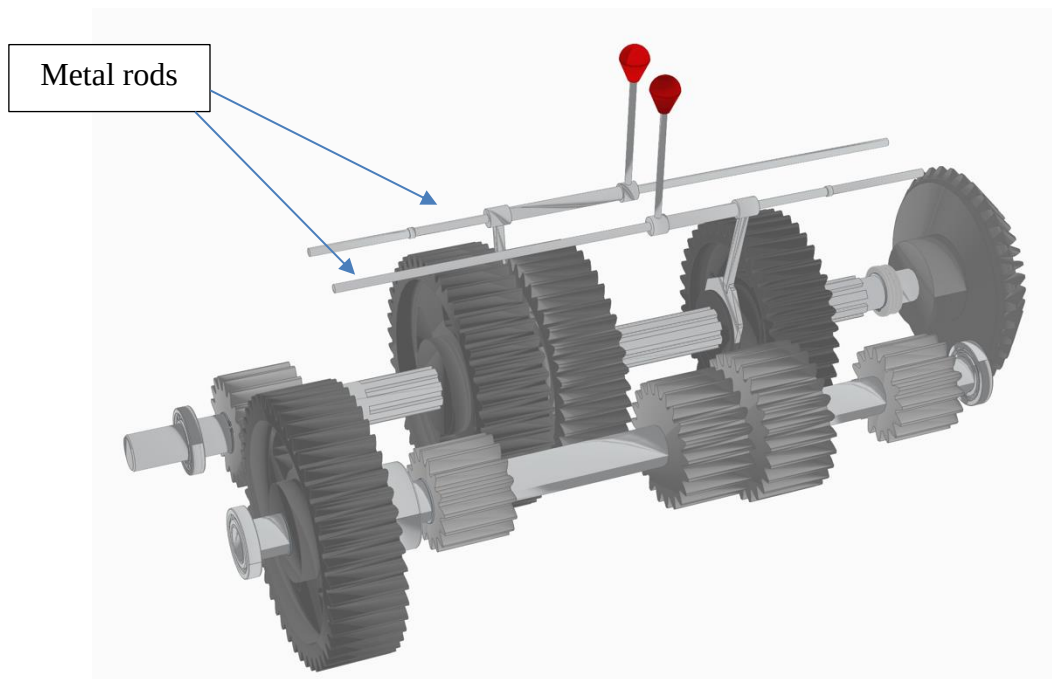


Figure 32:gear shifting mechanism.

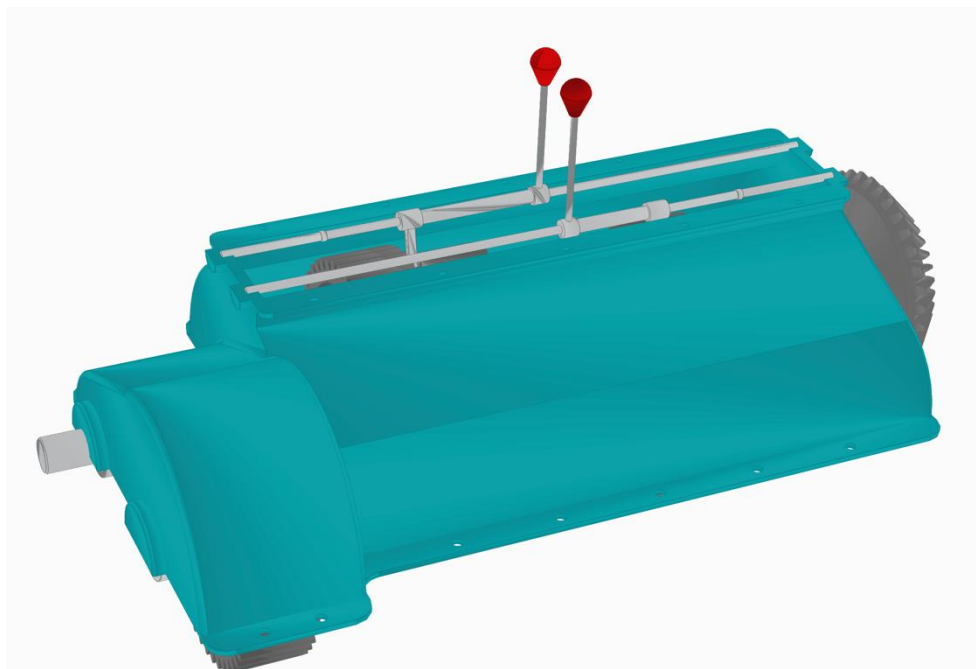


Figure 33:gear shifting mechanism with middle housing



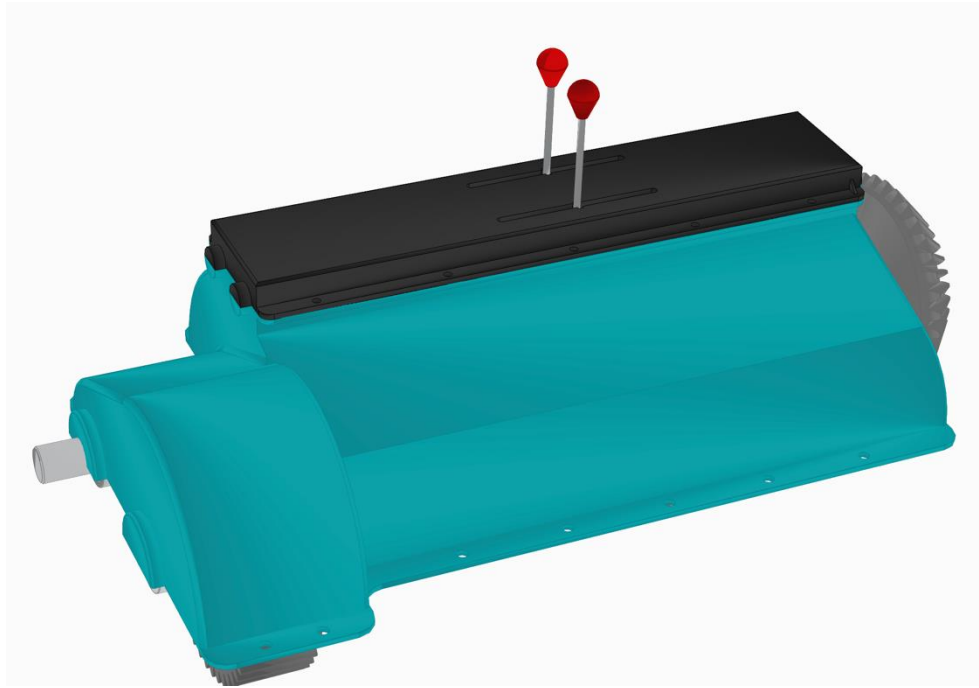


Figure 34:shifting mechanism with middle and top housings

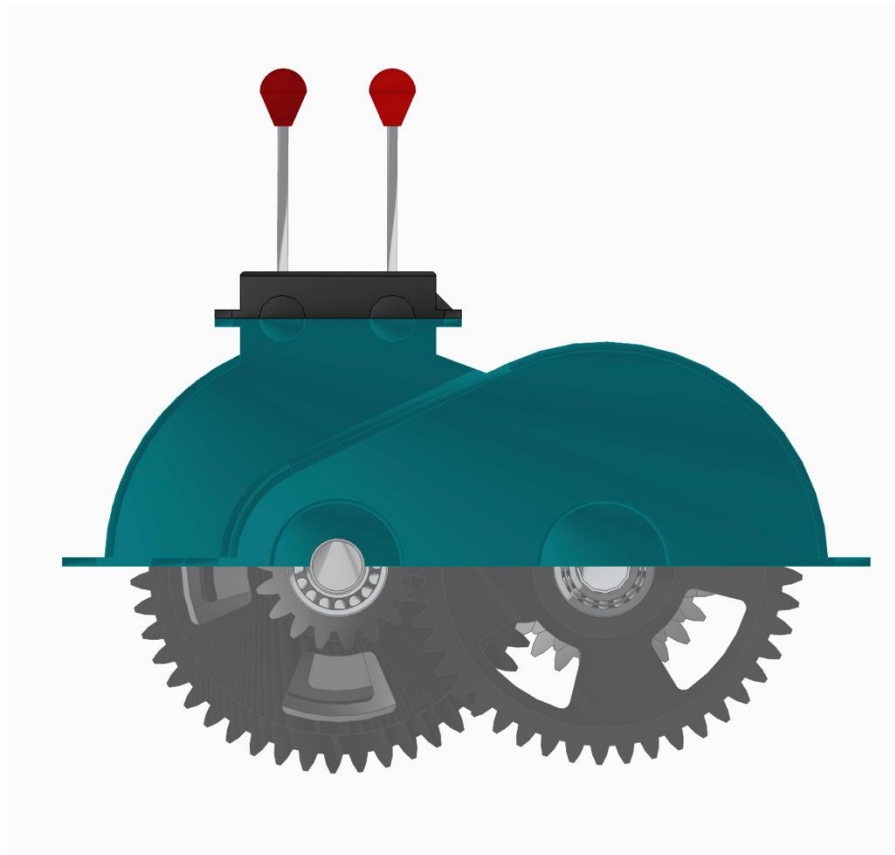
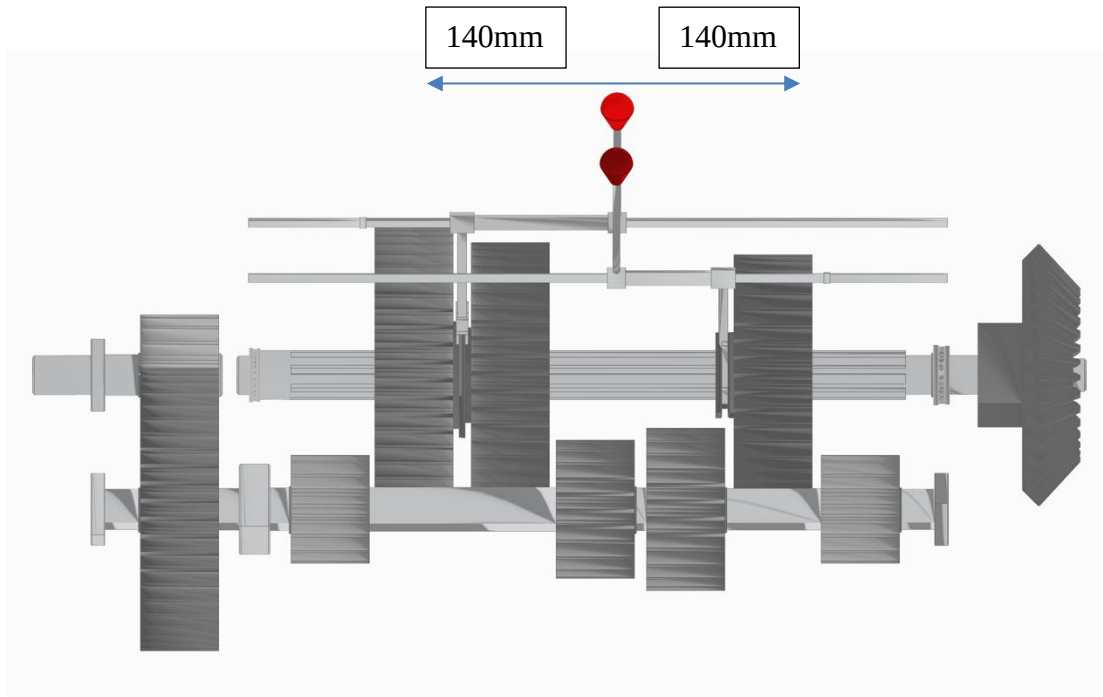
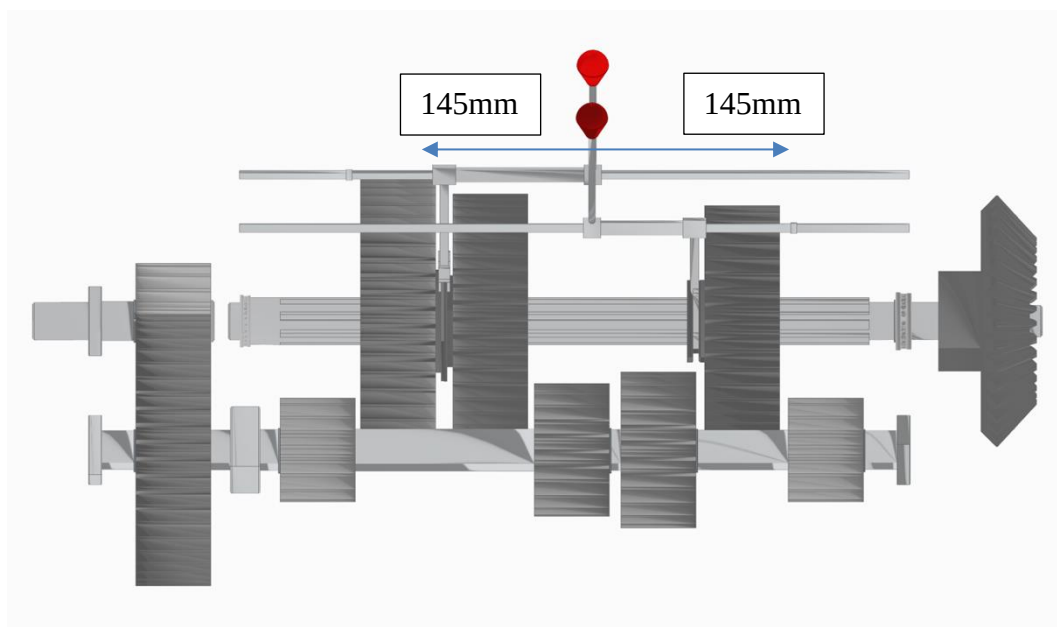


Figure 35:side view of shifting mechanism

The travel distance to the 1st gear position from neutral position is 140mm and there is a same distance for the second gear position.



The travel distance to the 3rd gear position from neutral position is 145mm and there is a same distance for the reverse gear position.



8.6 Casing Design

The casing has a thickness of 20 millimetres and is produced using a casting process. The bottom section is constructed from Iron:0.6015, which is EN-GJL-150 Gray cast iron, while the top part is crafted from Iron:0.6010, specifically EN-GJL-100 Gray cast iron.

We created a side view sketch of the casing based on our assembly of gears and shafts.

The minimum distance between gears and casing is 20mm.

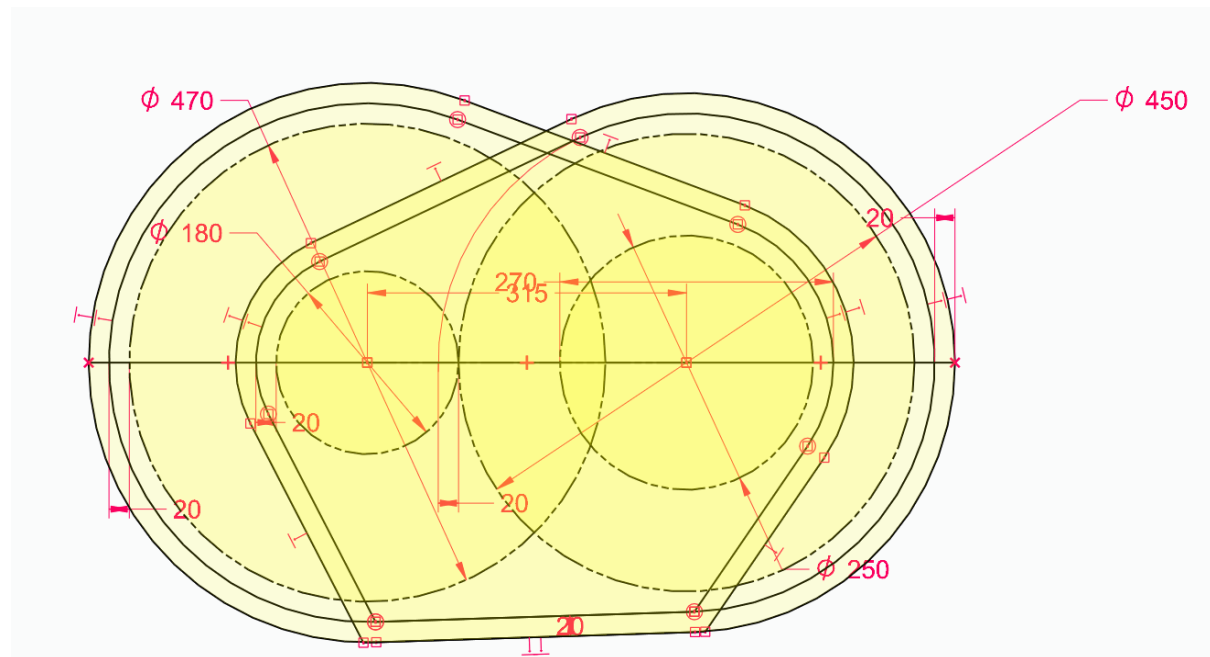


Figure 36 - Side view sketch of the casing

Given our specific application, we employ a splashing lubrication method. As a result, the oil needs to be changed at regular intervals. To facilitate this process effectively and efficiently, the lower section of the casing is at equal height.

Extrusion was used to make casing.

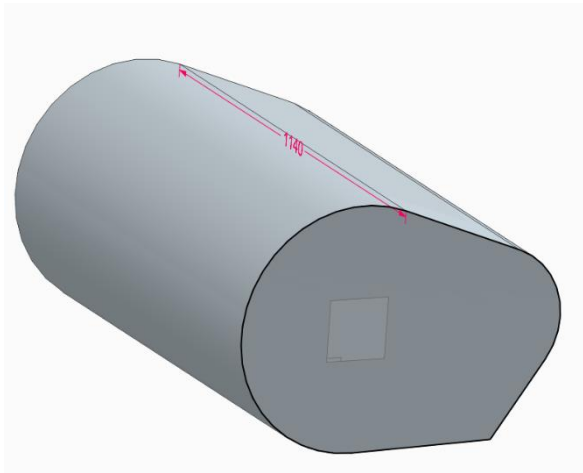


Figure 32 - First extrusion

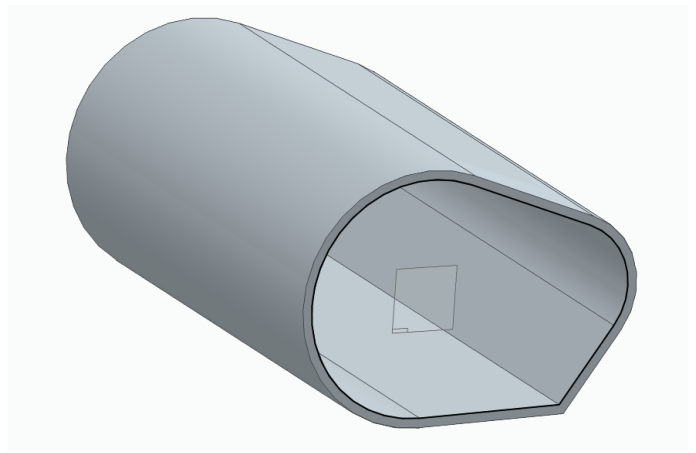


Figure 33 - First cutout

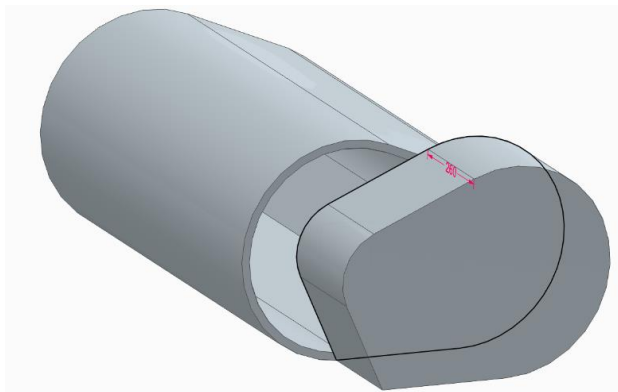


Figure 34 - Second extrusion

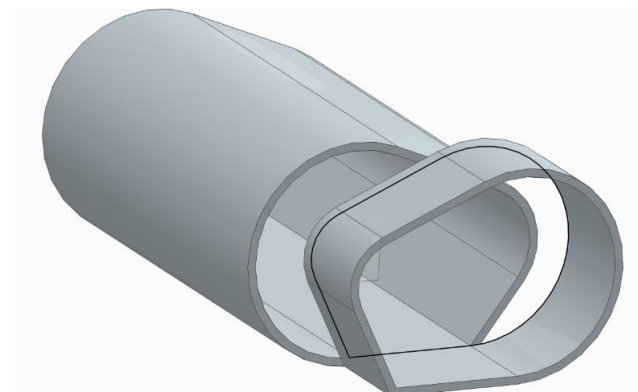


Figure 35 - Second cutout

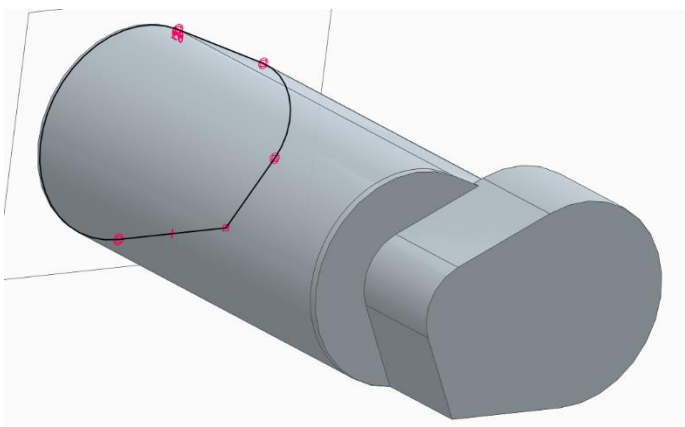


Figure 36 - Extrusion for walls

During disassembly, the initial step involves removing the upper portion of the casing. Consequently, the brackets for bearings were designed to accommodate specific bearing sizes, each with a thickness of 20 millimetres. Initially, in order to extract the bearings and shafts, our first step is to detach the upper section of the housing. Consequently, there is no need for bearing caps.

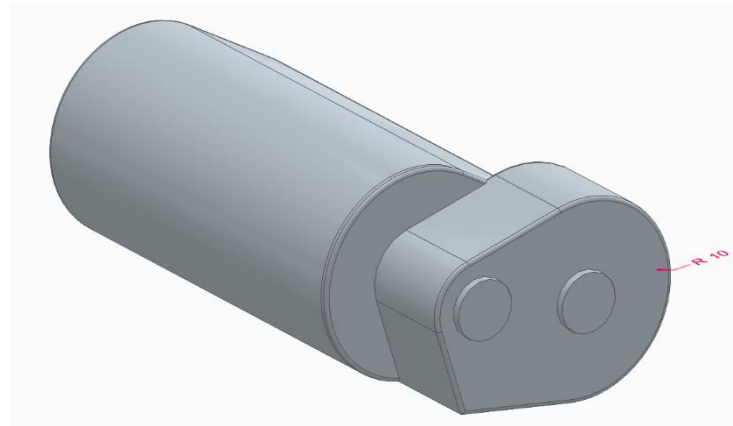


Figure 37: brackets for bearings

There are two parts of the casing.

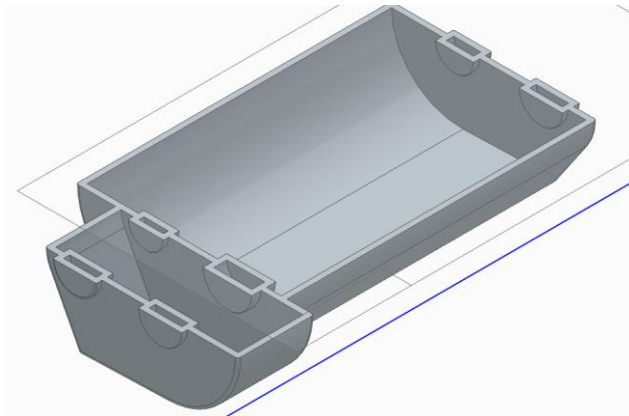


Figure 38 - Bottom part

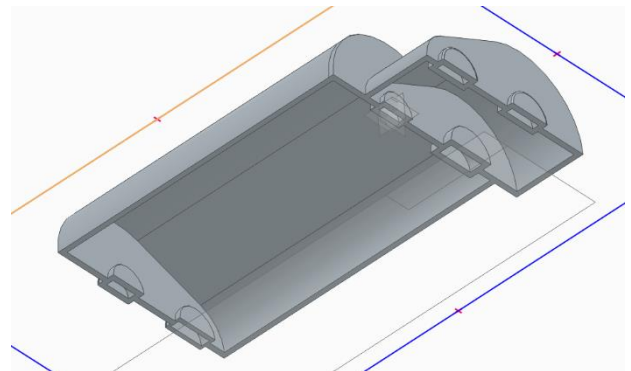


Figure 39 - Top part

To secure the supporters in the machine, we utilized three of them, fastened with six bolts. The thickness of each support was designed to 40 millimetres to enhance its strength. These supporters were strategically positioned near the bearings to ensure an even distribution of loads.

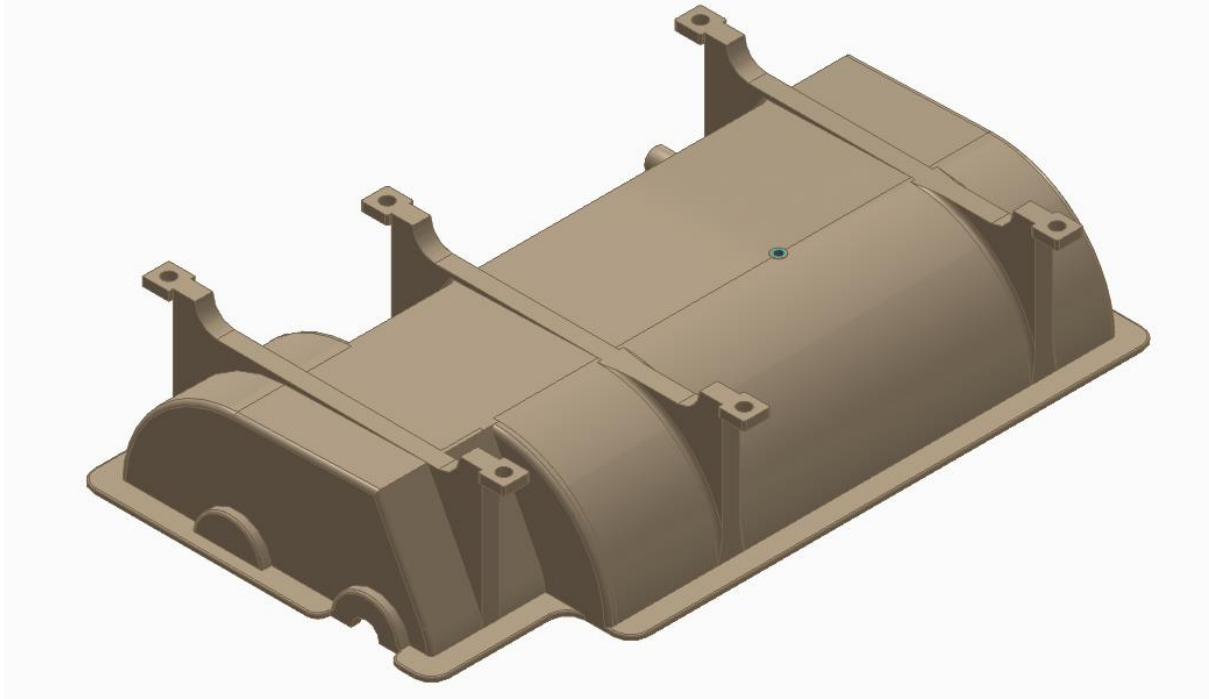


Figure 40: Bottom part from bottom view

The presence of the bevel gear results in external forces exerting shear stress on the six bolts.

Maximum external force = 1408.769 N

Given the assumption that the bolts are M30, we can determine the maximum shear force they can endure,

$$P = \tau \times n \times A$$

$$P = 42.5 \times 6 \times \pi \times \left(\frac{30}{2}\right)^2$$

$$P = 180248.87 \text{ N}$$

With safety factor 3, maximum external force,

$$F = 1408.769 \times 3$$

$$F = 4226.307 \text{ N}$$

$F < P$. Hence, the system is safe.

8.6.1 Calculation for the number of bolts required to fix two parts

To select the appropriate screw size, we opted for a method that involves comparing the required quantity of screws based on the specific screw size. Let's suppose the screw size is M20.

Permissible tensile stress in stainless steel = 85 Mpa

Permissible shear stress in stainless steel = 42.5 Mpa

By considering upward forces acting on housing,

$$P = \sigma_0 \times n \times A$$

$$P = 3.62 + 19.5 + 8.44 + 35.7 + 5.32 + 7.36 = 79.94 \text{ kN}$$

Let's consider safety factor 3 due to this machine is subjected to very high vibrations.

From maximum horizontal loads acting on bearings,

$$P = 7.48 + 14.1 + 23 + 17.6 + 9.46 + 9.2 = 80.84 \text{ kN}$$

$$n = \frac{P \times 2}{\sigma_0 \times A}$$

$$n = \frac{79940 \times 3}{85 \times \pi \times \left(\frac{20}{2}\right)^2}$$

$$n = 8.9808$$

Because of the horizontal forces exerted on the bearings, shear forces can propagate through the bolts. Let's assume that the shear force is uniformly distributed across each bolt. Therefore, we need to determine the necessary quantity of bolts.

$$P = \tau \times n \times A$$

$$n = \frac{P \times 2}{\tau \times A}$$

$$n = \frac{80840 \times 3}{42.5 \times \pi \times \left(\frac{20}{2}\right)^2}$$

$$n = 18.1638$$

Therefore, the required minimum number of bolts = 19.

8.6.2 Oil Drain Nut

At the lower part of the housing, there is a hole for the purpose of oil drainage. An oil removal bolt, equipped with a washer, is employed to guarantee a flawless seal. The bolt has a nominal diameter of 20 millimeters, and the hole in the bottom is specially designed with threads to ensure proper sealing.

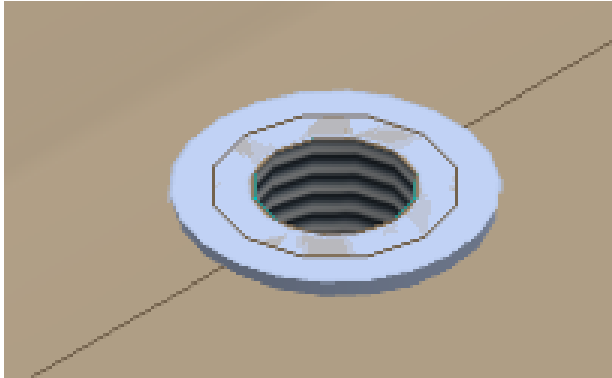


Figure 41 - Hole in the housing

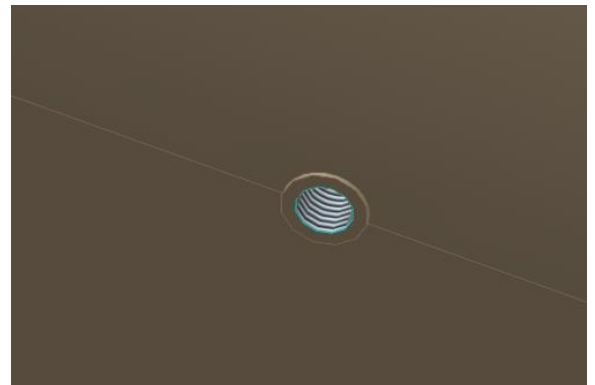


Figure 42 - Hole with washer

The gear box is mounted on the machine with a bracket. And the height of the bottom and the mounting surface is 50mm.

There is a hole for oil insertion into the housing, eliminating the need to disassemble the top portion of the casing. The bolt used for this purpose has a nominal diameter of 20 millimeters, and to ensure a secure seal, a washer is employed.

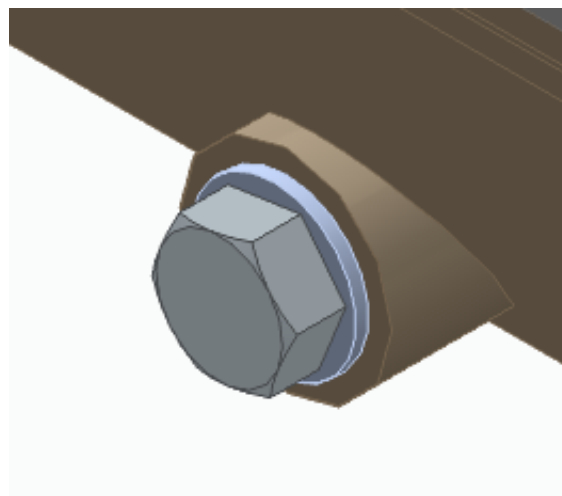


Figure 43 - Lubrication oil insertion hole with bolt

We have omitted the use of bearing caps in our design, as the disassembly process involves starting with the removal of the upper part of the housing.

9 Future work

Improvements on following points are determined to be covered in future design iterations.

Examine the existing lubrication system with fluid simulations to determine where improvements can be done. Further work should be carried out to determine the ideal operating conditions regards to lubrication for the gearbox.

Initial gear ratio from input shaft to layshaft is selected by norm. Efficiency and performance can be further increased by modifying gear ratios to better suit the application.

Further studies on vibrations has to be done. It is possible to increase machine comfort and extend component life by implementing different types of mounts, sound-absorbing materials, and vibration dampers if needed.

Topology optimization can be done on gears and casing to further decrease the material volume, hence decrease costs.

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